HOPE Under the Microscope

An Independent Executable Audit in Twelve Parts

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Reader's guide

This report collects the twelve public articles from *HOPE Under the Microscope* in one document. It is an independent executable audit of HOPE v1, with bounded diagnostics and a proposed corrected reference design. This is the version 1.0.0 report.

The work is not author code and does not claim an exact reproduction of the paper's ResNet-50/ImageNet or complete DEFT experiments. Each chapter distinguishes paper claims, verified facts, independent inferences, counterexamples, proposed repairs, and open questions.

The companion repository contains the standalone reproduction programs, focused tests, and authenticated CSV, JSON, and figure artifacts:

https://github.com/basharelnashef/hope-under-the-microscope

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}
```

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Questions, corrections, and contributions

Email Bashar Elnashef at `elnashefb@gmail.com` with questions, corrections, reproducibility reports, or proposals. Public problems and contributions can also use the repository's issue tracker and contribution guide.

Chapter 1

A neuron is not a filter

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `invariance_metrics.csv`
- Evidence: `summary.json`

From the repository root:

```
python -m unittest discover -s experiments/audits/part01_positive_
rescaling -p 'test_*.py' -v
```

Scope: A deterministic local-function audit, not a trained-model compression experiment.

Figure status: This part's three PNGs are static editorial visualizations of the nearby derivations and authenticated results, not computational outputs. Their exact published bytes are authenticated by `artifacts/SHA256SUMS`.

1.1 Why two wildly different parameter sets can represent the same contribution

Part 1 of *HOPE Under the Microscope*

Estimated reading time: 15 minutes

Take one trained ReLU neuron. Multiply its effective incoming affine map by 1,000. Divide every connection leaving that neuron by 1,000.

The incoming side is now enormous. The outgoing side is tiny. Yet the neuron computes exactly the same contribution for every input.

Which version is more important?

Functionally, neither. They are the same object written with different parameters. A pruning score based on only one side can nevertheless rank them very differently.

This rescaling symmetry is a useful place to begin understanding HOPE v1, the Hilbert Operator for Progressive Encoding framework proposed by Hossein Mobahi and Peter L. Bartlett. Before HOPE asks which neuron to prune or which pair to merge, it asks a more basic question:

> What should count as a neuron when the goal is to measure function rather than parameter size?

Its answer is simple enough to remember:

> A filter detects. BatchNorm calibrates. The activation gates. Outgoing weights route. The local function needs all four.

That change in the unit of analysis removes a parameterization ambiguity. It is also the foundation on which every later part of HOPE is built.

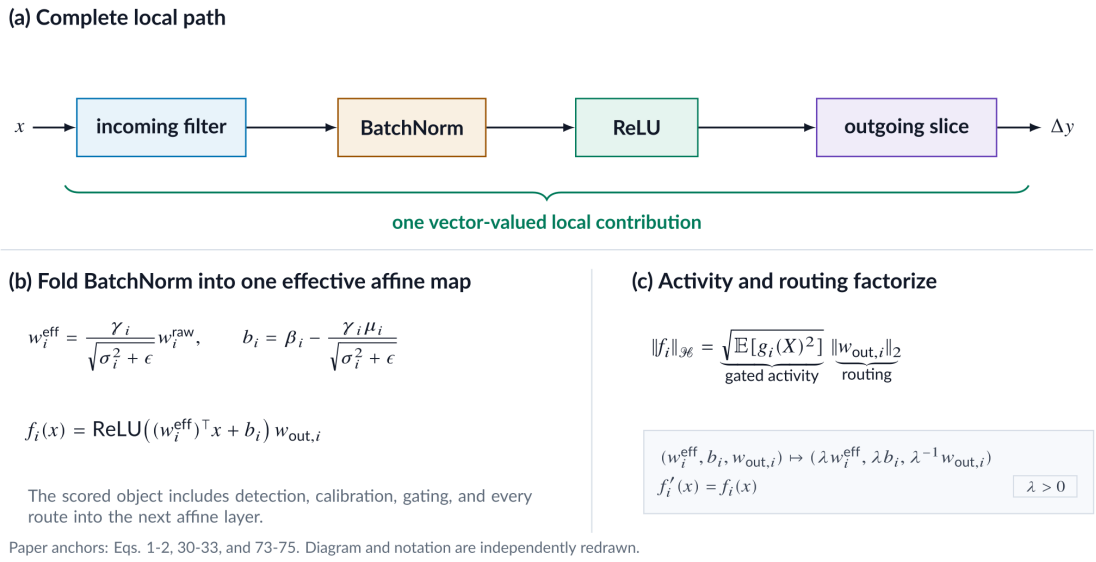


Figure 1.1: Three panels show the complete local path, BatchNorm folding, and capacity factorization.

Figure 1. A vector-source interpretation of HOPE Sections 3 and 5. The object being measured is the full local contribution to the next affine layer, not the incoming filter alone.

1.2 The filter is an incomplete compression atom

Structured pruning often talks about removing filters or channels. That language is practical because deleting a channel can remove an incoming filter, its BatchNorm state, and a slice of the next layer. But the parameter tensor we delete is not necessarily the right object to score.

Three small examples expose the missing information.

1.2.1 The loud but disconnected detector

Suppose an incoming filter has a very large norm, but every outgoing connection from its channel is zero.

The filter may produce a large activation. Nothing consumes it. Its contribution to the next affine layer is exactly zero.

If the scalar activation is $g_i(x)$ and the outgoing vector is $w_{\text{out},i}$, then

$$f_i(x) = g_i(x) w_{\text{out},i}.$$

Therefore,

$$w_{\text{out},i} = 0 \quad \Longrightarrow \quad f_i(x) = 0 \quad \text{for every } x.$$

An incoming-weight norm cannot see the disconnection.

1.2.2 The zero filter with a live bias

Now reverse the situation. Let both the raw incoming filter and its effective incoming vector be zero, but let the effective bias be positive:

$$w_i^{\text{raw}} = 0, \qquad w_{\text{in},i}^{\text{eff}} = 0, \qquad b_i > 0.$$

For ReLU,

$$f_i(x) = \text{ReLU}(b_i) w_{\text{out},i},$$

which is a nonzero constant contribution whenever $w_{\text{out},i} \neq 0$. An incoming-filter norm reports zero even though the local function is alive.

1.2.3 The same function with different scales

Consider the dense or one-channel 1x1-convolutional contribution

$$f(x) = \mathrm{ReLU}(w^\top x + b)v.$$

For any $\lambda > 0$, define

$$\widetilde{w} = \lambda w, \qquad \tilde{b} = \lambda b, \qquad \tilde{v} = \frac{v}{\lambda}.$$

ReLU is positively homogeneous of degree one:

$$\mathrm{ReLU}(\lambda z) = \lambda\,\mathrm{ReLU}(z) \qquad \text{for } \lambda \geq 0.$$

So

$$\begin{aligned}\tilde{f}(x) &= \mathrm{ReLU}\left(\lambda w^\top x + \lambda b\right)\frac{v}{\lambda} \\ &= \lambda\,\mathrm{ReLU}(w^\top x + b)\frac{v}{\lambda} \\ &= f(x).\end{aligned}$$

The equality is pointwise. It needs no dataset, Gaussian assumption, or asymptotic argument.

The incoming norm changes by a factor of λ. The outgoing norm changes by $1/\lambda$. The function does not move.

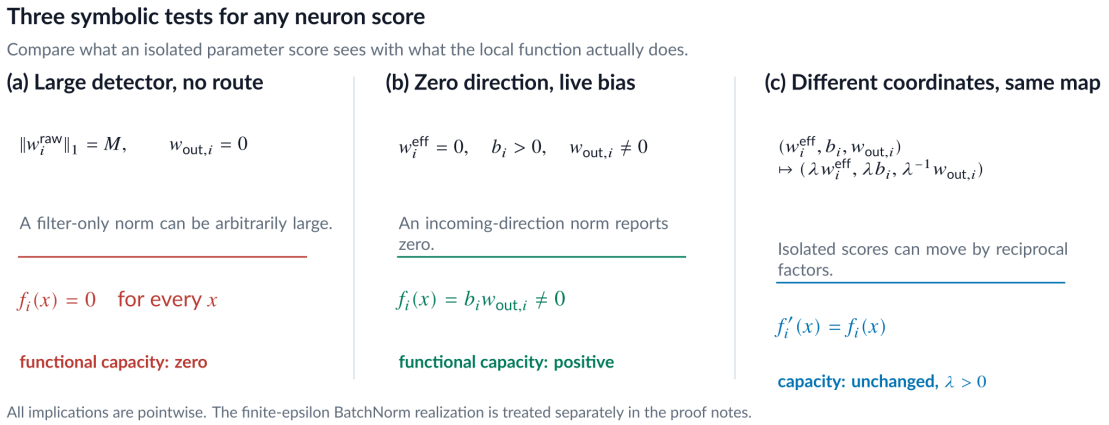


Figure 1.2: Three cards show a disconnected large filter, a live bias-only channel, and an exact positive rescaling.

Figure 2. Derived examples from the accompanying experiment. The numerical values are simulated, not reported by the paper.

These cases do not show that incoming-weight norms are always useless. They show that an incoming tensor does not contain enough information to identify a channel's local functional contribution.

1.3 HOPE folds the pieces together

Section 3 of the paper starts with a fully connected presentation. Neuron i has raw incoming weights w_i^{raw}, BatchNorm affine parameters (γ_i, β_i), stored mean and variance (μ_i, σ_i^2), and outgoing weights $w_{\text{out},i}$.

For the bias-free pre-BN affine map used in paper Eq. 1, BatchNorm can be absorbed into effective parameters at inference:

$$w_{\text{in},i}^{\text{eff}} = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} w_i^{\text{raw}}, \qquad b_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}.$$

The scalar preactivation and vector-valued contribution are

$$y_i(x) = \left(w_{\text{in},i}^{\text{eff}}\right)^\top x + b_i,$$

$$f_i(x) = \Psi(y_i(x)) w_{\text{out},i}.$$

If the Conv or Dense layer has a raw affine bias, that bias also contributes to the folded b_i. The equations here match the paper's bias-free definition and the accompanying reference simulation.

This is HOPE's neuron.

The paper restricts Ψ to positively homogeneous degree-one, or PH-1, activations for its rescaling arguments. ReLU, Leaky ReLU, PReLU, and linear activations are named examples. GELU, sigmoid, SiLU, and tanh do not have this property.

Each component now has a clear job:

- $w_{\text{in},i}^{\text{eff}}$ selects the input direction.
- b_i decides where the nonlinear gate opens.
- $\Psi(y_i)$ says how strongly the neuron fires.
- $w_{\text{out},i}$ says where that scalar signal is sent.

The object is still local. It is the contribution reaching the next affine layer, not the neuron's complete effect after every downstream nonlinearity. That distinction will matter later.

1.4 What does "outgoing weight" mean for a convolution?

Appendix B.1 translates the definition to an ordinary convolutional chain.

For filter i in layer A, the incoming object is the flattened filter slice

$$\mathrm{vec}\left(K_A[:,:,:,i]\right).$$

That feature map becomes input channel i of the next convolution. Its outgoing object is therefore the flattened downstream slice

$$\mathrm{vec}\left(K_B[:,:,i,:]\right).$$

This is a crucial detail. The outgoing influence is not generally one scalar per output channel. For a spatial downstream kernel, it includes the complete spatial slice that consumes channel i.

The paper assumes spatial stationarity when converting this convolutional object into its analytical functional norm. The basic rescaling cancellation itself is simpler and pointwise: if a channel's activation is multiplied by λ, every consumer of that channel must be divided by λ.

This means the clean proof applies directly to an ordinary chain. Networks with multiple consumers, grouped convolutions, concatenations, or identity skips need graph-aware handling. Scaling only one outgoing branch would change the network.

1.5 A function needs a ruler

Once the neuron is treated as a function, saying that it is "large" requires a ruler.

HOPE places the vector-valued functions in

$$\mathcal{H} = L_2(\mathcal{X}, P_X; \mathbb{R}^c).$$

The notation sounds heavier than the idea. It means:

- inputs x are weighted according to a distribution P_X;
- the neuron produces a vector in $\mathbb{R}^c$;
- its average squared output must be finite.

The inner product from Appendix A, Eq. 30, is

$$\langle f_i, f_j \rangle_{\mathcal{H}} = \mathbb{E}_{x \sim P_X}\left[f_i(x)^\top f_j(x)\right].$$

The resulting norm is

$$\|f_i\|_{\mathcal{H}} = \sqrt{\mathbb{E}_{x\sim P_X}\left[\|f_i(x)\|_2^2\right]}.$$

In plain language, this is the root-mean-square size of the neuron's local output under the inputs that P_X considers important.

That last phrase is a dependency, not a detail. Different choices of P_X can assign different geometry to the same functions. Part 3 of this series will examine how HOPE constructs P_X without the original data. For now, we only need a fixed distribution to understand the representation.

1.6 Why the operator has rank at most one

Write the scalar gated activity as

$$g_i(x) = \Psi\left(\left(w_{\text{in},i}^{\text{eff}}\right)^\top x + b_i\right).$$

Then

$$f_i(x) = g_i(x) w_{\text{out},i}.$$

The input changes the scalar $g_i(x)$. The vector direction $w_{\text{out},i}$ stays fixed. Every local output therefore lies in $\text{span}\{w_{\text{out},i}\}$, which is at most one-dimensional.

HOPE writes this separation as

$$f_i = g_i \otimes w_{\text{out},i},$$

and identifies the simple tensor with a Hilbert-Schmidt operator. When both g_i and $w_{\text{out},i}$ are nonzero, that operator has rank one. If either factor is zero, f_i is the zero operator and has rank zero.

In the nonzero rank-one case, the formal terminology reduces to two factors. One describes how the neuron reacts to inputs. The other describes how that reaction is broadcast.

The tensor structure makes the inner product factor:

$$\langle f_i, f_j\rangle_{\mathcal{H}} = \underbrace{\mathbb{E}[\Psi(y_i)\Psi(y_j)]}_{K(i,j)} \langle w_{\text{out},i}, w_{\text{out},j}\rangle.$$

For one neuron, Eqs. 73 through 75 give

$$\|f_i\|_{\mathcal{H}} = \sqrt{\mathbb{E}[\Psi(y_i)^2]}\, \|w_{\text{out},i}\|_2.$$

This is the central intuition of Part 1:

Functional capacity is RMS gated activity multiplied by downstream routing strength.

Neither factor is enough by itself.

1.7 The exact invariance proof

We can now prove that HOPE's full functional capacity ignores the harmless positive rescaling from the opening puzzle.

Let

$$\tilde{y}_i(x) = \lambda y_i(x), \qquad \widetilde{w}_{\text{out},i} = \frac{w_{\text{out},i}}{\lambda}, \qquad \lambda > 0.$$

For a PH-1 activation,

$$\Psi(\tilde{y}_i) = \lambda \Psi(y_i).$$

Therefore,

$$\tilde{f}_i(x) = \Psi(\tilde{y}_i)\widetilde{w}_{\text{out},i} = \lambda \Psi(y_i)\frac{w_{\text{out},i}}{\lambda} = f_i(x).$$

The self-kernel transforms as

$$\widetilde{K}(i,i) = \mathbb{E}[\Psi(\lambda y_i)^2] = \lambda^2 K(i,i).$$

Meanwhile,

$$\left\|\widetilde{w}_{\text{out},i}\right\|_2 = \frac{\|w_{\text{out},i}\|_2}{\lambda}.$$

Combining the two:

$$\begin{aligned} \|\tilde{f}_i\|_{\mathcal{H}} &= \frac{\|w_{\text{out},i}\|_2}{\lambda}\sqrt{\lambda^2 K(i,i)} \\ &= \|w_{\text{out},i}\|_2 \sqrt{K(i,i)} \\ &= \|f_i\|_{\mathcal{H}}. \end{aligned}$$

This is not merely a score that happens to be stable. The underlying local function is identical.

1.8 How the exact rescaling appears in a real BN network

The clean physical realization leaves the raw filter and stored BatchNorm moments fixed, then applies

$$\gamma_i \mapsto \lambda\gamma_i, \qquad \beta_i \mapsto \lambda\beta_i, \qquad w_{\mathrm{out},i} \mapsto \frac{w_{\mathrm{out},i}}{\lambda}.$$

The folded definition from paper Eq. 1 immediately gives

$$w^{\mathrm{eff}}_{\mathrm{in},i} \mapsto \lambda w^{\mathrm{eff}}_{\mathrm{in},i}, \qquad b_i \mapsto \lambda b_i.$$

This transformation is exact at finite BatchNorm ϵ, provided the scale is positive and every outgoing consumer is transformed.

It also shows why a BN-scale score can change while the function stays fixed. Under this transformation, $|\gamma_i|$ grows by λ. The outgoing norm shrinks by $1/\lambda$. The full capacity remains constant.

1.9 Technical note: scaling the raw filter through BatchNorm

Readers focused on the central functional-neuron idea can skip this note on a first pass. It separates exact post-BN positive rescaling from the related raw-filter normalization claim.

The paper also discusses normalization invariance. If the raw filter is scaled by $\alpha > 0$, its stored mean and variance should naturally transform as

$$w^{\mathrm{raw}} \mapsto \alpha w^{\mathrm{raw}}, \qquad \mu \mapsto \alpha\mu, \qquad \sigma^2 \mapsto \alpha^2\sigma^2.$$

Ignoring ϵ, BatchNorm cancels α. At finite ϵ, however,

$$\frac{\alpha(z-\mu)}{\sqrt{\alpha^2\sigma^2+\epsilon}} = \frac{z-\mu}{\sqrt{\sigma^2+\epsilon/\alpha^2}}.$$

This is not exactly equal to the original normalized signal unless $\epsilon = 0$, $\alpha = 1$, or both σ^2 and $\alpha^2\sigma^2$ dominate ϵ.

An exact inference reparameterization can compensate through γ:

$$\tilde{\gamma} = \gamma\frac{\sqrt{\alpha^2\sigma^2+\epsilon}}{\alpha\sqrt{\sigma^2+\epsilon}}.$$

With this compensation, both effective parameters in paper Eq. 1 remain exactly fixed.

This qualification matters. Exact post-BN positive rescaling and raw-filter normalization are related, but they are not the same transformation. The first is exact for positive scales. The second needs the finite-ϵ caveat or compensation.

1.10 The framework-independent numerical check

The accompanying NumPy reference models the algebra of a one-channel 1x1 Conv-BN-ReLU-Conv path. It is a framework-independent simulation, not a PyTorch or TensorFlow module test. A 1x1 convolution is a matrix multiplication at every spatial position, which keeps every term of the proof visible.

The experiment uses:

- float64 NumPy;
- 8,192 seeded standard-Gaussian input positions in nine dimensions;
- 25 logarithmically spaced scales from 10^{-3} through 10^{3};
- the exact finite-ϵ variance from paper Eq. 77;
- exact post-BN rescaling and exact finite-ϵ raw-filter rescaling.

The BN buffers are estimated from those same positions. For every scale, the script records the local output, isolated parameter scores, empirical RMS contribution, the finite-ϵ Gaussian-surrogate capacity, and the paper's epsilon-free approximation. It also runs the natural uncorrected BN scaling as a negative control.

The largest relative output error across all 50 invariant transformed configurations on the evaluation sample was

$$2.42 \times 10^{-16}.$$

The largest deviation of the finite-ϵ extension from ratio one was

$$2.22 \times 10^{-16}.$$

The isolated parameter scores moved by as much as a factor of 1,000 in either direction. Under the exact raw-filter transformation, the paper's printed epsilon-free approximation drifted by 39.0 percent at the most extreme scale. The figure now displays both formulas.

Figure 3. Derived float64 simulation. Panel A tests exact post-BN PH-1 rescaling. Panel B compares the paper's epsilon-free capacity with an explicit finite-ϵ extension under an exact raw-filter reparameterization. Panel C reports sampled output differences for both exact transformations and the uncorrected BN negative control. Errors are measured on 8,192 seeded inputs; the separate algebraic proof is pointwise. No sampling uncertainty is claimed for the identity itself.

The core transformation in the implementation is short:

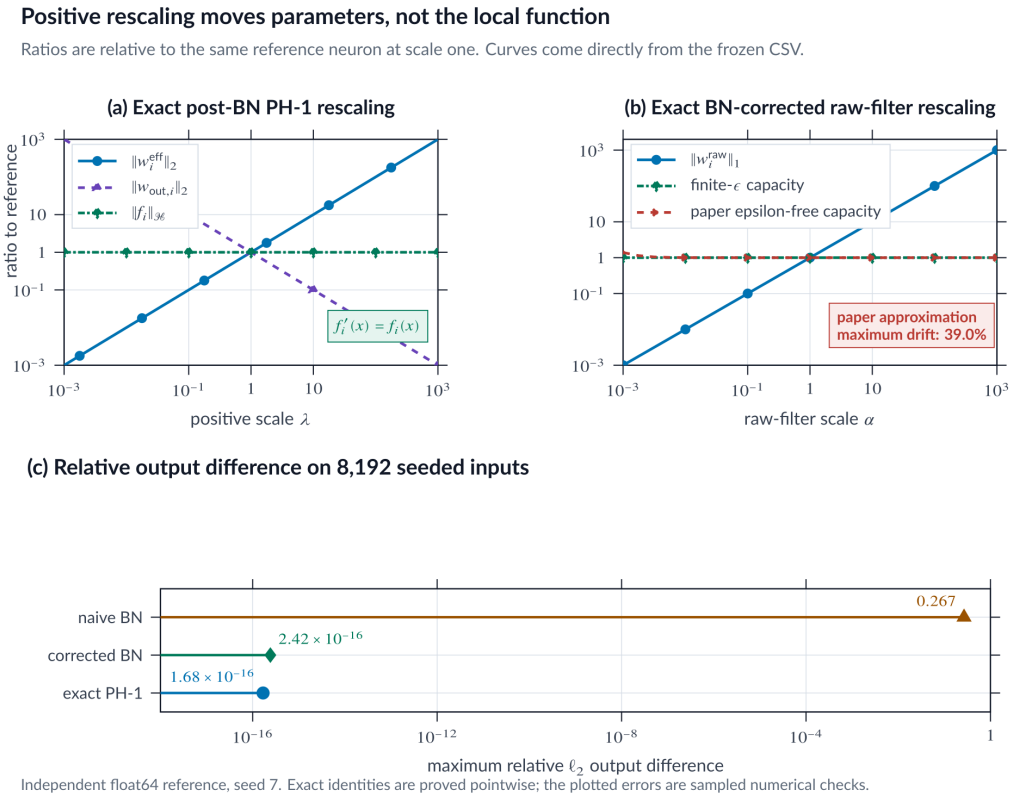


Figure 1.3: Two scale sweeps and one residual panel show parameter movement, invariant capacity, and a finite-epsilon negative control.

```
def ph1_rescale(self, scale):
    if scale <= 0:
        raise ValueError("PH-1 rescaling requires a positive scale")
    return replace(
        self,
        gamma=self.gamma * scale,
        beta=self.beta * scale,
        outgoing=self.outgoing / scale,
    )
```

The full script, result table, and tests are linked in the package README.

1.11 What this viewpoint buys us

The functional-neuron definition does several useful things at once.

1.11.1 It removes an important coordinate choice

A score should not change merely because activation scale was shifted from one side of a ReLU channel to the other. HOPE's paper capacity passes this exact post-BN test.

1.11.2 It notices disconnected features

If the outgoing slice is zero, capacity is zero regardless of how visually dramatic the incoming filter looks.

1.11.3 It includes the gate

The effective bias and activation statistics decide whether a channel is usually silent, occasionally active, or constantly on. Raw filter norm does not encode that behavior.

1.11.4 It provides a common geometry under one fixed input model

Exact local functions evaluated under the same fixed P_X share lengths and alignments in one function-space geometry. This fact alone does not establish that separately constructed pairwise surrogates assemble into one global positive-semidefinite geometry; that question is deferred to a later part.

1.11.5 It gives compression a better first question

Instead of asking, "Which tensor is small?" we can ask, "Which local vector-valued contribution is small under the chosen input model?"

That question removes a parameterization ambiguity before later assumptions are judged.

1.12 The questions this creates

It is important to stop exactly where the proof stops.

This post verifies a narrow but useful result:

- the complete local function is invariant to exact positive PH-1 rescaling;
- its L_2 norm is invariant under a fixed input distribution;
- isolated tensor scores can change while that function does not.

It leaves three natural questions for the rest of the series:

1. Which input distribution should define the RMS activity?
2. Can many pairwise neuron comparisons share one consistent geometry?
3. How should an abstract functional operation map back to a physical network edit?

Part 2 derives the closed-form capacity under HOPE's Gaussian marginal. Part 3 then examines where that marginal comes from.

1.13 The change in the unit of thought

HOPE's first contribution is not a complicated pruning formula. It is a change in what we agree to call the object.

A filter is a sensor. Its numerical size says little without its calibration, gate, and wiring. Once those pieces are treated as a single local function, some apparent differences disappear and some invisible differences become measurable.

The compact takeaway is:

> If an importance score changes while the function does not, it has measured the parameterization rather than the functional contribution.

In Part 2, we will open the RMS term in the capacity factorization. For ReLU under HOPE's Gaussian marginal, it has a closed form. At zero BatchNorm shift, the resulting capacity collapses to a surprisingly familiar score.

Evidence records:

- `P01-FUNCTIONAL-ATOM`: `paper_claim`
- `P01-POSITIVE-RESCALING`: `verified_fact`

- `P01-FINITE-EPSILON-CAVEAT`: `independent_inference`, verified on the frozen reference experiment

Primary source: HOPE v1, arXiv:2607.21366v1

Primary anchors: paper Sections 3, 3.1, and 5; Figure 1; Eqs. 1, 2, 30 through 33, 73 through 75, and 77; Appendix B.1.

Code and numerical results: Part 1 positive-rescaling audit

Chapter 2

The closed-form capacity inside HOPE

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `metrics.csv`
- Evidence: `summary.json`

From the repository root:

```
python -m unittest discover -s experiments/audits/part02_capacity -p
'test_*.py' -v
```

Scope: Analytical and simulated diagnostics only; no paper checkpoint is claimed.

Figure status: This part's four PNGs are static editorial visualizations of the nearby derivations and authenticated results, not computational outputs. Their exact published bytes are authenticated by `artifacts/SHA256SUMS`.

2.1 A truncated Gaussian moment, two revealing boundaries, and the simpler score contained inside it

Part 2 of *HOPE Under the Microscope*

Estimated reading time: 18 minutes

Status: published in the `v1.0.0` series from the frozen Part 2 audit package. Publication does not broaden the evidence scope stated below.

Seventy pages of Hilbert geometry eventually produce a surprisingly familiar pruning statistic.

That is not a criticism. A good mathematical framework should often reduce to something simple when its assumptions are specialized. The useful questions are whether the reduction is correct, what information survives it, and which baseline should be used to judge the extra machinery.

For HOPE's ReLU neuron, the answer begins with one scalar random variable. Under the paper's negligible-epsilon Gaussian marginal,

$$Y_i \sim \mathcal{N}(\beta_i, \gamma_i^2),$$

and the neuron's squared gated activity is

$$K(i,i) = \mathbb{E}[\mathrm{ReLU}(Y_i)^2].$$

The paper gives a closed form for this expectation. We will derive it from the one-dimensional integral, inspect the cases the displayed formula does not cover directly, and connect it to the pruning cost.

The most revealing specialization is $\beta_i = 0$:

$$\|f_i\|_{\mathcal{H}} = \frac{|\gamma_i|}{\sqrt{2}} \|w_{\mathrm{out},i}\|_2.$$

So zero-shift HOPE capacity is not BN scale alone. It is BN scale multiplied by the strength of every outgoing route, with a common $1/\sqrt{2}$ factor.

2.2 Chapter contract and evidence legend

This chapter answers a narrow question:

> Under HOPE's one-dimensional Gaussian marginal, what exactly is the ReLU self-kernel, how should it be evaluated, and what does it imply for solitary within-layer pruning?

Every central conclusion uses one of the repository's evidence classes:

- `paper_claim`: what HOPE v1 prints, tied to an exact PDF page and equation.
- `verified_fact`: what this project independently proves or checks.
- `independent_inference`: an interpretation assembled from those premises, with its scope stated.

The claim registry keeps the paper's formula separate from our verification:

- `P02-RELU-SELF-KERNEL`: `paper_claim`

- P02-RELU-SELF-KERNEL-VERIFIED: verified_fact
- P02-ZERO-SHIFT-CAPACITY: verified_fact
- P02-DEGENERATE-GAMMA-LIMIT: verified_fact
- P02-PRUNE-RANK-MONOTONICITY: verified_fact
- P02-REFINED-BN-BASELINE: independent_inference

The exact source is HOPE v1, arXiv:2607.21366v1. Its versioned PDF fingerprint is recorded in the paper manifest. The paper traceability summary above records every page used here.

2.3 The strongest fair version of the paper's claim

Evidence: paper_claim, P02-RELU-SELF-KERNEL.

The strongest fair statement is not that HOPE knows a neuron's true activity without data. It is conditional:

> If the preactivation marginal is modeled by the paper's Gaussian Eq. 78, then a ReLU neuron's expected squared activity is available in closed form, and the complete local capacity also includes the outgoing weight norm.

Paper Section 5 on PDF pages 7 and 8 defines the vector-valued functional neuron and its Hilbert norm. Paper Eqs. 73 through 75 on PDF page 52 then separate its capacity into

$$\boxed{\|f_i\|_{\mathcal{H}} = \|w_{\text{out},i}\|_2 \sqrt{K(i,i)}}.$$

Paper Eq. 73 defines the scalar kernel as

$$K(i,j) = \mathbb{E}_{x\sim P_X}\left[\Psi(y_i)\Psi(y_j)\right].$$

For one neuron and ReLU,

$$K(i,i) = \mathbb{E}[\max(0, Y_i)^2].$$

Paper Eq. 3 on PDF page 9 and Eq. 79 on PDF page 53 print

$$\boxed{K(i,i) = (\gamma_i^2 + \beta_i^2)\Phi\left(\frac{\beta_i}{|\gamma_i|}\right) + \beta_i|\gamma_i|\phi\left(\frac{\beta_i}{|\gamma_i|}\right)}$$

for the positive-variance setting stated in Theorem E.2.

The claim is useful for two reasons. The high-dimensional input has disappeared from the final integration, and the output routing remains visible as a separate multiplier. Once the marginal parameters are available, each self-capacity is a constant-size scalar computation plus an outgoing-vector norm.

This chapter verifies that scalar computation. It does not verify how well the Gaussian marginal describes a trained network. That empirical question belongs to Part 3.

2.4 From a functional norm to one truncated moment

Write the gated scalar response as

$$g_i(x) = \mathrm{ReLU}(y_i(x)),$$

and the local vector contribution as

$$f_i(x) = g_i(x) w_{\mathrm{out},i}.$$

Then

$$\begin{aligned} \|f_i\|_{\mathcal{H}}^2 &= \mathbb{E}\left[\|g_i(x) w_{\mathrm{out},i}\|_2^2\right] \\ &= \mathbb{E}[g_i(x)^2] \|w_{\mathrm{out},i}\|_2^2 \\ &= K(i,i) \|w_{\mathrm{out},i}\|_2^2. \end{aligned}$$

Taking the nonnegative square root recovers paper Eq. 75.

In plain language:

> Capacity is RMS gated activity multiplied by downstream routing strength.

The first factor asks how much scalar signal crosses the ReLU threshold under the chosen marginal. The second asks how strongly the next affine layer uses that signal.

Figure 1. Capacity factorization from paper Eqs. 73 through 75. This is a derived explanatory visual, not an experimental result.

2.5 The Gaussian assumption enters before the closed form

Evidence: `paper_claim`.

The order of Eqs. 77 and 78 matters.

On PDF page 52, paper Eq. 77 retains finite BatchNorm epsilon:

$$\mathrm{Var}(Y_i) = \gamma_i^2 \frac{\sigma_i^2}{\sigma_i^2 + \epsilon}.$$

The paper then says that epsilon is assumed negligible compared with the raw pre-BatchNorm variance. Under that approximation, Eq. 78 writes

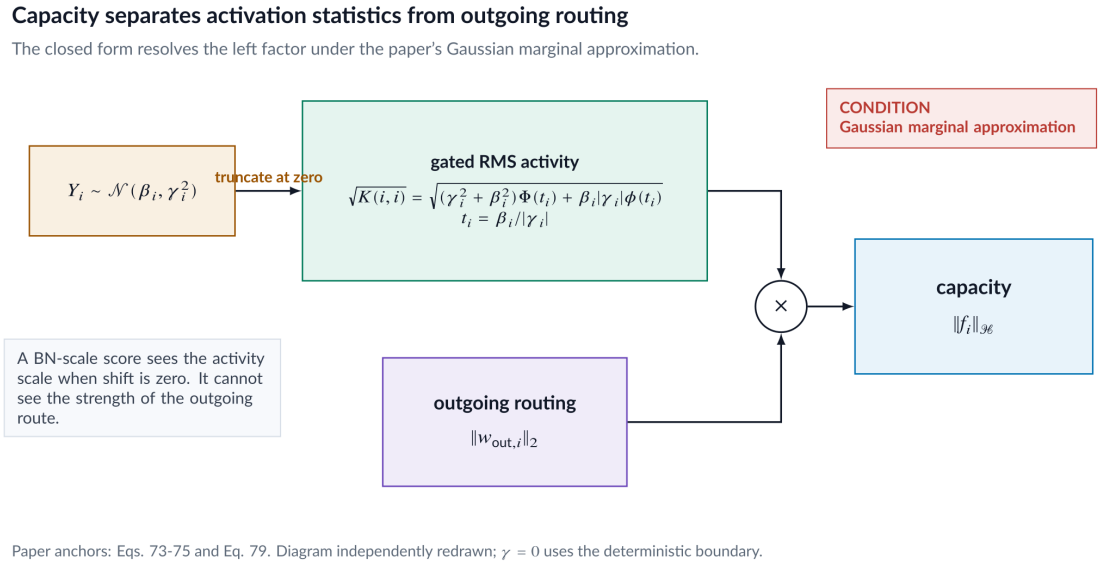


Figure 2.1: A product diagram separates gated RMS activity from outgoing routing strength.

$$Y_i \sim \mathcal{N}(\beta_i, \gamma_i^2).$$

The closed form in Eq. 79 follows from this latter marginal. It is not a finite-epsilon formula printed by the paper.

For now, follow Eq. 78 exactly. Let

$$s_i = |\gamma_i| > 0, \qquad a_i = \frac{\beta_i}{s_i}.$$

The absolute value is not cosmetic. A Gaussian standard deviation is nonnegative, while the BatchNorm affine scale γ_i can have either sign. The marginal variance depends on γ_i^2.

2.6 Deriving the closed form

Evidence: `verified_fact`, `P02-RELU-SELF-KERNEL-VERIFIED`. The frozen executable audit passed its declared implementation gates.

The derivation starts from the density rather than the printed answer:

$$K(i,i) = \int_0^\infty y^2 \frac{1}{s_i} \phi\left(\frac{y - \beta_i}{s_i}\right) dy.$$

Only $y > 0$ remains because ReLU zeros the negative half-line.

Standardize with

$$z = \frac{y - \beta_i}{s_i}, \qquad y = s_i z + \beta_i.$$

When $y = 0$, the lower limit becomes $-a_i$, so

$$K(i,i) = \int_{-a_i}^{\infty} (s_i z + \beta_i)^2 \phi(z)\, dz.$$

Expanding the square reduces the problem to three standard-normal tail moments:

$$\int_{-a}^{\infty} \phi(z)\, dz = \Phi(a),$$

$$\int_{-a}^{\infty} z\phi(z)\, dz = \phi(a),$$

and

$$\int_{-a}^{\infty} z^2 \phi(z)\, dz = \Phi(a) - a\phi(a).$$

The first identity uses Gaussian symmetry. The second follows from $\phi'(z) = -z\phi(z)$. The third follows by integration by parts.

Insert all three:

$$\begin{aligned} K(i,i) = s_i^2[\Phi(a_i) - a_i\phi(a_i)] \\ + 2\beta_i s_i \phi(a_i) + \beta_i^2 \Phi(a_i). \end{aligned}$$

Because $s_i a_i = \beta_i$, one of the two cross terms cancels:

$$-s_i^2 a_i \phi(a_i) + 2\beta_i s_i \phi(a_i) = \beta_i s_i \phi(a_i).$$

Therefore,

$$K(i,i) = (s_i^2 + \beta_i^2)\Phi(a_i) + \beta_i s_i \phi(a_i),$$

which is exactly the formula printed in Eq. 79 after substituting $s_i = |\gamma_i|$.

The intimidating part of the expression is mostly bookkeeping. It is the second moment of a one-dimensional Gaussian after its negative half has been cut away.

2.7 A normalized view of the gate

For $s_i > 0$, write $\beta_i = s_i a_i$. Then

$$K(i,i) = s_i^2 h(a_i),$$

with

$$h(a) = (1 + a^2)\Phi(a) + a\phi(a).$$

The dimensionless function h says how the ReLU gate changes the second moment. The scale s_i^2 says how large the underlying Gaussian is.

Three regimes are easy to read:

- $a \ll 0$: the mean sits below the threshold, so only a small positive tail contributes.
- $a = 0$: the symmetric Gaussian sends half its squared energy through ReLU.
- $a \gg 0$: almost all mass lies above zero, so the moment approaches the untruncated value $\beta_i^2 + s_i^2$.

The next surface plots $\sqrt{K(i,i)}$ with unit outgoing norm, then divides every value by the maximum on its declared display grid. That grid normalization is only for visualization.

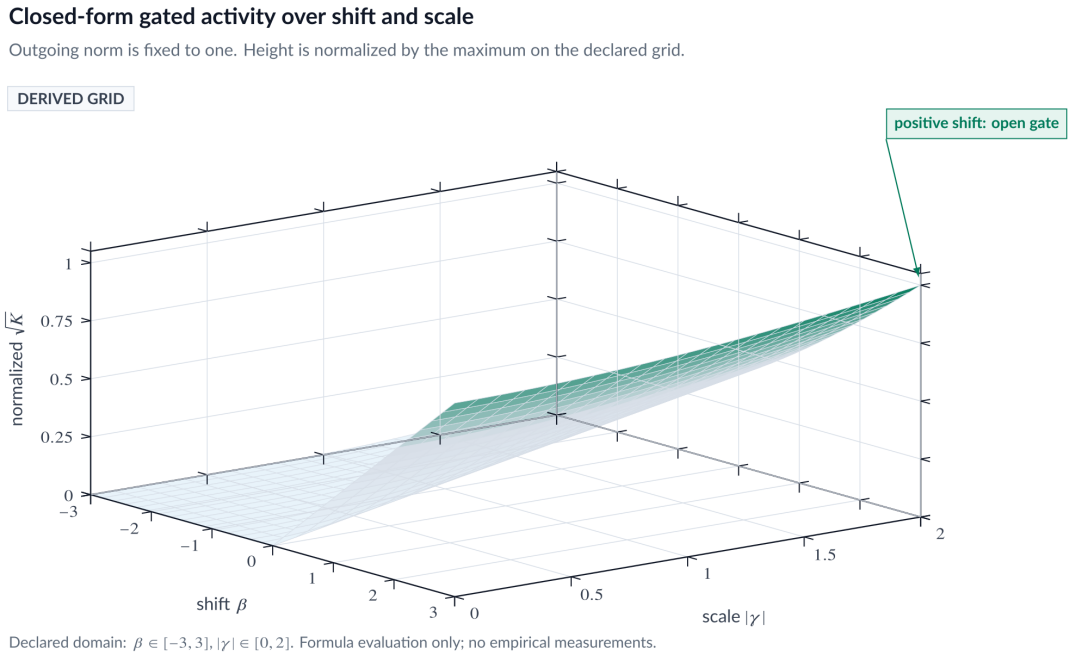


Figure 2.2: A surface shows normalized gated RMS activity across shift and scale, including the deterministic boundary.

Figure 2. Analytical gated RMS under the Eq. 78 marginal, normalized to the maximum on the displayed grid. The zero-scale edge is evaluated by its deterministic definition, not by dividing by zero.

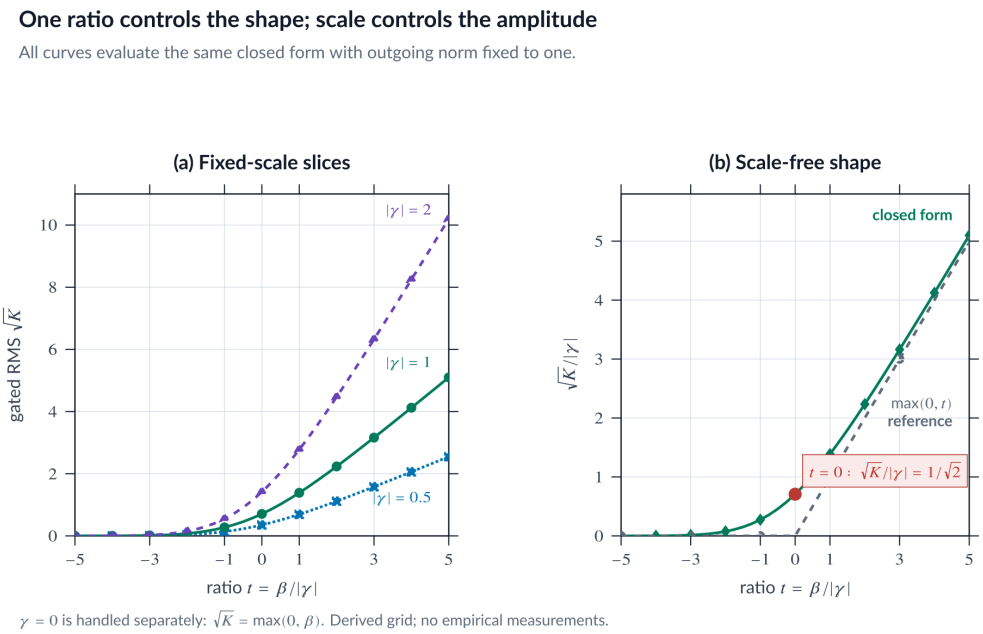


Figure 2.3: Directly labeled curves show gated RMS and its scale-free form against beta over absolute gamma.

Figure 3. Analytical gated-RMS slices; the right panel divides out absolute gamma to expose the dimensionless $\sqrt{h(a)}$ curve. These curves visualize the formula; they do not measure a trained checkpoint.

2.8 Special case 1: zero shift

Evidence: `verified_fact`, `PO2-ZERO-SHIFT-CAPACITY`.

Set $\beta_i = 0$. Then $a_i = 0$, $\Phi(0) = 1/2$, and the density term is multiplied by zero:

$$K(i,i) = \gamma_i^2 \Phi(0) = \frac{\gamma_i^2}{2}.$$

Capacity becomes

$$\boxed{\|f_i\|_{\mathcal{H}} = \frac{|\gamma_i|}{\sqrt{2}} \|w_{\text{out},i}\|_2}.$$

This identity is a project derivation from the paper's formula. The paper does not print it as a separate theorem.

The result is simple, but it preserves an important idea from Part 1. A large BN scale does not imply a large local contribution if the outgoing slice is small. If the outgoing slice is zero, capacity is exactly zero.

Conversely, two neurons with equal $|\gamma|$ can have very different capacities because their downstream routing norms differ.

2.9 What the outgoing norm changes

Compare three zero-shift scores:

$$S_{\mathrm{BN}}(i) = |\gamma_i|,$$

$$S_{\mathrm{out}}(i) = \|w_{\mathrm{out},i}\|_2,$$

and

$$S_{\mathrm{refined}}(i) = |\gamma_i| \|w_{\mathrm{out},i}\|_2.$$

The first sees marginal activation scale but not whether the feature is used. The second sees routing but not how strongly the feature fires. The product keeps both.

It also respects the positive-rescaling cancellation introduced in Part 1. If the post-BN affine response is multiplied by a positive λ and every outgoing route is divided by λ, then $|\gamma|$ grows by λ, the outgoing norm shrinks by $1/\lambda$, and their product stays fixed.

At nonzero shift, the full $h(\beta/|\gamma|)$ factor adds information about where the Gaussian lies relative to the ReLU threshold. The refined product is then a comparator, not an algebraically identical replacement.

2.10 Special case 2: zero variance

Evidence: `verified_fact`, `P02-DEGENERATE-GAMMA-LIMIT`.

Theorem E.2 on PDF page 53 assumes $|\gamma_i| > 0$. Its displayed formula contains $\beta_i/|\gamma_i|$, so substituting `gamma = 0` directly is undefined.

The probability model is not undefined. It becomes deterministic:

$$Y_i = \beta_i \quad \text{almost surely.}$$

Therefore,

$$\boxed{K(i,i) = \max(0, \beta_i)^2}.$$

This boundary includes three intuitive cases:

- $\beta_i > 0$: an always-on constant feature with activity β_i^2;
- $\beta_i = 0$: a zero feature;
- $\beta_i < 0$: an always-closed ReLU with zero activity.

The boundary is also the continuous $s_i \to 0^+$ limit of the integral. But a numerical implementation should not try to discover it by evaluating an infinite ratio. It should dispatch the deterministic case explicitly.

2.11 Why a literal transcription is numerically fragile

The closed form is exact on paper and still deserves a careful implementation.

2.11.1 Division at `gamma = 0`

A literal `beta / abs(gamma)` produces an infinity or `nan`. The correct value depends on the deterministic ReLU output, so the zero-scale branch must run before division.

2.11.2 Cancellation in the negative tail

For a large negative $a = \beta/|\gamma|$, the two closed-form terms are individually small and nearly equal in magnitude:

$$(1 + a^2)\Phi(a) \quad \text{and} \quad a\phi(a).$$

Their sum is a still smaller positive number. Ordinary floating-point subtraction can discard significant digits or produce a tiny negative value, even though a squared moment cannot be negative.

The stable implementation factors out the Gaussian exponential in the moderately negative tail. Beyond the declared switch, it evaluates an adaptively truncated Mills-ratio asymptotic series in log space. It checks nonnegativity and does not treat an arbitrary negative answer as zero.

2.11.3 Extreme positive ratios

For large positive a, the correct limit is

$$K(i,i) \to \beta_i^2 + \gamma_i^2.$$

The implementation evaluates a scale-normalized form, rejects nonfinite inputs, and raises on an unrepresentable output instead of silently sending an `inf` into a ranking.

2.11.4 Vectorized boundary behavior

Real audits evaluate arrays of neurons. The dispatcher must preserve broadcast shape, handle positive and negative γ identically, and return the exact deterministic boundary element by element.

The tested source is the authority. The publication article carries two contiguous excerpts copied from that implementation, not a second implementation. The first makes both exact boundaries explicit:

```
        zero_scale = flat_scale == 0.0
        if np.any(zero_scale):
            positive_beta = np.maximum(flat_beta[zero_scale], 0.0)
            with np.errstate(over="ignore", invalid="ignore"):
                deterministic = positive_beta * positive_beta
            if not np.all(np.isfinite(deterministic)):
                raise FloatingPointError(
                    "deterministic ReLU self-kernel exceeds float64 range"
                )
            result[zero_scale] = deterministic
        zero_beta = (~zero_scale) & (flat_beta == 0.0)
        if np.any(zero_beta):
            with np.errstate(over="ignore", invalid="ignore"):
                exact_zero_shift = (0.5 * flat_scale[zero_beta]) * flat_scale[zero_
beta]
            if not np.all(np.isfinite(exact_zero_shift)):
                raise FloatingPointError("zero-shift ReLU self-kernel exceeds
float64 range")
            result[zero_beta] = exact_zero_shift
```

The second shows the cancellation-safe negative-tail path and the final fail-closed checks:

```
            negative_ratio = ~nonnegative_ratio
            if np.any(negative_ratio):
                local_ratio = ratio[negative_ratio]
                local_scale = active_scale[negative_ratio]
```

```
                tail_logs = np.fromiter(
                    (
                        _negative_tail_log_second_moment(float(-value))
                        for value in local_ratio
                    ),
                    dtype=np.float64,
                    count=local_ratio.size,
                )
                log_kernel = 2.0 * np.log(local_scale) + tail_logs
                if np.any(log_kernel > LOG_FLOAT_MAX):
                    raise FloatingPointError("ReLU self-kernel exceeds float64
range")
                underflow = log_kernel < LOG_FLOAT_MIN_SUBNORMAL
                values = np.zeros_like(log_kernel)
                with np.errstate(under="ignore"):
                    values[~underflow] = np.exp(log_kernel[~underflow])
                active_result[negative_ratio] = values
            if not np.all(np.isfinite(active_result)):
                raise FloatingPointError("ReLU self-kernel produced a non-finite
value")
            if np.any(active_result < 0.0):
                raise FloatingPointError("ReLU self-kernel produced a negative
value")
            result[general] = active_result
```

Each safeguard exists for a distinct mathematical reason:

1. `abs(gamma)` represents the Gaussian standard deviation.
2. The zero-scale mask implements the degenerate distribution before any ratio.
3. The zero-shift mask returns the exact half-variance identity without a CDF call.
4. The positive-ratio branch uses scale normalization to avoid overflowed intermediates.
5. The negative-tail branch prevents subtractive cancellation.
6. Finiteness and strict nonnegativity checks prevent corrupted ranking values.

The complete executable source is the Part 2 capacity audit, and its fail-closed tests are in test_run.py.

2.12 Finite epsilon: a qualification, not a hidden replacement

Evidence: Eq. 77 and Eq. 78 are `paper_claim`; the following substitution is an `independent_inference` that reuses `P01-FINITE-EPSILON-CAVEAT`.

If finite epsilon must be retained, Eq. 77 gives the marginal standard deviation

$$s_{i,\epsilon} = |\gamma_i| \sqrt{\frac{\sigma_i^2}{\sigma_i^2 + \epsilon}}.$$

The general truncated-normal derivation remains valid when this value is used as s_i. But that is our finite-epsilon extension of the moment. It is not what the paper prints in Eq. 79, because Eq. 79 is stated after the Eq. 78 approximation.

This distinction matters near an epsilon-dominated variance. When $\sigma_i^2 \gg \epsilon$, $s_{i,\epsilon} \approx |\gamma_i|$. When the raw variance is comparable to epsilon, the difference can be material and must be reported with its parameter domain.

Part 1 already established the related lesson for BatchNorm rescaling: an epsilon-free approximation should not be called exact at finite epsilon.

2.13 Eq. 6 does not create a new same-layer ranking

Evidence: paper formula is `paper_claim`; derivative and ordering are `verified_fact`, `P02-PRUNE-RANK-MONOTONICITY`.

Paper Eq. 6 on PDF page 13 gives solitary pruning cost as

$$J_{\text{prune}}(c) = \frac{Nc}{E_a - c},$$

where $c = \|f_i\|_{\mathcal{H}}$, N is the layer width factor used by the cost, and E_a is the pre-action layer capacity.

Now compare candidates in one fixed pre-action layer. Hold $N > 0$ and $E_a > 0$ constant, and require $0 \le c < E_a$. Then

$$\begin{aligned} \frac{dJ_{\text{prune}}}{dc} &= N \frac{(E_a - c) + c}{(E_a - c)^2} \\ &= \frac{NE_a}{(E_a - c)^2} \\ &> 0. \end{aligned}$$

So, under exactly those conditions, pruning-cost order is capacity order. The rational transformation changes the spacing between scores but not which candidate is smallest.

The scope is important:

- It applies to candidates in the same unchanged layer.
- It assumes their capacities lie below the common layer capacity.
- It concerns solitary pruning.

- It does not compare layers with different N or E_a.
- It does not cover merge costs, block eviction, parameter payoff, or the next greedy state after an edit.

Turning this derivative into a global selection claim would be an unsupported extrapolation.

2.14 The refined BN baseline

Evidence: `independent_inference`, `P02-REFINED-BN-BASELINE`.

Combine two verified premises:

1. At zero shift, $c_i = |\gamma_i| \|w_{\text{out},i}\|_2 / \sqrt{2}$.
2. Within one fixed layer, Eq. 6 is strictly increasing in c_i.

It follows that the same solitary pruning ranking is produced by

$$|\gamma_i| \|w_{\text{out},i}\|_2.$$

Paper Section 11.1 on PDF page 20 says that its compression plot compares HOPE against incoming-L_1, joint incoming-outgoing-L_1, and BN-scale structured pruning. The listed BN baseline uses γ as its proxy. The paper does not list the product above.

That makes the product a direct baseline for a future controlled comparison. It would isolate a precise question:

> Does performance come from including outgoing routing, from the nonzero shift correction, or from other HOPE actions and modeling choices?

This interpretation should not be sharpened into "the Hilbert construction is useless." The functional definition still explains why outgoing routing belongs in the score. It handles disconnected routes. It gives the correct nonzero-shift moment. It also supports other questions that the zero-shift solitary-pruning reduction does not answer.

The fair conclusion is narrower:

> Sophisticated framing does not remove the need to compare against the simple baseline contained inside the method.

2.15 Independent analytical and numerical agreement

The new audit uses three routes to the same one-dimensional moment.

2.15.1 Analytical evaluation

The tested vectorized function evaluates the closed form with explicit boundary and tail handling. The audit records the combined method origin as `paper_formula_independent_`

`stable_implementation`: Eq. 79 supplies the paper formula, while the numerical dispatcher and deterministic boundary are independent project work.

2.15.2 Deterministic quadrature

The audit independently integrates

$$\int_0^\infty y^2 \frac{1}{s\sqrt{2\pi}} \exp\left[-\frac{(y-\beta)^2}{2s^2}\right] dy$$

on a declared parameter grid. Because this route does not reuse the closed-form algebra and has no sampling noise, it is the primary numerical reference.

The frozen 256-node Gauss-Legendre audit crossed ratios $\{-8, -4, -1, 0, 1, 4, 8\}$ with signed gamma scales $\{-3, -0.25, 0.25, 3\}$, for 28 cases on the active part of $[-12, 12]$. Every case passed the combined gate

$$|K_{\text{quad}} - K_{\text{analytical}}| \le 5 \times 10^{-14} + 5 \times 10^{-11} |K_{\text{analytical}}|.$$

The maximum absolute discrepancy was $3.410605131648481 \times 10^{-13}$; the maximum scaled error was $1.3079565430043264 \times 10^{-4}$ tolerance multiples. Repeating the quadrature produced zero maximum absolute difference.

2.15.3 Seeded Monte Carlo

The secondary check estimates

$$\frac{1}{M} \sum_{m=1}^{M} \text{ReLU}(\beta + sZ_m)^2, \qquad Z_m \sim \mathcal{N}(0, 1).$$

It records a seed and configurable sample count. Its acceptance rule must allow for sampling variability, especially in rare positive tails.

All six frozen Monte Carlo cases passed with 200,000 samples per case and seed `20260801`. The acceptance tolerance was 5×10^{-14} plus seven estimated standard errors. The maximum absolute error was 0.013082247044492235, and the maximum scaled error was 0.19622037094506206 tolerance multiples.

The full audit README defines the grid, tolerances, environment record, and output schema. The frozen summary is the source for the final numerical sentences, while the artifact checksum index authenticates the published result files.

Monte Carlo agreement is not the proof. The derivation above is the proof; quadrature and simulation test that the implementation matches it.

2.16 What the simulated ranking comparison can show

The audit also scores one generated neuron table four ways:

1. incoming L_1 norm;
2. absolute BN scale $|\gamma|$;
3. refined zero-shift score $|\gamma|\|w_{\text{out}}\|_2$;
4. full capacity $\|w_{\text{out}}\|_2\sqrt{K(i,i)}$.

At zero shift, scores 3 and 4 must have the same ranking. The common factor $1/\sqrt{2}$ cannot reorder anything. With nonzero shifts, the full score can move neurons whose Gaussian mass lies differently relative to zero.

The outgoing factor can also reorder equal-$|\gamma|$ neurons, and it sends a disconnected neuron to zero capacity.

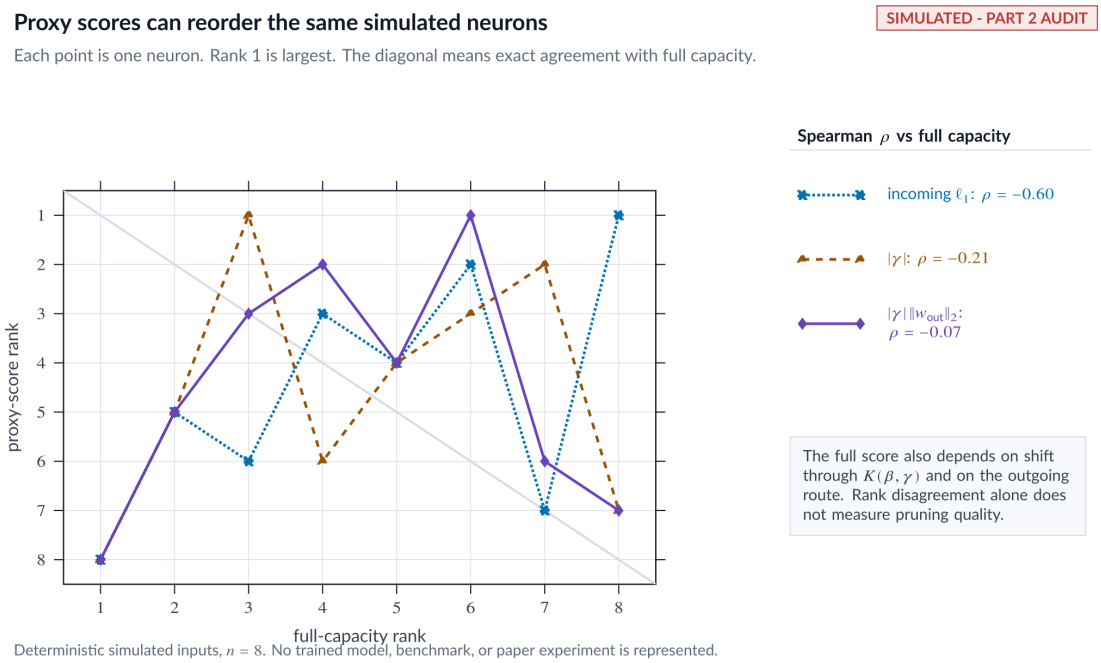


Figure 2.4: A rank scatter compares three simulated proxy scores with full capacity.

Figure 4. Simulated ranking comparison. The inputs are generated by this project and do not come from the paper's model, plot, or checkpoint.

On the frozen eight-neuron simulated table, the four methods produced four distinct descending score orders, and no proxy order matched full capacity. Kendall's tau against full capacity was -0.5 for incoming L_1, -0.21428571428571427 for $|\gamma|$, and -0.07142857142857142 for $|\gamma|\|w_{\text{out}}\|_2$. Every method's eight score values were distinct, so no tie rule affected these ranks.

This is a mathematical and diagnostic comparison, not a pruning benchmark. It does

not measure accuracy loss, latency, FLOPs, or the quality of a real checkpoint. It cannot establish that one score is empirically superior.

2.17 What the paper's experiment says, and what we did not reproduce

Evidence: `paper_claim` for the following description of Section 11.1.

On PDF page 20, the paper says it evaluates a public Keras ResNet-50 checkpoint pretrained on ImageNet. It lists three structured magnitude baselines and says the displayed HOPE curve has superior accuracy in that plot.

Part 2 does not reproduce that experiment. The independent audit has no ResNet-50 checkpoint, ImageNet evaluation, structural surgery, fine-tuning, or accuracy measurement. Its ranking table is simulated and addresses only how the scalar scores order controlled inputs.

The inherited review also records that the Section 11.1 plot lacks the exact numeric table and protocol detail required for an exact reproduction. The legacy adapted compression outputs in this repository have mismatched checkpoint provenance and are excluded. None of those values appears here.

2.18 What checks out

The frozen executable audit passed, and the Part 2 result is constructive:

- The printed ReLU self-kernel follows from a standard truncated-normal second moment.
- The high-dimensional input integral really does reduce to one scalar marginal once that marginal is assumed.
- The outgoing norm has a clear functional role.
- The zero-shift identity exposes an interpretable direct baseline.
- The deterministic zero-variance boundary is simple and continuous.
- Eq. 6 preserves capacity order within one fixed layer.

The formula's simplicity is a strength. It makes the assumptions and baseline comparisons easier to see.

2.19 What remains outside this result

The proof stops at the self-kernel.

It does not establish:

- that real preactivations are Gaussian;
- that BN moments alone determine the right input model;

- any cross-neuron kernel;
- global positive-semidefiniteness of a multi-neuron Gram matrix;
- correctness of the maximum-entropy construction;
- physical merge semantics;
- block-eviction correctness;
- DEFT guarantees;
- compression accuracy from any simulated ranking.

Those are Parts 3 through 12. Cross-neuron geometry and maximum entropy appear here only as boundary labels so that a one-neuron result is not accidentally presented as a global theorem.

2.20 Source and reproduction matrix

Statement	Evidence class	Exact source or project evidence
Functional capacity is outgoing norm times square root of self-kernel	`paper_claim`	PDF page 8, Section 5; PDF page 52, Eqs. 73 through 75
Biased ReLU self-kernel is the Eq. 79 closed form	`paper_claim`	PDF page 9, Eq. 3; PDF page 53, Eq. 79 and Theorem E.2
Eq. 78 uses the negligible-epsilon marginal	`paper_claim`	PDF page 52, Eqs. 77 and 78 plus intervening sentence
Truncated-normal derivation agrees with Eq. 79	`verified_fact`	Calculations above and `python -m unittest discover -s experiments/audits/part02_capacity -p 'test_*.py' -v`
Zero shift gives $K = \gamma^2/2$	`verified_fact`	Calculations above, frozen summary, and boundary tests
Zero variance gives $K = \max(0, \beta)^2$	`verified_fact`	Calculations above, frozen summary, and boundary tests
Eq. 6 is increasing in capacity within a fixed layer	`verified_fact`	PDF page 13, Eq. 6; derivative here; audit ranking test
Product score is the direct zero-shift comparator	`independent_inference`	Verified premises above; PDF page 20 baseline list
Analytical, quadrature, and Monte Carlo values agree within declared gates	`verified_fact`	Audit, summary, and focused tests

Statement	Evidence class	Exact source or project evidence
Four-way ranking comparison	`independent_inference` for the diagnostic design; `verified_fact` only for frozen calculations on the generated table	Metrics CSV; explicitly simulated

2.21 The capacity hidden in plain sight

HOPE's self-kernel is analytically correct under its stated marginal. Its derivation is a clean truncated-Gaussian calculation, and its boundary cases make the statistic easier to interpret.

At zero shift, the full solitary capacity is simply

$$\frac{1}{\sqrt{2}} \times \text{absolute BN scale} \times \text{outgoing L2 norm}.$$

That is more informative than BN scale alone and much simpler than the Hilbert terminology might suggest. Both facts can be true at once.

The right response is not to dismiss the framework. It is to test the exact incremental value of each layer of structure against the baseline that the framework itself reveals.

The compact takeaway is:

> A sophisticated derivation earns trust by exposing its simplest special case, then competing against it directly.

Part 3 will examine the assumption upstream of this formula: what HOPE's "data-free" Gaussian marginal knows, what it reconstructs, and what remains untested.

Evidence records:

- `P02-RELU-SELF-KERNEL`: `paper_claim`
- `P02-RELU-SELF-KERNEL-VERIFIED`: `verified_fact`, frozen audit passed
- `P02-ZERO-SHIFT-CAPACITY`: `verified_fact`
- `P02-DEGENERATE-GAMMA-LIMIT`: `verified_fact`
- `P02-PRUNE-RANK-MONOTONICITY`: `verified_fact`
- `P02-REFINED-BN-BASELINE`: `independent_inference`

Primary source: HOPE v1, arXiv:2607.21366v1

Primary anchors: PDF pages 7 through 9, 13, 20, 52, and 53; Section 5; Section 11.1; Eqs. 3, 6, 73 through 75, and 77 through 79; Proposition E.1; Theorem E.2.

Code and numerical evidence: Part 2 capacity audit

Chapter 3

What data-free actually means

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `layer_summary.csv`
- Evidence: `pair_summary.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part03_data_free -p
'test_*.py' -v
```

Scope: The included results support a bounded checkpoint study. A fresh dataset-backed run requires external checkpoint, data, and software dependencies that are not included in this release.

Part 3 of HOPE Under the Microscope

A data-free method still needs a model of the data.

In HOPE, some of that model is already inside a trained BatchNorm checkpoint. Each normalized channel carries a running mean and variance accumulated during training. Those numbers are real information about the signals the network once saw.

But they are not the whole distribution. They do not record how channels move together. They do not record skewness, heavy tails, multimodality, spatial dependence, or class-conditional structure. And they do not prove that a trained feature direction is Gaussian.

HOPE bridges that gap with a Gaussian surrogate. This is the right place to be precise about the phrase “data-free”:

> Data-free is not assumption-free. The checkpoint supplies per-channel running means and variances; the surrogate supplies the unobserved distributional structure needed by the analytical kernel.

That distinction does not make the surrogate useless. It tells us what must be tested.

This article is bound to arXiv:2607.21366v1. The exact 70-page PDF fingerprint is recorded in the paper manifest. The exact anchors are physical PDF pages 2, 6, 7, 52, 53, and 55; Eq. 83 on page 55 supplies the exact biased pair moment used by the diagnostic. The paper traceability summary above separates the paper's statements from the algebra and diagnostics developed here.

3.1 Evidence labels used here

This article uses four labels deliberately:

- `paper_claim`: what the paper says.
- `verified_fact`: algebra proved independently here, or a checkpoint fact backed by a frozen audit.
- `independent_inference`: our interpretation of stated premises.
- `open_question`: something neither the source nor the current evidence resolves.

The empirical checkpoint audit is frozen. Its measurements are a `verified_fact` about one declared VGG-8 checkpoint and sample, not a universal Gaussian-fit verdict.

3.2 The strongest fair reading of HOPE's idea

`paper_claim` - HOPE needs expectations such as

$$K(i,j) = \mathbb{E}\left[\mathrm{ReLU}(Y_i)\,\mathrm{ReLU}(Y_j)\right]$$

to compare the functions computed by neurons. If the original training data are unavailable, those expectations cannot be evaluated directly over the true layer-input distribution.

The paper's answer is not to pretend that no distribution is needed. Section 4 explicitly says the true input law is inaccessible and constructs a surrogate. For a BatchNorm network, the checkpoint already contains empirical constraints on each raw preactivation. HOPE uses those constraints to choose a continuous Gaussian model, then evaluates the required ReLU moments analytically.

This is an appealing computational exchange:

```
stored moments + distributional model -> analytical expectations
```

It avoids repeatedly passing a dataset through the network during progressive encoding. It also makes the assumptions inspectable. That is stronger than a black-box score, provided we keep the stored evidence and the added model separate.

On PDF page 2, the paper calls the method data-free for networks with BatchNorm. Footnote 1 immediately gives the scope qualifier: architectures without BatchNorm require a one-time calibration pass on a small data batch. So even in the paper's own terminology,

"data-free" describes the evaluation procedure available for a particular kind of checkpoint. It does not mean that the checkpoint contains no trace of its training distribution, and it does not mean distribution-free.

3.3 What BatchNorm actually stores

Consider one layer. Flatten the input patch to a vector $X \in \mathbb{R}^n$, and collect the current raw channel weights as rows of $W \in \mathbb{R}^{c \times n}$. In the bias-free model used for this derivation, the raw preactivation vector is

$$Z = WX.$$

For each channel i, BatchNorm keeps a running mean μ_i and running variance σ_i^2. In an idealized stationary setting where those values equal exact moments under the same current map,

$$\mu_i = \mathbb{E}[w_i^\top X], \qquad \sigma_i^2 = \mathrm{Var}(w_i^\top X).$$

Equivalently, with input mean m_X and covariance Σ_X,

$$W m_X = \mu_{\mathrm{BN}}, \qquad w_i^\top \Sigma_X w_i = \sigma_i^2.$$

`paper_claim` - These are the constraints that Section 4 uses to construct one shared surrogate for a layer.

They are useful. The mean constraint locates the center as seen through each trained filter. The variance constraint measures the scale of each raw projection. After the learned BatchNorm affine transformation, those moments lead to the effective preactivation parameters used in the self-kernel.

But notice the shape of the information: one mean and one variance per scalar projection. BatchNorm observes a collection of one-dimensional shadows.

3.4 What BatchNorm does not store

Suppose two channels have centered raw preactivations Z_i and Z_j. Knowing

$$\mathbb{E}[Z_i], \quad \mathrm{Var}(Z_i), \quad \mathbb{E}[Z_j], \quad \mathrm{Var}(Z_j)$$

does not give

$$\mathrm{Cov}(Z_i, Z_j).$$

Nor does it determine the joint distribution. Separate marginals can be coupled in many ways.

`verified_fact` - Let U and V be independent standard normal random variables. Compare three pairs:

$$(U, U), \qquad (U, -U), \qquad (U, V).$$

Every coordinate in all three pairs is standard normal. Each channel therefore has the same mean, variance, and ReLU self-energy:

$$\mathbb{E}[\mathrm{ReLU}(U)^2] = \frac{1}{2}.$$

Yet the ReLU cross-moment is

$$\frac{1}{2}, \qquad 0, \qquad \frac{1}{2\pi},$$

respectively. Identical one-dimensional marginals can support maximally aligned, anti-aligned, or independent pair behavior.

This is not a claim that any of those three toy couplings is the trained network's joint distribution. It is a proof of non-identifiability: separate channel statistics alone cannot tell us which dependence is present.

That boundary matters because HOPE uses both kinds of quantities:

- The self-kernel $K(i, i)$ depends only on one channel's marginal.
- The pair kernel $K(i, j)$ depends on a joint law.

BatchNorm can support the first more directly than the second.

3.5 A marginal Gaussian is enough for the self-kernel

After BatchNorm and its learned affine transform, write

$$Y_i = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}}(w_i^\top X - \mu_i) + \beta_i.$$

If the stored moments agree with the moments of the current raw projection, then

$$\mathbb{E}[Y_i] = \beta_i$$

and

$$\mathrm{Var}(Y_i) = \gamma_i^2 \frac{\sigma_i^2}{\sigma_i^2 + \epsilon}.$$

`paper_claim` - Eq. 77 prints the finite-epsilon variance. The text then takes epsilon to be negligible and Eq. 78 writes

$$Y_i \sim \mathcal{N}(\beta_i, \gamma_i^2).$$

Under that scalar Gaussian premise, the ReLU self-kernel is a one-dimensional truncated-normal moment. Appendix E.2 evaluates it in closed form:

$$K(i,i) = (\gamma_i^2 + \beta_i^2)\Phi\left(\frac{\beta_i}{|\gamma_i|}\right) + \beta_i|\gamma_i|\phi\left(\frac{\beta_i}{|\gamma_i|}\right).$$

Part 2 independently verified this identity under its stated marginal model. Part 3 asks the upstream question: how well does that marginal model describe captured checkpoint preactivations?

The distinction is important. A poor empirical fit would not make the integral wrong. It would show that its premise is a poor approximation for the evaluated channel and sample. A close fit would support the approximation in that bounded setting, not establish a universal theorem.

3.6 A pair kernel needs dependence too

The cross-kernel is

$$K(i,j) = \mathbb{E}[\mathrm{ReLU}(Y_i)\,\mathrm{ReLU}(Y_j)].$$

Even if both scalar marginals are exactly Gaussian, this expectation remains undetermined until their dependence is specified. Under a bivariate Gaussian model, a correlation parameter completes the pair law. But BatchNorm does not store that correlation.

`paper_claim` - Section 4 proposes a shared maximum-entropy covariance under the per-channel variance constraints, then says that practical pairwise operations bypass the full high-dimensional covariance. Appendix E develops the pair construction.

Part 3 stops at the information boundary: the dependence is reconstructed, not read directly from separate BatchNorm buffers. Whether the entropy problem is well-posed belongs to Part 4. Whether separately constructed pair kernels form one global positive-semidefinite Gram matrix belongs to Part 5.

This separation prevents one valid criticism from being stretched into a different claim. "BatchNorm does not identify the joint law" is exact. "The paper's selected joint surrogate is wrong on every network" is not established.

3.7 What the Gaussianity argument does and does not prove

`paper_claim` - On PDF page 6, HOPE gives a two-step motivation.

First, a neuron observes a high-dimensional input through a scalar weighted sum. The paper invokes the Central Limit Theorem and the Diaconis-Freedman effect to argue that high-fan-in preactivations become Gaussian. It adds that although ReLU disrupts Gaussianity, later high-dimensional linear maps smooth the signal back toward Gaussian preactivations.

Second, it chooses a multivariate Gaussian input surrogate because every linear combination of a multivariate Gaussian is Gaussian. In the "Post-ReLU Support Paradox" box, it argues that the kernel depends only on a two-dimensional preactivation slice, so replacing the nonnegative ambient post-ReLU support can remain asymptotically accurate.

This is a coherent motivation. It is not itself a measurement; the frozen checkpoint audit below supplies one bounded empirical test.

`independent_inference` - Central-limit intuition has conditions. Contributions must not be dominated by a small number of terms, and their dependence must be controlled. Typical-projection results concern most or randomly oriented directions under their hypotheses. A trained filter is not drawn independently of the data. Optimization selects directions that detect structure.

Modern convolutional layers add concrete reasons to measure rather than assume:

- a receptive field can be small;
- neighboring spatial values can be strongly correlated;
- post-ReLU inputs can be sparse and asymmetric;
- a few weights can dominate a projection;
- mixtures of classes or activation regimes can remain multimodal;
- early, middle, and late layers need not behave alike.

None of these bullets proves non-Gaussianity. Together they explain why an asymptotic story should be checked in the actual trained directions used by the method.

`open_question` - For a fixed checkpoint and sample, which channels are close enough to the Gaussian surrogate for the resulting kernel estimates, and which are not?

3.8 Eq. 76: the width objection that does not work

The review history matters here because an initially tempting criticism was wrong.

On PDF page 7, HOPE defines the layer surrogate mean as

$$\hat{\mu}_X = W^+ \mu_{\mathrm{BN}}.$$

Appendix E.1 then uses

$$W\hat{\mu}_X = WW^+\mu_{\mathrm{BN}} = \mu_{\mathrm{BN}}$$

to obtain Eq. 76.

The equality holds exactly when μ_{BN} lies in the column space of W, because WW^+ is the orthogonal projector onto that space.

It can look as if an expanding layer, with more output channels than input coordinates, should make an arbitrary μ_{BN} miss that subspace. But the relevant vector is not arbitrary. Under the same bias-free linear model,

$$\mu_{\mathrm{BN}} = \mathbb{E}[WX] = W\mathbb{E}[X].$$

It is therefore in the column space by construction, independent of whether the layer expands or contracts.

`verified_fact` - Width alone does not break Eq. 76.

That conclusion must survive every rewrite of Part 3. A synthetic counterexample that samples W and μ_{BN} independently violates the premise it claims to test.

Real checkpoints can still depart from the ideal equality. Running statistics are exponential moving averages accumulated while weights change. A raw affine bias might be omitted from the equation. The extraction graph might not match the algebra. Convolutional spatial aggregation can import a stationarity approximation. Numerical conditioning also matters.

Those are hypotheses about causes, not automatic findings. The audit should measure the normalized projection residual

$$R_\mu = \frac{\|(I - WW^+)\mu_{\mathrm{BN}}\|_2}{\max(\|\mu_{\mathrm{BN}}\|_2, \tau)}$$

with a declared floor τ, then report what happens for the specified checkpoint. It must not relabel a residual as a width failure.

3.9 The frozen checkpoint audit

`P03-CHECKPOINT-GAUSSIAN-DIAGNOSTIC` is a bounded composite claim record from one VGG-8/CIFAR-10 `architecture_adaptation`. Its raw checkpoint measurements and algebraic decomposition identities are `verified_fact`; its interpretation of Gaussian-surrogate

fit errors is `independent_inference`. The exact checkpoint fingerprint is recorded in the frozen summary.

The audit used 2,000 CIFAR-10 test examples selected by a sorted, seeded, near-equal class-stratified sample without replacement. The selected-example fingerprint is recorded in the same frozen summary.

It captured 16 deterministic channels and eight pairs at each graph point:

Layer	Module	Shape and rank	Observations
`conv0`	`features.1`	128 by 27, rank 27	32,000
`conv2`	`features.9`	512 by 2,304, rank 512	32,000
`conv5`	`features.19`	512 by 4,608, rank 512	32,000
`dense`	`bn_b`	512 by 512, rank 512	2,000

`conv0` selected channels 0, 8, 16, 25, 33, 42, 50, 59, 67, 76, 84, 93, 101, 110, 118, and 127. Each other graph point selected channels 0, 34, 68, 102, 136, 170, 204, 238, 272, 306, 340, 374, 408, 442, 476, and 511.

The full channel lists, selection rules, output fingerprints, and configuration fingerprint are recorded in the frozen summary. The artifact checksum index authenticates the published result files.

The checkpoint's historical training provenance is incomplete. Its bytes and strict-loaded state schema are bound, but epoch history, optimizer and scheduler state, training-data hashes, and the original generating command and environment are unavailable. The audit uses no legacy compression result and does not reproduce HOPE's ResNet-50/ImageNet experiment.

3.9.1 What the selected one-dimensional distributions looked like

Across the four layer-summary rows, per-layer median absolute skewness ranged from `0.4381046015109638` to `0.916664923200222`; the largest selected-channel value was `2.3082733334908636`. Median absolute excess kurtosis ranged from `0.5276760976728814` to `4.1728378058823345`, with maximum `9.07941365824211`. Median fitted-normal KS distance ranged from `0.04980412829317285` to `0.13194043073030828`, with maximum `0.2124892081316252`.

Those values show substantial variation across the selected channels. They do not produce one network-wide yes-or-no Gaussian label. Spatial positions and repeated values from each image are correlated, so these are descriptive effect sizes, not normality p-values.

3.9.2 Separating three self-kernel errors

Let $G(m, s)$ denote the exact Gaussian ReLU second moment. The audit compares the empirical moment E_i, a Gaussian fitted to captured moments G_i^{emp}, the checkpoint prediction

with finite epsilon $G_i^{\mathrm{BN},\epsilon}$, and the paper's negligible-epsilon prediction $G_i^{\mathrm{BN},0}$. The signed error telescopes:

$$E_i - G_i^{\mathrm{BN},0} = \underbrace{E_i - G_i^{\mathrm{emp}}}_{\text{shape}} + \underbrace{G_i^{\mathrm{emp}} - G_i^{\mathrm{BN},\epsilon}}_{\text{moment state}} + \underbrace{G_i^{\mathrm{BN},\epsilon} - G_i^{\mathrm{BN},0}}_{\text{finite epsilon}}.$$

Across the four layer summaries, median Gaussian-fit self-kernel absolute error ranged from `0.00013562328773035452` to `0.001469023748179522`; the largest selected absolute error was `0.008144367516825735`. Median bounded symmetric error ranged from `0.018853763884460188` to `0.5458623797033268`. Near-dead channels reached the symmetric metric ceiling `2.0`, so that ceiling must be read beside the small absolute-error scale.

The decomposition identity is a fail-closed gate. The sizes of the Gaussian fit errors are not pass gates; they are the scientific observations.

3.9.3 Pair moments and dependence

Across 32 selected pairs, the empirical-versus-exact-biased-Gaussian absolute error ranged from `3.222170944350448e-10` to `0.007075261127352704`. Bounded symmetric error ranged from `0.000633394514674313` to `2.0`, again with near-zero pair moments at the ceiling.

Independently shuffling one channel preserved the measured marginal summaries to maximum absolute difference `5.551115123125783e-17`, while per-layer scaled dependence change ranged from `0.2981947166796576` to `0.6397540775253039`. The shuffle is an identifiability diagnostic, not a plausible replacement network distribution.

3.9.4 What Eq. 76 measured

The 128 by 27 rank-27 `conv0` expansion had the only nontrivial stored-mean projection residual: `0.000551072218291825`. It measures mismatch with the ideal current common-map condition. It does not show that expansion width caused the mismatch.

The other maps were full row rank. Their column spaces equal their output spaces, so machine-scale residuals from `1.082752994927010 6e-15` to `1.4707953601090622e-15` are non-diagnostic: any stored mean vector projects back in those cases. They do not prove that running statistics are current.

The separate same-sample common-map residual ranged from `5.718194556149604e-10` to `1.1232923462430606e-8`, and all four declared implementation gates passed. That supports the implementation of the ideal algebra without converting width into a causal explanation.

3.9.5 The bounded conclusion

The selected channels do not share one uniform Gaussian-fit profile. Some absolute ReLU-moment errors are small while normalized errors saturate on near-dead signals; pair behavior also changes when marginals are held fixed. This is evidence about the declared specimen and estimands. It is not "neural preactivations are Gaussian" or "HOPE's Gaussian premise is false."

Whether these discrepancies change pruning or merge rankings remains an `open_question`.

3.10 So what does data-free mean here?

`independent_inference` - For a BatchNorm checkpoint, HOPE can compute its surrogate-based scores without loading real or synthetic examples during the encoding loop. That is a meaningful operational advantage.

But the computation still depends on:

1. historical moments stored from training;
2. an idealized relationship between those moments and current weights;
3. a Gaussian marginal premise;
4. a reconstructed dependence model for pair operations;
5. architectural assumptions about the observed projections and graph.

The better phrase is not "no data model." It is "a checkpoint-carried, analytically completed data model."

That description is both fairer and more useful. It shows why the method can be efficient, why BatchNorm is central, where calibration enters without BatchNorm, and where empirical validation should focus.

3.11 What the audit establishes

Several conclusions now have frozen support.

`paper_claim` - HOPE intentionally replaces the inaccessible true layer-input distribution with a Gaussian surrogate constrained by BatchNorm statistics.

`verified_fact` - BatchNorm's separate channel means and variances do not identify cross-channel dependence or a full joint distribution.

`verified_fact` - The Eq. 76 mean reconstruction is exact under one common bias-free linear model. Width alone is not a counterexample.

`independent_inference` - CLT and typical-projection intuition motivate a surrogate but do not substitute for measurement in trained directions.

`verified_fact` - On the declared VGG-8 checkpoint and sample, the bounded raw measurements, exact self-kernel decomposition identities, shuffle-preserved marginals, and Eq. 76 projection measurements are as reported above.

`independent_inference` - The measured marginal-shape, self-kernel, and pair-kernel discrepancies quantify fit to the chosen Gaussian surrogate; their variation by layer and signal scale does not by itself establish behavior for other checkpoints or the paper's target architecture.

`open_question` - Are any measured errors large enough to alter HOPE's action rankings, and do the patterns transfer to other checkpoints?

The final question is the constructive one. Data-free methods do not need to hide their assumptions. They can expose them, test them on provenance-bound checkpoints, and report exactly where analytical convenience remains faithful to the learned signals.

That is the standard Part 3 applies.

3.12 Reproduction and release status

The executable audit lives under `experiments/audits/part03_data_free/`; its selected frozen outputs live under `artifacts/part_03/results/`.

This article is published in the `v1.0.0` twelve-part package. Its frozen evidence boundary is unchanged. Part 4 examines the maximum-entropy construction as a separate mathematical question.

Chapter 4

When pure maximum entropy is unbounded

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `escape_path.csv`
- Evidence: `tau_sweep.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part04_maximum_
entropy -p 'test_*.py' -v
```

Scope: The included tests cover deterministic offline fixtures; no checkpoint or paper PDF is required.

Part 4 of HOPE Under the Microscope

If constraints observe only part of a space, entropy can escape through the directions they never see.

That sentence is the core of Part 4, but it needs two qualifications every time it is used:

1. the projection-variance constraints must admit at least one positive-definite covariance;
2. the constrained directions must fail to span the finite-dimensional input space.

Under those hypotheses, HOPE's pure maximum-differential-entropy covariance problem is unbounded. When the same feasible constraints do span the input space, the objective is bounded and has one unique positive-definite maximizer. The objection is decisive in its domain, not universal.

The second question is what to make of Appendix E. Section 4 describes a pure entropy objective, whose stationarity expression contains only weighted filter outer products. Appendix

E.3.1 instead uses an identity precision term:

$$\Sigma^{-1} = I + W\Lambda W^{\top}.$$

That identity makes the local covariance well behaved outside the constrained span. It is also the stationarity signature of a different objective: a relative-entropy projection onto an isotropic Gaussian reference. That interpretation is ours, not wording supplied by the paper.

The constructive repair is therefore simple to state: if a reference Gaussian is intended, name it, give its scale, and test sensitivity to that scale.

This article is bound to arXiv:2607.21366v1. Its exact 70-page PDF fingerprint is recorded in the paper manifest. The exact anchors are physical PDF pages 6, 7, 54, and 55. The paper traceability summary above separates statements printed by the paper from the independent mathematics developed here.

4.1 Evidence labels used here

- **paper_claim**: what HOPE v1 states or prints.
- **verified_fact**: algebra independently proved in the calculations above.
- **independent_inference**: an interpretation assembled from the printed formulas and verified algebra, but not stated by the paper.
- **proposed_repair**: a replacement objective that is not presented as HOPE v1.
- **open_question**: an issue not resolved by the source or current evidence.

Exact matrices, determinants, and the cosine-one-half result below are algebraic fixtures. A separate frozen-diagnostics section reports values from an independent deterministic audit, not the paper's experiments. The focused tests validate those values, and the artifact checksum index authenticates the published files.

4.2 The strongest fair version of the paper's idea

Evidence: **paper_claim**, **P04-PURE-MAXENT-FORMULATION**.

HOPE needs a probability model for a layer input because its Hilbert inner products are expectations. For a network with BatchNorm, the checkpoint gives one mean and variance for each observed preactivation direction. Section 4 uses those stored moments as constraints on a Gaussian input surrogate.

Maximum entropy is an attractive organizing principle here. Given only a declared set of moment constraints, choose the distribution that adds the least extra structure. For a Gaussian with fixed mean μ and covariance $\Sigma \succ 0$,

$$h(\mathcal{N}(\mu, \Sigma)) = \frac{d}{2}\log(2\pi e) + \frac{1}{2}\log\det\Sigma.$$

The covariance problem is therefore

$$\max_{\Sigma \succ 0} \log\det\Sigma \quad \text{subject to} \quad w_i^\top \Sigma w_i = v_i, \qquad i = 1, \dots, m.$$

This is a clean question. It is concave, its constraints are linear in Σ, and it has a unique answer whenever a maximizer exists.

The missing issue is existence.

4.3 Start with feasibility, not rank alone

Let

$$W = [w_1, \dots, w_m] \in \mathbb{R}^{d \times m}, \qquad r = \operatorname{rank}(W),$$

where d is finite. Define

$$\mathcal{F} = \left\{\Sigma \in \mathbb{S}^d_{++} : w_i^\top \Sigma w_i = v_i \text{ for every } i\right\}.$$

Every result in this article begins with the hypothesis

$$\mathcal{F} \neq \emptyset.$$

That condition is not clerical. Rank tells us which directions the constraints can observe. It does not make contradictory target variances consistent. Likewise, if $w_i \neq 0$, then a positive-definite covariance cannot satisfy $w_i^\top \Sigma w_i = 0$.

So there are two separate questions:

1. Does any positive-definite covariance satisfy the targets?
2. If one does, is pure entropy bounded on that feasible set?

Part 4 answers the second question conditional on the first.

4.4 The non-spanning theorem

Evidence: **verified_fact**, **P04-NONSPANNING-UNBOUNDED**.

Assume:

- the input dimension d is finite;

- $\mathcal{F}$ contains a covariance $\Sigma_0 \succ 0$;
- $r < d$.

Because the constrained directions do not span the input, the nullspace of $W^\top$ contains a nonzero vector. Choose a unit vector z such that

$$W^\top z = 0.$$

For every $t \geq 0$, define

$$\Sigma_t = \Sigma_0 + tzz^\top.$$

The update is positive semidefinite, so every Σ_t remains positive definite. More importantly, the constraints cannot see it:

$$\begin{aligned} w_i^\top \Sigma_t w_i &= w_i^\top \Sigma_0 w_i + t\, w_i^\top z z^\top w_i \\ &= v_i + t(w_i^\top z)^2 \\ &= v_i. \end{aligned}$$

Thus the whole path remains feasible.

Entropy does see the update. The matrix determinant lemma gives

$$\det \Sigma_t = \det \Sigma_0 \left(1 + tz^\top \Sigma_0^{-1} z\right).$$

Since $\Sigma_0^{-1} \succ 0$ and $z \neq 0$,

$$z^\top \Sigma_0^{-1} z > 0.$$

Therefore

$$\log\det \Sigma_t = \log\det \Sigma_0 + \log\left(1 + tz^\top \Sigma_0^{-1} z\right) \longrightarrow \infty.$$

There is no finite maximizer.

This proof does not depend on a solver, an initialization, a tolerance, or an ill-conditioned numerical inversion. It supplies an explicit feasible path whose objective diverges.

4.5 The smallest example

Take a two-dimensional input and constrain only

$$w_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad v_1 = 1.$$

The constraint observes the horizontal variance and nothing else. The family

$$\Sigma_t = \begin{bmatrix} 1 & 0 \\ 0 & 1+t \end{bmatrix}, \qquad t \geq 0,$$

is positive definite and satisfies

$$w_1^\top \Sigma_t w_1 = 1$$

for every t. But

$$\log\det \Sigma_t = \log(1+t) \longrightarrow \infty.$$

Geometrically, the covariance ellipse keeps the same horizontal width while stretching without limit in the unobserved vertical direction.

This example is not meant to imitate a trained layer. It is the smallest exact witness to the mathematical mechanism.

4.6 Why the Section 4 inverse exposes the same boundary

Evidence: **paper_claim** for the printed expression; **verified_fact** for the rank consequence.

Section 4 conceptually writes the pure stationarity form as

$$\widehat{\Sigma} = \left(\sum_i \lambda_i w_i w_i^\top \right)^{-1}.$$

The matrix inside the inverse has rank at most r. If $r < d$, it is singular for every choice of multipliers.

That observation alone says the displayed inverse cannot exist in the non-spanning case. The nullspace construction proves the stronger fact: under positive-definite feasibility, this is not merely a defective formula for a finite answer. The pure optimization itself is unbounded.

4.7 What changes when the constraints span the input

Evidence: **verified_fact**, **P04-FULL-SPAN-BOUNDARY**.

Now keep the finite-dimensional and positive-definite feasibility hypotheses, but suppose

$$r = d.$$

Define

$$C = WW^\top = \sum_i w_i w_i^\top.$$

Full span makes $C \succ 0$. Also define

$$S = \sum_i v_i.$$

Every feasible covariance obeys

$$\begin{aligned} \operatorname{tr}(C\Sigma) &= \operatorname{tr}\left(\sum_i w_i w_i^\top \Sigma\right) \\ &= \sum_i w_i^\top \Sigma w_i \\ &= S. \end{aligned}$$

Set

$$A = C^{1/2}\Sigma C^{1/2} \succ 0.$$

Then $\operatorname{tr}(A) = S$. Applying the arithmetic-geometric mean inequality to the d positive eigenvalues of A yields

$$\det A \le \left(\frac{S}{d}\right)^d.$$

Since

$$\det A = \det C \, \det \Sigma,$$

we obtain the finite bound

$$\log\det \Sigma \le d \log\left(\frac{S}{d}\right) - \log\det C.$$

The trace identity also bounds the feasible covariances:

$$\lambda_{\min}(C)\,\mathrm{tr}(\Sigma) \le \mathrm{tr}(C\Sigma) = S.$$

The closure of the feasible set in the positive-semidefinite cone is therefore compact. Singular boundary points have extended objective $\log\det\Sigma = -\infty$, while the feasibility hypothesis supplies at least one positive-definite point with a finite objective. A maximizer exists in the positive-definite interior.

Finally, $\log\det$ is strictly concave and the feasible set is convex, so that maximizer is unique.

The boundary is exact:

- feasible and $r < d$: pure entropy is unbounded above;
- feasible and $r = d$: pure entropy is bounded and has one positive-definite maximizer.

This is why the title says *when* pure maximum entropy is unbounded.

4.8 In two dimensions, pure entropy removes projection correlation

Suppose the complete ambient problem is two-dimensional and two independent constraint directions span it. Set

$$Y = W^\top X, \qquad K = \mathrm{Cov}(Y) = W^\top \Sigma W.$$

Because W is invertible in this two-dimensional setting,

$$\log\det\Sigma = \log\det K - 2\log|\det W|.$$

The change of coordinates adds only a constant. The two target variances fix the diagonal of K:

$$K = \begin{bmatrix} v_1 & c \\ c & v_2 \end{bmatrix}.$$

Its determinant is

$$\det K = v_1 v_2 - c^2.$$

Pure maximum entropy therefore chooses

$$c = 0.$$

The constrained projections are uncorrelated. A nonzero correlation based on the geometry of the raw directions needs structure beyond the two target variances.

4.9 An exact feasible contrast

Evidence: **verified_fact**, exact algebraic fixture.

Choose

$$w_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad w_2 = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}, \qquad v_1 = v_2 = 1.$$

The two unit directions are non-collinear and span the full two-dimensional input. The pure optimizer is

$$\Sigma_{\text{pure}} = \begin{bmatrix} 1 & -1/\sqrt{3} \\ -1/\sqrt{3} & 5/3 \end{bmatrix}.$$

Direct multiplication gives

$$W^\top \Sigma_{\text{pure}} W = I.$$

Both target variances equal one and the projected cross-covariance is zero. Also,

$$\det \Sigma_{\text{pure}} = \frac{4}{3}.$$

Now consider the identity covariance

$$\Sigma_{\text{ref}} = I.$$

It is also feasible because both raw directions have unit norm:

$$w_1^\top \Sigma_{\text{ref}} w_1 = w_2^\top \Sigma_{\text{ref}} w_2 = 1.$$

But its two projected coordinates retain the raw cosine:

$$w_1^\top \Sigma_{\text{ref}} w_2 = \frac{1}{2},$$

and

$$\det \Sigma_{\text{ref}} = 1.$$

The identity answer is feasible, but it is not the pure entropy maximizer. This contrast holds in a full-spanning problem, so no nullspace escape is involved. The two answers differ because they optimize different objectives.

4.10 The identity precision in Appendix E

Evidence: **paper_claim**, **P04-LOCAL-IDENTITY-PRECISION**.

Appendix E.3.1 builds a local pairwise covariance and uses

$$\Sigma^{-1} = I + W\Lambda W^{\top}.$$

The identity term is not present in the Section 4 pure stationarity expression. Its effect is concrete: on any direction orthogonal to the local pair, the added precision remains one rather than zero. The covariance is therefore anchored instead of free to expand.

That makes the Appendix construction mathematically different from the pure problem. It does not make it incoherent. It means the reference structure needs to be named.

4.11 The pair formula has its own rank domain

Evidence: **verified_fact**, **P04-PAIR-RANK-DOMAIN**.

The layer-wide rank boundary compares the span of every constrained direction with the ambient input dimension. The local Appendix formula has a separate domain condition.

For a selected pair, write

$$B = [w_i, w_j], \qquad G = B^{\top}B.$$

The printed proof uses G^{-1}, while Eqs. 80 and 81 divide by one minus the squared raw correlation. These steps require

$$\operatorname{rank}(B) = 2,$$

or equivalently:

- both directions are nonzero;
- they are not collinear;
- their absolute cosine is strictly below one.

Two nonzero collinear directions do not satisfy this domain. Nonzero norms alone are insufficient. An implementation should reject or separately define that boundary rather than manufacture a finite inverse.

This pair-rank condition is not the same as full span of a high-dimensional layer. A rank-two pair can still occupy only a tiny part of an ambient space with $d > 2$.

4.12 A reference-prior interpretation

Evidence: **independent_inference**.

Claim: `P04-IDENTITY-PRIOR-INTERPRETATION`.

Let the reference covariance be

$$P_\tau = \tau^2 I, \qquad \tau > 0.$$

Consider the Gaussian relative-entropy projection

$$\min_{\Sigma \in \mathcal{F}} \mathrm{KL}\left(\mathcal{N}(0,\Sigma) \middle\| \mathcal{N}(0,\tau^2 I)\right).$$

Ignoring terms independent of Σ, the objective is

$$\frac{1}{2}\left(\tau^{-2}\,\mathrm{tr}\,\Sigma - \log\det\Sigma\right).$$

Its stationarity condition is

$$\Sigma^{-1} = \tau^{-2} I + W \Lambda W^\top.$$

At $\tau = 1$, this is exactly the identity-precision shape used in Appendix E.3.1.

The conclusion has two evidence levels:

- **paper_claim**: Appendix E prints the identity term.
- **independent_inference**: the term is exactly explained by an isotropic Gaussian reference-prior objective.

We should not collapse those statements into one attribution.

4.13 Why the reference scale matters

Evidence: **verified_fact**, **P04-PRIOR-SCALE-SENSITIVITY**, bounded to the declared deterministic rank-two fixture.

Assume a valid rank-two pair. Let the raw directions have cosine ρ, norms a and b, and target standard deviations s_1 and s_2. For the KL projection, let q_τ denote the resulting projected correlation. The calculations above derive

$$\frac{q_\tau}{1-q_\tau^2} = \kappa_\tau,$$

where

$$\kappa_\tau = \frac{1}{\tau^2}\frac{\rho}{1-\rho^2}\frac{s_1 s_2}{ab}.$$

The stable root is

$$q_\tau = \frac{2\kappa_\tau}{1+\sqrt{1+4\kappa_\tau^2}}.$$

The target marginal variances remain fixed, yet the chosen pair correlation depends on τ. In the exact full-spanning fixture above, $\rho = 1/2$, $a = b = s_1 = s_2 = 1$, and $\tau = 1$ gives

$$q_1 = \frac{1}{2}.$$

By contrast, pure maximum entropy gives zero projected correlation. In this full-spanning fixture, the KL solution approaches the pure answer as τ grows without bound.

The exact formula proves that the scale is substantive rather than cosmetic. The numerical sweep below checks a declared finite grid; it does not replace that analytic statement.

4.13.1 Frozen deterministic fixture diagnostics

The authenticated independent audit passes all 17 declared checks. These are deterministic fixture diagnostics, not paper results or trained-network measurements:

- the rank-deficient escape path reaches $t = 1{,}000{,}000$ and records a log-determinant gain of `13.815511557963774`;
- the incompatible duplicate-constraint control records a minimum linear residual of `0.5000000000000002`, or about `0.5`, and is classified as infeasible rather than passed into the unboundedness theorem;
- the prior-scale fixture uses $\tau \in \{0.5, 0.75, 1, 2, 4, 10\}$ and records a correlation range of `0.8232599168595268` while the two target marginal variances remain fixed;
- the maximum constraint residual on that tau sweep is `1.5543122344752192e-15`; and
- the maximum stationarity residual is `2.220446049250313e-16`.

The exact 60-degree geometry, cosine 1/2, determinants 4/3 and 1, and correlations 0 and 1/2 remain algebraic results. The decimal values above only document the finite float64 fixtures declared in the summary and its configuration.

4.14 Proposed repair: state the real objective

Evidence: **proposed_repair**, **P04-RELATIVE-ENTROPY-REPAIR**.

If the intended construction is relative to an isotropic Gaussian, state:

1. the exact KL direction;
2. the reference covariance $\tau^2 I$;
3. the requirement $\tau > 0$;
4. the feasibility requirements on the projection targets;
5. the rank-two domain of any inverse-Gram pair formula;
6. how outputs change over a declared range of τ.

The repair does not turn inconsistent constraints into consistent ones. It does not retroactively make the Section 4 pure objective bounded when $r < d$. It defines a different, explicit problem whose identity precision prevents the nullspace escape.

That is a constructive correction because it preserves the useful goal: build an analytical Gaussian surrogate from checkpoint moments while making every additional modeling choice visible.

4.15 What checks out

The maximum-entropy framing is meaningful when its domain is stated.

- Under positive-definite feasibility and full span, the pure covariance problem is bounded and has a unique answer.
- In a genuinely two-dimensional, rank-two problem, pure entropy sets the projection cross-covariance to zero.
- The identity-precision form has a coherent mathematical interpretation as a relative-entropy projection onto a Gaussian reference.
- The exact two-direction fixture lets us compare the pure and identity-reference answers without optimizer behavior or empirical noise.

These points are part of a fair reconstruction, not caveats added to soften a predetermined verdict.

4.16 What does not follow

- Feasibility does not follow from rank.
- Boundedness does not follow from feasibility when $r < d$.
- Unboundedness does not follow when $r = d$ and positive-definite feasibility holds.
- The identity term does not follow from the pure Section 4 stationarity condition.
- A reference scale is not determined by the two target marginal variances.
- Nonzero pair norms do not guarantee that the pair Gram matrix is invertible.

The smallest correction is not to discard maximum entropy. It is to specify which maximum-entropy or relative-entropy problem is actually being solved.

4.17 Source and evidence matrix

Statement	Evidence label	Source or proof
Section 4 maximizes Gaussian differential entropy under projection-variance constraints	**paper_claim**	PDF pages 6 and 7
Section 4 conceptually inverts a weighted sum of filter outer products	**paper_claim**	PDF page 7
Feasible non-spanning constraints make the pure objective unbounded	**verified_fact**	Calculations above, non-spanning theorem
Feasible full-spanning constraints give a bounded objective and unique positive-definite maximizer	**verified_fact**	Calculations above, full-spanning boundary
Pure entropy gives zero projection correlation in the complete rank-two problem	**verified_fact**	Calculations above, local two-dimensional comparison
Appendix E.3.1 uses identity precision	**paper_claim**	Proposition E.3 proof, PDF page 54
The printed pair inverse requires two non-collinear directions	**verified_fact**	Calculations above, pair-rank domain
Identity precision is the stationarity form of a Gaussian KL projection	**independent_inference**	Calculations above, reference-prior derivation
Introduce and disclose τ explicitly	**proposed_repair**	Calculations above, proposed repair
Prior-scale sensitivity on the declared rank-two fixture	**verified_fact**	Frozen summary, `P04-PRIOR-SCALE-SENSITIVITY`

4.18 The bounded conclusion

A maximum-entropy claim needs more than a list of observed projections. It needs a feasible constraint set and enough constrained directions to control every entropy-carrying direction.

When those feasible directions do not span a finite-dimensional input, variance can escape through their orthogonal complement and pure differential entropy is unbounded. When they do span, the escape closes and the strict concavity of $\log\det$ selects one covariance.

Appendix E closes the escape in another way: it adds identity precision. The mathematics becomes transparent once that term is read as a Gaussian reference measure. The

constructive next step is to say so explicitly and expose its scale.

> A maximum-entropy construction must constrain every entropy-carrying direction or state the reference measure that anchors the rest.

4.19 Reproduction status

The calculations and paper anchors are included above. The artifact checksum index authenticates these six published outputs:

- summary.json;
- escape_path.csv;
- bounded_control.csv;
- pair_comparison.csv;
- tau_sweep.csv; and
- rank_cases.csv.

From the repository root, validate the declared numerical contract with `python -m unittest discover -s experiments/audits/part04_maximum_entropy -p 'test_*.py' -v`. The outputs are independent mathematical fixtures, not measurements from a checkpoint and not a reproduction of a paper experiment.

This article is published in the `v1.0.0` twelve-part package. Publication does not extend the frozen fixtures beyond the scope stated above.

Chapter 5

When valid pairs do not form a Hilbert space

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `spectra.csv`
- Evidence: `negative_direction.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part05_global_psd
-p 'test_*.py' -v
```

Scope: A fixed three-neuron counterexample and controls, not an estimate of trained-network frequency.

Part 5 of HOPE Under the Microscope

Every pair in a similarity table can look legal while the full table remains geometrically impossible.

That sentence sounds like a criticism of Hilbert spaces. It is not. The correct starting point is the theorem that makes Hilbert geometry useful: when all functions are measured under one common input distribution, their Gram matrix is automatically positive semidefinite.

Let

$$K_{ij} = \mathbb{E}_{X \sim P_X}[g_i(X) g_j(X)].$$

Then, for every coefficient vector c,

$$c^\top K c = \mathbb{E}_{X \sim P_X}\left[\left(\sum_i c_i g_i(X)\right)^2\right] \geq 0.$$

That proof is one line because all entries refer to the same random input X and the same measure P_X.

HOPE's scalable cross-kernel takes a different route. Instead of constructing one full layer covariance, Appendix E.3.1 solves a local Gaussian surrogate for each selected pair. Every isolated 2×2 result can be valid. The question is whether the collection of those local answers must still equal one global Gram matrix.

The fixed three-neuron witness in this article says no.

It starts from a valid correlation matrix whose smallest eigenvalue is approximately 0.126005. After the paper's pairwise warped-correlation map, every isolated pair remains legal, but the assembled three-neuron correlation matrix has smallest eigenvalue approximately -0.250680. After applying the normalized zero-bias ReLU formula pairwise, every isolated pair is again legal, but the full kernel still has smallest eigenvalue approximately -0.045070.

That is enough to refute a general guarantee of one global Hilbert geometry. It is not enough to claim that every layer fails, to estimate a failure rate, or to invalidate the paper's local two-neuron derivation.

This article is bound to arXiv:2607.21366v1. Its exact 70-page PDF fingerprint is recorded in the paper manifest. The exact anchors are physical PDF pages 7 through 9, 35 through 36, and 52 through 56. The paper traceability summary above separates the paper's common kernel and pair-local formulas from the independent PSD theorem and counterexample.

5.1 Evidence labels used here

- **paper_claim**: what HOPE v1 states or prints.
- **verified_fact**: algebra proved independently or a fixed numerical result reproduced by the deterministic audit.
- **counterexample**: an explicit construction that satisfies the declared local assumptions and contradicts a claim with general scope.
- **proposed_repair**: a modified construction not presented as HOPE v1.
- **open_question**: an issue not settled by the paper or this fixed witness.

The matrices printed below come from a deterministic mathematical fixture, not from the paper's experiments and not from a random sweep. The derivations above give the unrounded values and scope.

5.2 Begin with the strongest fair theorem

Evidence: **paper_claim**, `P05-COMMON-PX-KERNEL`, and **verified_fact**, `P05-COMMON-GRAM-PSD`.

HOPE defines a neuron as a vector-valued function. If

$$g_i(x) = \Psi(y_i(x))$$

is its scalar activated signal and $w_{\text{out},i}$ is the downstream weight vector, then

$$f_i(x) = g_i(x) w_{\text{out},i}.$$

Section 5 places these functions in an $L_2(P_X)$-based Hilbert space. Appendix A defines the expectation inner product, and Appendix E, Eq. 73, defines the scalar kernel

$$K(i,j) = \mathbb{E}_{X \sim P_X}[g_i(X) g_j(X)].$$

This construction is globally coherent when P_X is genuinely common. For any finite collection of scalar functions,

$$\begin{aligned} c^\top K c &= \sum_{i,j} c_i c_j \mathbb{E}[g_i(X) g_j(X)] \\ &= \mathbb{E}\left[\left(\sum_i c_i g_i(X)\right)^2\right] \\ &\geq 0. \end{aligned}$$

The full vector-valued neuron Gram is just as safe:

$$G_{ij} = \mathbb{E}[f_i(X)^\top f_j(X)],$$

so

$$c^\top G c = \mathbb{E}\left[\left\|\sum_i c_i f_i(X)\right\|_2^2\right] \geq 0.$$

Nothing about this proof depends on a Gaussian approximation. It does not even depend on ReLU. It depends on square-integrable functions being evaluated under one common probability measure.

This theorem is the ruler for Part 5:

> If a cached table has a materially negative eigenvalue, it cannot be the exact Gram matrix of those functions under one common P_X.

5.3 The scalable move is local

Evidence: **paper_claim**, `P05-PAIR-LOCAL-SURROGATE`.

The paper also recognizes the computational burden of a full layer covariance. Appendix E.3.1 says the global covariance is computationally prohibitive and restricts the maximum-entropy formulation to the two-dimensional subspace of a selected neuron pair.

For effective incoming directions w_i^{eff} and w_j^{eff}, define their cosine

$$\rho_{ij} = \frac{\langle w_i^{\text{eff}}, w_j^{\text{eff}} \rangle}{\|w_i^{\text{eff}}\|_2 \|w_j^{\text{eff}}\|_2}$$

and positive scale ratio

$$r_i = \frac{|\gamma_i|}{\|w_i^{\text{eff}}\|_2}.$$

Paper Eqs. 80 and 81 use

$$\kappa_{ij} = \frac{\rho_{ij}}{1 - \rho_{ij}^2} r_i r_j$$

to produce the pairwise warped correlation

$$\hat{\rho}_{ij} = \frac{2\kappa_{ij}}{1 + \sqrt{1 + 4\kappa_{ij}^2}}.$$

Eq. 82 gives the resulting bivariate Gaussian law for that pair. Eq. 83 gives an exact biased bivariate ReLU moment. For scalable use, Eq. 85 supplies the normalized zero-bias approximation

$$\mathcal{R}(q) = \frac{\sqrt{1 - q^2} + (\pi - \arccos q) q}{\pi}.$$

There is nothing internally contradictory about computing one valid bivariate Gaussian. The missing implication is global:

$$\text{valid pair } (1,2), \quad \text{valid pair } (1,3), \quad \text{valid pair } (2,3)$$

does not by itself give one valid joint law for $(1,2,3)$.

5.4 Three kinds of validity

The whole issue can be hidden by the word "valid." It helps to split that word into three tests.

5.4.1 Isolated pair validity

A unit-diagonal pair matrix

$$\begin{bmatrix} 1 & q \\ q & 1 \end{bmatrix}$$

is PSD exactly when $|q| \leq 1$. This is the pairwise Cauchy-Schwarz check. It says that one bivariate correlation or one normalized pair kernel is legal.

5.4.2 Joint correlation validity

An assembled $n \times n$ unit-diagonal matrix must be PSD to be the correlation matrix of one n-dimensional random vector. For $n \geq 3$, pairwise bounds do not check every quadratic form.

5.4.3 ReLU Gram validity

The matrix of all activation pair moments must be PSD to be one common-P_X Gram matrix. Applying a valid bivariate moment formula separately to the entries of an invalid joint correlation table carries no automatic global guarantee.

The fixed witness passes Test 1 for every pair, fails Test 2, and still fails Test 3 after the ReLU map.

5.5 A valid starting geometry

Evidence: **counterexample**, `P05-GLOBAL-GRAM-COUNTEREXAMPLE`.

Begin with the effective-weight correlation matrix

$$R = \begin{bmatrix} 1 & -0.173508 & 0.590994 \\ -0.173508 & 1 & 0.513747 \\ 0.590994 & 0.513747 & 1 \end{bmatrix}.$$

Its eigenvalues, computed from exactly those six-decimal entries, are approximately

$$0.1260054039, \qquad 1.1716390523, \qquad 1.7023555438.$$

Therefore

$$\lambda_{\min}(R) \approx 0.126005 > 0.$$

This first matrix is not a broken input. It is a positive-definite correlation matrix and therefore the Gram matrix of three unit directions.

Use the positive scale ratios

$$r = \begin{bmatrix} 1.106838 & 1.793881 & 1.634949 \end{bmatrix}.$$

Each pair has $|\rho_{ij}| < 1$, so the denominator in the warped-correlation formula is nonzero. Every ratio is positive. These are the fixed assumptions the local computation needs.

5.6 Every warped pair remains legal

Evidence: **verified_fact**, `PO5-LOCAL-PAIR-VALIDITY`.

Applying Eqs. 80 and 81 independently to the three pairs produces the unrounded values

$$\hat{\rho}_{12} \approx -0.3190441425, \qquad \hat{\rho}_{13} \approx 0.7410265449, \qquad \hat{\rho}_{23} \approx 0.7851452758.$$

Rounded for display, the assembled table is

$$\widehat{R} = \begin{bmatrix} 1 & -0.319044 & 0.741027 \\ -0.319044 & 1 & 0.785145 \\ 0.741027 & 0.785145 & 1 \end{bmatrix}.$$

Every off-diagonal value lies strictly between -1 and 1. The three isolated 2×2 determinants, using the unrounded computed values, are approximately

$$0.8982108351, \qquad 0.4508796597, \qquad 0.3835468959.$$

All are positive. If the audit stopped after checking each pair, every check would pass.

That is precisely why the witness matters. The problem is not visible inside any one pair.

5.7 The joint correlation matrix is impossible

Evidence: **counterexample**, `P05-GLOBAL-GRAM-COUNTEREXAMPLE`.

The full eigenvalue spectrum of $\widehat{R}$, computed before display rounding, is approximately

$$-0.2506797684, \qquad 1.3183990152, \qquad 1.9322807532.$$

So

$$\lambda_{\min}(\widehat{R}) \approx -0.250680 < 0.$$

There is no three-dimensional random vector whose correlation matrix is this $\widehat{R}$. Equivalently, the three valid pair-local Gaussian laws cannot all be the bivariate marginals of one joint Gaussian with the assembled entries.

Notice what has and has not failed:

- The original effective-weight geometry R is valid.
- Each local transformed pair is valid.
- The collection is not jointly valid.

This is the gap between local and global consistency.

5.8 The ReLU map does not repair this witness

Evidence: **counterexample**, `P05-GLOBAL-GRAM-COUNTEREXAMPLE`.

Perhaps applying the activation kernel entry by entry could restore a valid matrix. It does not do so here.

Using the unrounded warped correlations in the normalized Eq. 85 map gives

$$K_{12} \approx 0.1751298794, \qquad K_{13} \approx 0.7811123043, \qquad K_{23} \approx 0.8153656368.$$

The displayed normalized kernel is

$$K_{\text{norm}} = \begin{bmatrix} 1 & 0.175130 & 0.781112 \\ 0.175130 & 1 & 0.815366 \\ 0.781112 & 0.815366 & 1 \end{bmatrix}.$$

Again, every pair looks legal. The unrounded pair determinants are approximately

$$0.9693295253, \qquad 0.3898635681, \qquad 0.3351788783.$$

All are positive.

But the complete spectrum is approximately

$$-0.04506976335, \qquad 0.8250394189, \qquad 2.2200303444.$$

Therefore

$$\lambda_{\min}(K_{\text{norm}}) \approx -0.045070 < 0.$$

The bivariate ReLU formula is valid for each bivariate Gaussian supplied to it. The failure comes from pretending that three separately supplied bivariate laws necessarily belong to one joint distribution.

5.9 A shared-surrogate control stays positive definite

Evidence: **verified_fact**, `P05-SHARED-SURROGATE-CONTROL`, bounded to the same fixed input.

To distinguish the ReLU map from the pair-dependent assembly, apply the same normalized zero-bias map directly to the valid joint correlation matrix R. This gives

$$K_{\text{shared}} = \begin{bmatrix} 1 & 0.2363593803 & 0.6712159349 \\ 0.2363593803 & 1 & 0.6181972853 \\ 0.6712159349 & 0.6181972853 & 1 \end{bmatrix}.$$

Its spectrum is approximately

$$0.1973982261, \qquad 0.7645613281, \qquad 2.0380404458.$$

Thus

$$\lambda_{\min}(K_{\text{shared}}) \approx 0.197398 > 0.$$

This control does not claim that the paper's unavailable full covariance would equal R. It makes a narrower comparison: on the declared fixture, the same ReLU map applied to one valid joint correlation stays positive definite, while the pair-dependent warped assembly is indefinite.

5.10 A negative squared norm

Evidence: **verified_fact**, scoped to the fixed counterexample.

An approximate unit eigenvector for the negative eigenvalue is

$$c \approx \begin{bmatrix} -0.4656516215 \\ -0.4945479776 \\ 0.7338875018 \end{bmatrix}.$$

Reversing its sign gives the same quadratic form. Direct evaluation gives

$$c^\top K_{\text{norm}} c \approx -0.04506976335.$$

If the cached entries were inner products

$$K_{ij} = \langle g_i, g_j \rangle_{\mathcal{H}},$$

then the same quadratic form would be

$$\left\| \sum_i c_i g_i \right\|_{\mathcal{H}}^2.$$

A squared Hilbert norm cannot be negative. So no real Hilbert space contains vectors whose exact Gram matrix is this K_{norm}.

This is a statement about the assembled cached table. It is not a statement that the actual network activations, evaluated together under a real input distribution, have a negative Gram matrix. Their exact empirical or population Gram would be PSD by construction.

5.11 Even the induced triangle breaks

Evidence: **verified_fact**, scoped to the same matrix.

A normalized Gram matrix would induce pair distances

$$d_{ij} = \sqrt{K_{ii} + K_{jj} - 2K_{ij}} = \sqrt{2 - 2K_{ij}}.$$

For the unrounded fixed kernel,

$$d_{12} \approx 1.2844221429, \qquad d_{13} \approx 0.6616459713, \qquad d_{23} \approx 0.6076748525.$$

But

$$d_{12} - (d_{13} + d_{23}) \approx 0.0151013191 > 0.$$

So this proposed set of three Hilbert distances violates the triangle inequality.

Not every indefinite matrix must reveal itself through this particular three-point triangle check. This fixed witness does. The negative quadratic form remains the general PSD certificate.

5.12 What the counterexample proves

It proves one narrow but important statement:

> Pairwise validity of the local surrogate correlations and pairwise ReLU moments does not guarantee that the assembled multi-neuron table is one positive-semidefinite Gram matrix.

The consequence is structural. If an implementation uses the assembled table as though it were one exact Hilbert inner-product matrix, then global quantities built from arbitrary linear combinations need not behave like Hilbert norms or distances.

The proof requires only one fixed witness because the targeted guarantee is universal. One counterexample settles that universal implication.

5.13 What the counterexample does not prove

The boundaries are as important as the negative eigenvalue.

- It does not invalidate an isolated pair covariance from Proposition E.3.
- It does not contradict the common-P_X Gram theorem.
- It does not claim that every layer or checkpoint yields an indefinite cache.
- It does not report the probability of failure under any random model.
- It does not show that the exact Eq. 83 biased kernel fixes or worsens the issue across real checkpoints.
- It does not establish a task-accuracy consequence.
- It does not show that one proposed repair is preferable in every use case.

The project has no authenticated random failure-rate study for Part 5. None is invented here.

5.14 Repair 1: derive every pair from one shared covariance

Evidence: **proposed_repair**.

The most literal repair is to make the probability model genuinely common. Choose one layer covariance $\Sigma \succeq 0$, define

$$Y_i = w_i^\top X + b_i$$

for all neurons under the same X, and derive every pair from the joint preactivation covariance

$$C_Y = W^\top \Sigma W \succeq 0.$$

Then compute the activation moments under that one joint law. Their exact Gram is PSD by the theorem at the start of this article.

This repair has a cost: the shared covariance must be defined, feasible, and computable. It inherits the modeling choices around a layer-wide Gaussian surrogate. Global consistency is restored only after those choices are made explicitly.

5.15 Repair 2: calibrate one activation Gram

Evidence: **proposed_repair**.

If a calibration batch is available, evaluate all m activations together on the same n inputs. Let $A \in \mathbb{R}^{n \times m}$ hold those values and set

$$G_{\text{cal}} = \frac{1}{n} A^\top A.$$

Then

$$c^\top G_{\text{cal}} c = \frac{1}{n} \|Ac\|_2^2 \geq 0.$$

This is one empirical Gram, not a table of independently fitted pairs. A positive-diagonal congruence can normalize it if unit diagonal is needed.

The tradeoff is clear: calibration restores a shared observation basis but abandons a strictly data-free computation. The sample source, size, preprocessing, and handling of zero-activity channels become part of the method.

5.16 Repair 3: clip the spectrum, then renormalize the diagonal

Evidence: **proposed_repair**, `P05-PSD-PROJECTION-REPAIR`.

If the pairwise table is retained, symmetrize it and compute

$$K_s = Q \operatorname{diag}(\lambda_i) Q^\top.$$

The audit's `plain_eigenvalue_clipping` branch uses

$$K_+ = Q \operatorname{diag}(\max\{\lambda_i, 0\}) Q^\top.$$

This removes the negative spectrum, but it generally changes the diagonal. For a normalized kernel, that is not the end of the repair.

When every diagonal entry of K_+ is positive, define

$$D = \operatorname{diag}(K_+), \qquad K_{\text{renorm}} = D^{-1/2} K_+ D^{-1/2}.$$

The result has unit diagonal. It remains PSD because it is a congruence of a PSD matrix. The audit names this branch `psd_renormalized_correlation`.

This package deliberately does not call the two-step procedure a nearest correlation method. Eigenvalue clipping and diagonal renormalization define a specific repair, but the second step moves the matrix again. The final Frobenius change, maximum entrywise change, spectrum, and any change in pair ranking should be measured rather than hidden behind a name.

5.17 Repair 4: call the pair values heuristics

Evidence: **proposed_repair**.

The least invasive computational option is to keep the pair values and narrow their interpretation:

> These are pair-local heuristic scores, not certified entries of one global Hilbert Gram matrix.

That wording avoids a false geometric guarantee, but it has consequences. Any operation that requires global squared norms, projections, angles, or metric inequalities needs a new justification. A useful pair score can remain useful without being an inner product.

5.18 None of the repairs is paper-literal

The paper supplies the common-P_X definition and the pair-local scalable formula. It does not present the four-branch repair menu above as its implemented method.

That boundary matters:

- shared covariance changes what is computed;
- calibration changes the data requirements;

- clipping and renormalization change the cached values; and
- heuristic reframing changes the mathematical claim.

Each repair should be evaluated on the quantity the implementation ultimately uses. Part 5 establishes the consistency issue and the algebraic properties of the repair constructions. It does not choose a winner by downstream performance.

5.19 Open questions

Evidence: **open_question**.

The fixed counterexample leaves several empirical questions unanswered:

1. How often do real checkpoint layers produce materially negative eigenvalues under the pair-local cache?
2. How does the result change with layer width, bias, finite precision, or use of exact Eq. 83 rather than approximate Eq. 85?
3. Which repair best preserves the pair ordering and the behavior an implementation cares about?
4. What numerical tolerance should separate floating-point noise from a material PSD violation?

Those questions need a declared checkpoint study with frozen inputs and results. They cannot be answered from the one witness, and the one witness does not need them to refute a universal guarantee.

5.20 Evidence matrix

Statement	Evidence label	Source or proof
HOPE defines the scalar kernel as an expectation under one P_X	**paper_claim**	Section 5, PDF page 8; Eq. 73, PDF page 52
Any genuine common-P_X Gram matrix is PSD	**verified_fact**	Calculations above, common-distribution theorem
HOPE restricts the scalable surrogate to a selected pair	**paper_claim**	Appendix E.3.1, PDF page 54
The warped correlation and scalable ReLU pair kernel are given by Eqs. 80, 81, and 85	**paper_claim**	PDF pages 54 and 56
Every isolated pair in the fixed witness is valid	**verified_fact**	`pair_checks.csv` in the frozen Part 5 audit

Statement	Evidence label	Source or proof
The fixed warped-correlation table and ReLU table are indefinite	**counterexample**	`spectra.csv` and calculations above
Direct ReLU mapping of the shared valid R gives a positive-definite control	**verified_fact**, bounded to the fixed input	`P05-SHARED-SURROGATE-CONTROL`, `spectra.csv`
The negative direction gives a negative quadratic form	**verified_fact**, bounded to the fixed matrix	`negative_direction.csv` and calculations above
Use a shared covariance, calibrated Gram, or heuristic framing	**proposed_repair**	Calculations above, repair menu
Clip eigenvalues and apply PSD diagonal renormalization	**proposed_repair**	`P05-PSD-PROJECTION-REPAIR`
Checkpoint incidence and repair impact	**open_question**	Requires a separate empirical protocol

5.21 Reproduction contract

The audit README specifies the fixed inputs, method origins, and tolerances. The artifact checksum index authenticates these six published numerical outputs:

- summary.json;
- witness_matrices.csv;
- spectra.csv;
- pair_checks.csv;
- negative_direction.csv; and
- projection_comparison.csv.

The direct deterministic generator is:

```
python3 experiments/audits/part05_global_psd/run.py --result-dir <fresh-dir>
```

Run the focused tests shown at the top of this page to validate the numerical contract. The checksum index authenticates the result-file bytes.

5.22 The bounded conclusion

HOPE's common-P_X kernel has exactly the right global property: it is a Gram matrix, so it is PSD.

The scalable pair-local replacement needs a separate global check. A valid 2×2 result certifies one pair, not a whole table. In the fixed three-neuron construction, all pairs pass

while both the assembled warped correlation and the assembled normalized ReLU kernel fail PSD.

That makes the right conclusion precise:

> Pairwise consistency is weaker than global realizability.

The constructive response is equally precise: use one shared covariance, use one calibrated activation Gram, clip and PSD-renormalize the assembled matrix while reporting the perturbation, or present the values as heuristics rather than one Hilbert geometry.

This article is published in the `v1.0.0` twelve-part package. Its conclusions remain bounded to the declared constructions and controls.

Chapter 6

What the axioms derive, and what they do not

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `derivation_steps.csv`
- Evidence: `path_integrals.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part06_axioms -p
'test_*.py' -v
```

Scope: The audit distinguishes conditional derivations from independent counterexamples and proposed scoring choices.

Figure status: This part's four PNGs are static editorial visualizations of the nearby derivations and authenticated tables, not computational outputs. Their exact published bytes are authenticated by `artifacts/SHA256SUMS`.

6.1 A theorem, a change of objective, a bound, and a useful design choice

An elegant derivation stops being one derivation when its objective changes halfway through.

That does not make the final objective useless. It changes what can honestly be claimed for it.

HOPE needs one score that can compare structural edits across layers. The score should ignore a common rescaling, resist complete layer extinction, react to feature geometry, and

remain cheap enough to evaluate repeatedly. Those are sensible requirements. The final score,

$$J_{\text{final}} = N\frac{D(\Phi_a, \widetilde{\Phi}_b)}{E(\Phi_b)},$$

is also a sensible engineered proxy for them.

The narrower question in Part 6 is logical:

> Which pieces follow from the stated assumptions, and which pieces enter as additional modeling or algorithm-design choices?

The answer has two distinct layers of non-uniqueness.

1. The continuous capacity-drain theorem leaves a positive scale-invariant factor $c(\Phi)$ unspecified.
2. After the paper explicitly relaxes path additivity, the two retained axioms admit endpoint proxies besides the printed linear ratio.

This is not a claim that the paper hides the second transition. Section 6.2 states that Axiom 3 is intentionally relaxed. The issue is how much of the final formula should still be described as derived from all three axioms.

6.2 Evidence labels used here

- **paper_claim** reports what HOPE v1 states, defines, or proves.
- **verified_fact** is supported by an independent derivation and the scoped deterministic Part 6 audit.
- **independent_inference** separates consequences of printed statements from the paper's own wording.
- **counterexample** is an explicit construction that refutes a stated implication within its declared domain.
- **proposed_repair** changes the presentation or method and is not attributed to HOPE v1.
- **open_question** marks something this package does not establish.

The numerical examples below are deterministic mathematical fixtures, not a paper experiment and not a checkpoint study. The published result files give their exact machine-readable values, and the focused tests validate them.

6.3 The strongest fair version of the paper's argument

Evidence: **paper_claim** for the paper's capacity, geometric change, relaxation, and width choice; independently reconstructed **verified_fact** for the integral family and straight-path bound.

The paper begins with a layer state

$$\Phi = (f_1, \dots, f_N) \in \mathcal{H}^N$$

and assigns it a scalar capacity

$$E(\Phi) = \sum_{i=1}^{N} \|f_i\|_{\mathcal{H}}.$$

It motivates a transition cost through three goals:

1. **Magnitude neutrality.** A common positive rescaling of both endpoints should not change the cost.
2. **Connectivity preservation.** The cost should diverge as terminal layer capacity approaches zero.
3. **Infinitesimal capacity dependence.** Cost should accumulate additively along a continuous path and be driven by capacity drain.

Theorem C.2 turns those requirements into a relative capacity-drain integral. Section 6.2 then observes that scalar capacity drain cannot distinguish two transitions with the same lost capacity but different geometric movement. The paper defines a geometric rate, selects a straight path, bounds its varying denominator by terminal capacity, chooses $c(\Phi) = N$, and instantiates the resulting endpoint score.

That sequence contains useful mathematics. It also contains several kinds of step:

- a theorem under explicit assumptions;
- a restriction used to obtain an endpoint expression;
- a new geometric definition;
- a selected path;
- a conditional upper bound;
- an explicit relaxation of one axiom; and
- a width-normalization choice.

Part 6 keeps those categories visible.

6.4 Partition invariance selects the sum inside an L_p-style family

Evidence: **paper__claim**, `P06-L1-CAPACITY-CLAIM`; independent **verified__fact**, `P06-LP-PARTITION-RESULT`.

Let one functional neuron have capacity

$$e = \|f\|_{\mathcal{H}} > 0.$$

Consider

$$E_p(e_1, \dots, e_N) = \left(\sum_{i=1}^{N} e_i^p \right)^{1/p}, \qquad p > 0.$$

For $p < 1$, this is an L_p-style aggregator rather than a norm. The calculation does not require the triangle inequality.

Partition f into M identical fractional operators:

$$f \longmapsto \left(\frac{f}{M}, \dots, \frac{f}{M} \right).$$

Positive homogeneity gives each fraction capacity e/M. Therefore

$$\begin{aligned} E_p\left(\frac{e}{M}, \dots, \frac{e}{M} \right) &= \left(M \left(\frac{e}{M} \right)^p \right)^{1/p} \\ &= M^{1/p-1} e. \end{aligned}$$

Requiring equality with e for every positive integer M forces

$$\frac{1}{p} - 1 = 0, \qquad \text{so} \qquad p = 1.$$

Within this declared family, the result is exact:

$$E_1(\Phi) = \sum_i \|f_i\|_{\mathcal{H}}.$$

The frozen audit uses $e = 5$,

$$M \in \{1, 2, 3, 5\}, \qquad p \in \{0.5, 1, 2\}.$$

For $p = 0.5$, the split capacity grows as $5M$. For $p = 1$, it remains exactly 5. For $p = 2$, it decreases as $5/\sqrt{M}$. Only the $p = 1$ row is invariant on every declared split and, by the exponent proof, on every positive integer split.

Figure 1. The equal-split test inside the explicit L_p-style family. Only $p = 1$ remains fixed at capacity five.

6.5 The broader uniqueness statement needs narrower wording

Evidence: **independent_inference** based on direct proof inspection; `P06-LP-PARTITION-RESULT` is the bounded verified result.

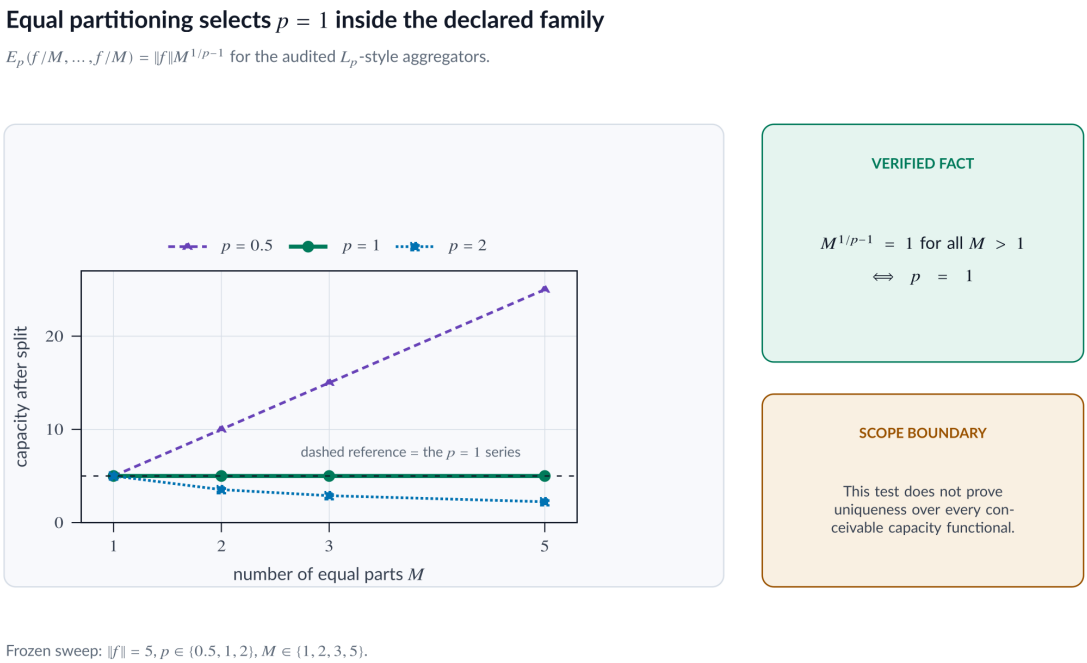


Figure 6.1: Equal-split capacity for the declared p and M values.

The main text summarizes Lemma C.1 as a uniqueness result for a symmetric, separable, homogeneous capacity. The appendix states a more specific quasi-arithmetic form:

$$E(\Phi) = h\left(\sum_{k=1}^{N} g(\|f_k\|_{\mathcal{H}})\right),$$

with g continuous and strictly monotone, plus identity and partition invariance conditions.

That is a narrower modeling class than symmetry and homogeneity alone. The printed proof identifies $h = g^{-1}$ on the image of g, then applies the inverse to summed arguments without separately establishing that those arguments remain in that image or that h is globally injective.

This package does not turn that proof-domain issue into a claim that the lemma is false. It takes a safer route:

- report the paper's broader uniqueness statement as a **paper_claim**;
- verify the equal-split exponent argument inside the explicit L_p-style family; and
- avoid saying that partition invariance plus a few informal adjectives rules out every other conceivable capacity functional.

"The sum is well motivated" is weaker than "the sum is uniquely forced over every reasonable functional class."

6.6 The three axioms, written precisely

Evidence: independently reconstructed **verified_fact**, `P06-CAPACITY-INTEGRAL-FAMILY`, anchored to the paper's three axioms and Theorem C.2.

Let $\Phi : [0,1] \to \mathcal{H}^N$ be a differentiable deformation with endpoints Φ_a and Φ_b.

Magnitude neutrality requires

$$J(k\Phi_a, k\Phi_b) = J(\Phi_a, \Phi_b), \qquad k > 0.$$

Connectivity preservation requires

$$\lim_{E(\Phi_b)\to 0^+} J(\Phi_a, \Phi_b) = \infty.$$

Infinitesimal capacity dependence writes

$$J(\Phi_a, \Phi_b) = \int_0^1 -\xi(\Phi(t))\dot{E}(t)\, dt.$$

where $\dot{E}(t) < 0$ on the idealized capacity-drain path and $\xi(\Phi) > 0$.

The third statement specifies both the local quantity that accumulates and an additive continuous-path model.

6.7 What Theorem C.2 determines

Evidence: **verified_fact** from independently reconstructed paper algebra, `P06-CAPACITY-INTEGRAL-FAMILY`.

Under common positive scaling,

$$E(k\Phi) = kE(\Phi), \qquad \dot{E}(k\Phi) = k\dot{E}(\Phi).$$

For the integral to remain scale neutral on every valid localized path, the density scales with degree negative one:

$$\xi(k\Phi) = k^{-1}\xi(\Phi).$$

Define

$$c(\Phi) = \xi(\Phi)E(\Phi).$$

Then $c(k\Phi) = c(\Phi)$, and Eq. 43 becomes

$$J_{\text{capacity}}(\Phi_a, \Phi_b) = \int_0^1 -c(\Phi(t)) \frac{\dot{E}(t)}{E(\Phi(t))}\, dt. \tag{A}$$

The theorem determines the relative capacity rate $-\dot{E}/E$. It does not determine one unique $c(\Phi)$.

For example, while active width is fixed along one continuous piece, both

$$c(\Phi) = 1 \qquad \text{and} \qquad c(\Phi) = N$$

are positive and scale invariant. Additional design criteria are needed to choose between them.

6.8 Connectivity requires more than pointwise positivity

Evidence: **counterexample**, `P06-CONNECTIVITY-LOWER-BOUND`.

The proof of Theorem C.2 obtains divergence by assuming

$$c(\Phi(t)) \geq c_{\min} > 0$$

along a path to extinction. Under that condition,

$$J \geq c_{\min} \log\left(\frac{E_a}{E_b}\right) \longrightarrow \infty.$$

Pointwise positivity and global scale invariance alone do not provide a uniform lower bound.

For $0 < E < 1$, define the two-coordinate state

$$\Phi(E) = (E - E^2, E^2)$$

and the globally scale-invariant factor

$$c(\phi_1, \phi_2) = \frac{\phi_2}{\phi_1 + \phi_2}.$$

For every $k > 0$,

$$c(k\phi_1, k\phi_2) = c(\phi_1, \phi_2).$$

Along the declared path only,

$$c(\Phi(E)) = \frac{E^2}{E - E^2 + E^2} = E.$$

Therefore

$$\int_{E_b}^{E_a} c(\Phi(E)) \frac{dE}{E} = \int_{E_b}^{E_a} dE = E_a - E_b,$$

which stays finite as $E_b \to 0^+$.

This factor does not satisfy Axiom 2 on the path. That is the point: the conditions "positive" and "scale invariant" do not make connectivity preservation automatic. Axiom 2 must constrain the family further.

6.9 Piecewise-constant c creates an endpoint expression

Evidence: reconstructed **verified_fact**, `P06-CAPACITY-INTEGRAL-FAMILY`.

The paper next restricts c to remain constant during one transition:

$$c(\Phi(t)) = c(\Phi_a).$$

Equation (A) then integrates exactly:

$$J_{\text{capacity}} = c(\Phi_a) \log\left(\frac{E_a}{E_b}\right).$$

The frozen scalar fixture uses

$$E_a = 5, \qquad E_b = 2, \qquad c = 3,$$

and therefore

$$J_{\text{capacity}} = 3 \log\left(\frac{5}{2}\right) \approx 2.7488721956.$$

This endpoint expression still cannot distinguish two edits with the same capacity drain but different motion in $\mathcal{H}^N$.

6.10 The central fork: capacity drain becomes geometric speed

Evidence: **paper_claim** and direct source inspection, `P06-GEOMETRIC-OBJECTIVE-CHANGE`.

For a path that shrinks one coordinate while holding all others fixed, geometric speed and negative capacity rate coincide:

$$\|\dot{\Phi}(t)\|_{\mathcal{H}^N} = -\dot{E}(t).$$

The paper calibrates on that special direction and defines

$$\dot{J}_{\mathrm{proj}}(t) \equiv c(\Phi(t))\frac{\|\dot{\Phi}(t)\|_{\mathcal{H}^N}}{E(\Phi(t))}. \tag{B}$$

Equation (B) is Eq. 44. It responds to geometric motion even when scalar capacity stays fixed.

This is the point where the analytical object changes. Equation (A) accumulates capacity drain. Equation (B) accumulates geometric arc length weighted by inverse capacity. They agree on the calibration path, but not on every deformation by algebra alone.

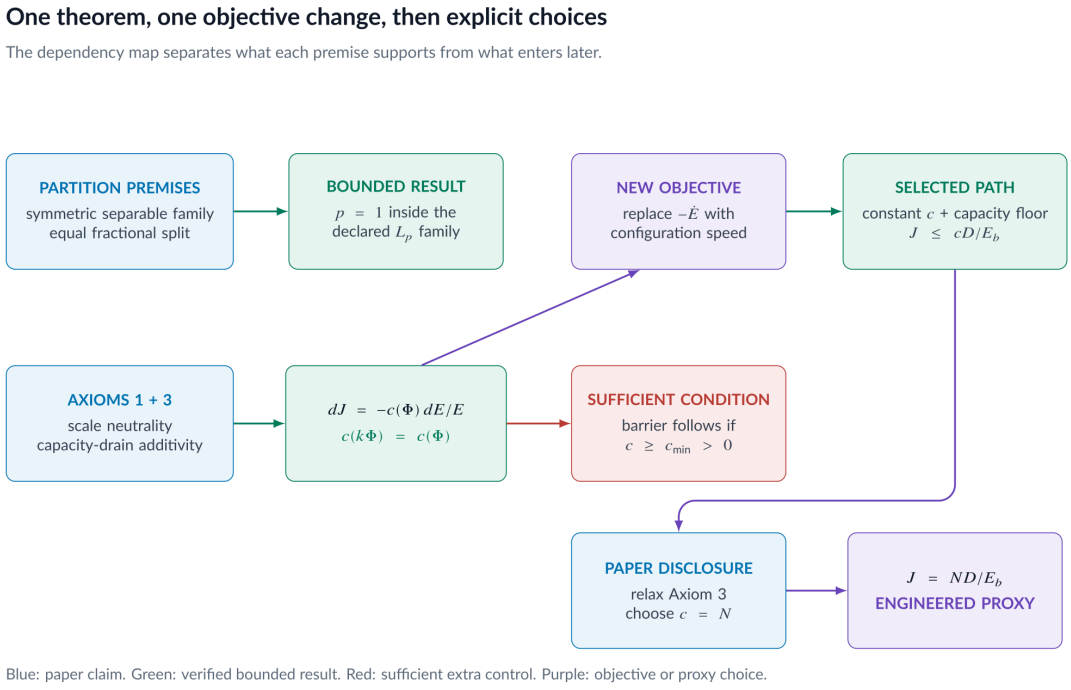


Figure 6.2: Dependency map from the partition assumptions and axioms to Eq. 6.

Figure 2. The derivation map labels theorem, restriction, definition, path choice, bound, relaxed axiom, and width-normalization choice separately.

6.11 The geometric integral is path dependent

Evidence: independent deterministic calculation supporting the distinction introduced by `P06-GEOMETRIC-OBJECTIVE-CHANGE`. The stable claim ID remains a **paper_claim** about Eq. 44; this fixture has no separate claim ID.

Use $\mathcal{H} = \mathbb{R}$, two nonnegative coordinates,

$$E(x, y) = x + y, \qquad c = N = 2.$$

Compare paths from

$$A = (3, 1) \qquad \text{to} \qquad B = (1, 1).$$

The straight path is

$$\Phi_s(t) = (3 - 2t, 1).$$

Its speed is 2 and its capacity is $4 - 2t$, so

$$\begin{aligned} I_s &= 2 \int_0^1 \frac{2}{4 - 2t}\, dt \\ &= 2 \log 2 \\ &\approx 1.3862943611. \end{aligned}$$

Now detour through

$$C = (1, 2.5).$$

On $A \to C$, speed is 2.5, capacity is $4 - 0.5t$, and the contribution is

$$10 \log\left(\frac{8}{7}\right).$$

On $C \to B$, speed is 1.5, capacity is $3.5 - 1.5t$, and the contribution is

$$2 \log\left(\frac{7}{4}\right).$$

Thus

$$I_d = 2\left[5\log\left(\frac{8}{7}\right) + \log\left(\frac{7}{4}\right)\right]$$
$$\approx 2.4545455021.$$

The endpoints are identical, but the costs differ. This is not a contradiction of the paper. Section 6.2 responds by selecting a straight path and bounding that selected path under a capacity-floor premise.

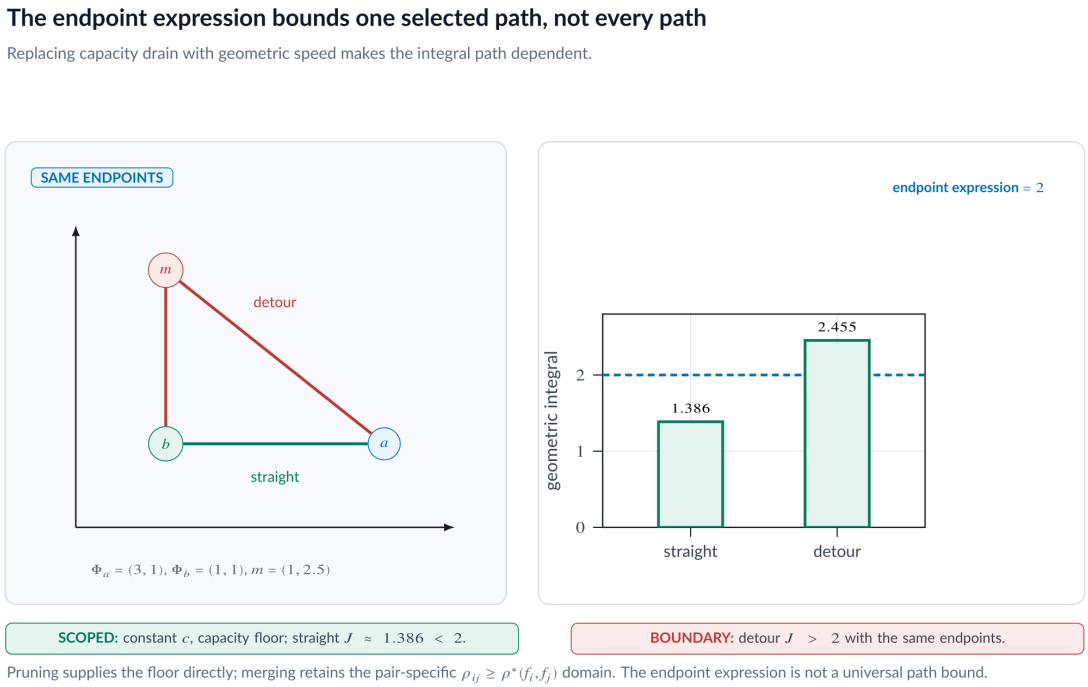


Figure 6.3: Straight and detour paths over the same capacity field.

Figure 3. Same endpoints, different Eq. 44 path integrals. The endpoint expression is a bound for the selected straight path under its stated premise, not for the displayed detour.

6.12 The straight-path upper bound

Evidence: independently reconstructed **verified_fact**, `P06-STRAIGHT-PATH-BOUND`.

Choose

$$\Phi(t) = (1-t)\Phi_a + t\widetilde{\Phi}_b.$$

Its speed is constant:

$$\|\dot{\Phi}(t)\|_{\mathcal{H}^N} = D(\Phi_a, \widetilde{\Phi}_b).$$

Under the stated capacity-floor condition

$$E(\Phi(t)) \geq E(\Phi_b) > 0,$$

and with c constant on the transition,

$$\begin{aligned} J_{\text{proj}}^{\text{straight}} &= c(\Phi_a) \int_0^1 \frac{D}{E(\Phi(t))}\, dt \\ &\leq c(\Phi_a) \frac{D}{E(\Phi_b)} \\ &\equiv J_{\text{bound}}. \end{aligned}$$

For pruning, the paper verifies this capacity floor directly along the linear shrink path. For the merge branch, Lemma C.3 supplies it only inside an existential, pair-specific neighborhood

$$\rho_{ij} \geq \rho^*(f_i, f_j).$$

Part 6 carries that condition with the theorem. It does not audit whether a runtime procedure enforces or encounters the threshold.

For the path fixture,

$$D(A, B) = 2, \qquad E(B) = 2, \qquad c = 2,$$

so the endpoint expression is

$$J_{\text{bound}} = 2.$$

The selected straight cost is $2 \log 2 < 2$. The detour cost is greater than two, which reinforces the scope: this endpoint expression does not bound an arbitrary path.

The bound is a configuration-space statement. It is not a demonstrated bound on final logits, loss, accuracy, or every continuous deformation.

6.13 The paper explicitly relaxes Axiom 3

Evidence: **paper_claim**, `P06-AXIOM3-RELAXATION`.

Proposition C.5 checks magnitude neutrality and connectivity preservation for the endpoint proxy on its stated structural-transition domain. Section 6.2 then says that infinitesimal capacity dependence and its path additivity are intentionally relaxed.

This disclosure controls the correct interpretation. The proxy can be judged by its retained properties, but it should not also be presented as the unique consequence of the axiom it no longer satisfies.

The loss of additivity is visible in a one-dimensional endpoint fixture. Let

$$J(E_a, E_b) = 2\frac{E_a - E_b}{E_b}.$$

Then the direct transition $4 \to 2$ costs

$$J(4, 2) = 2.$$

Splitting it at 3 gives

$$J(4, 3) + J(3, 2) = \frac{2}{3} + 1 = \frac{5}{3},$$

which is not two. This is a deterministic illustration of the relaxation, not a defect the paper denies.

6.14 Choosing $c(\Phi) = N$

Evidence: **paper_claim** for the printed proposal and rationale; **independent_inference** for non-uniqueness, `P06-WIDTH-NORMALIZATION-CHOICE`.

The paper proposes

$$c(\Phi) = N,$$

where N is live pre-action width. This normalizes by average terminal capacity.

Suppose all N active neurons have capacity $\bar{e}$, and prune one. Then

$$D = \bar{e}, \qquad E_b = (N-1)\bar{e},$$

so

$$J_{\text{prune}} = \frac{N}{N-1}.$$

For wide layers, this is close to one instead of growing with raw layer capacity. That is a coherent normalization goal. It is still a design criterion: neither Eq. 43 nor scale invariance uniquely selects N.

6.15 Eq. 6 follows from the selected proxy

Evidence: **paper_claim**, `P06-WIDTH-NORMALIZATION-CHOICE`, and reconstructed **verified_fact**, `P06-STRAIGHT-PATH-BOUND`.

With $c = N$,

$$J_{\text{final}} = N\frac{D}{E_b}.$$

For solitary pruning,

$$D = \|f_i\|_{\mathcal{H}}, \qquad E_b = E_a - \|f_i\|_{\mathcal{H}},$$

which gives

$$J_{\text{prune}}(i) = \frac{N\|f_i\|_{\mathcal{H}}}{E_a - \|f_i\|_{\mathcal{H}}}.$$

For the paper's two-coordinate pre-deletion target,

$$D = \sqrt{\|f_i - f_p\|_{\mathcal{H}}^2 + \|f_j - f_p\|_{\mathcal{H}}^2},$$

and

$$E_b = E_a - \|f_i\|_{\mathcal{H}} - \|f_j\|_{\mathcal{H}} + \|f_p\|_{\mathcal{H}}.$$

Substitution gives Eq. 6. Part 6 stops at that algebra. It does not test later structural execution or the global optimizer.

6.16 A family of endpoint proxies passes the retained tests

Evidence: independent **counterexample** to endpoint-proxy uniqueness, `P06-ENDPOINT-PROXY-NONUNIQUENESS`.

After Axiom 3 is relaxed, define

$$r = \frac{D}{E_b}$$

and consider

$$J_p = N r^p, \qquad p > 0. \tag{C}$$

The paper uses $p = 1$. Other values are independent countermodels to a uniqueness claim, not methods reported by HOPE v1 and not proposed repairs.

Scale every state by $k > 0$. Both D and E_b scale by k, so r is unchanged. Every J_p is magnitude neutral.

On the same fixed-initial structural-transition limit used by the endpoint proxy, $D/E_b \to \infty$ as $E_b \to 0^+$. Therefore

$$J_p \to \infty$$

for every $p > 0$.

Applying a fixed scalar power also preserves constant-operation evaluation once r is available. These facts do not restore path additivity. They show that after it is removed, the retained endpoint properties do not select one exponent.

6.17 The exponent can change a cross-layer ranking

Evidence: bounded independent **verified_fact**.

Claim: `P06-CROSS-LAYER-RANKING-SENSITIVITY`.

Use three synthetic equal-payoff candidates:

candidate	N	$r = D/E_b$
A	2	4
B	7	1
C	3	2

At $p = 0.5$, their scores are

$$J_{0.5}(A) = 4, \qquad J_{0.5}(B) = 7, \qquad J_{0.5}(C) = 3\sqrt{2} \approx 4.242641.$$

The ascending order is

$$A < C < B.$$

At $p = 1$,

$$J_1(A) = 8, \qquad J_1(B) = 7, \qquad J_1(C) = 6,$$

and the order is

$$C < B < A.$$

At $p = 2$,

$$J_2(A) = 32, \qquad J_2(B) = 7, \qquad J_2(C) = 12,$$

and the order is

$$B < C < A.$$

The fixture proves possibility, not prevalence or performance impact. It is not a checkpoint result and not a compression experiment.

There is also an important non-result. If candidates share the same fixed N, have equal payoff, and differ only through positive r, then $r \mapsto r^p$ is strictly increasing for every $p > 0$. The exponent does not change that restricted ordering.

Figure 4. The independent endpoint family preserves common-scale neutrality and the scoped fixed-initial extinction behavior when D retains a positive limit, but changes the declared cross-layer ranking as p changes. It does not measure compression performance.

6.18 Why $p = 1$ remains defensible

Evidence: **independent_inference** from the paper's construction and the verified family calculation.

Non-uniqueness is not the same as arbitrariness. The printed exponent has at least three coherent motivations:

1. It is linear in the geometric distance from the straight-line arc-length bound.
2. Combined with $c = N$, equal-capacity mean-field pruning costs $N/(N-1)$, nearly independent of width.
3. It preserves the direct interpretation of normalized configuration-space displacement.

Those are design reasons. A method can defend a design choice without claiming that every alternative violates the original axioms.

Holding $c = N$ fixed across the family gives equal-capacity behavior



Figure 6.4: Endpoint proxy families and the declared ranking fixture.

$$J_p = \frac{N}{(N-1)^p}.$$

Only $p = 1$ is approximately width neutral under that particular normalization. A different exponent could be paired with another width factor, which again exposes an added selection criterion.

6.19 What checks out

1. **paper_claim**, `P06-L1-CAPACITY-CLAIM`: HOPE states additive layer capacity and presents partition invariance as its justification.
2. **verified_fact**, `P06-LP-PARTITION-RESULT`: in the declared L_p-style family, equal-split invariance selects $p = 1$.
3. **verified_fact**, `P06-CAPACITY-INTEGRAL-FAMILY`: scale neutrality converts Axiom 3 into the Eq. 43 family with scale-invariant c.
4. **counterexample**, `P06-CONNECTIVITY-LOWER-BOUND`: the divergence proof requires additional behavior such as a positive lower bound on c.
5. **paper_claim**, `P06-GEOMETRIC-OBJECTIVE-CHANGE`: Eq. 44 changes the local rate from capacity drain to geometric speed.
6. **verified_fact**, `P06-STRAIGHT-PATH-BOUND`: under its stated capacity floor, the selected straight-path integral is bounded by cD/E_b.

7. **paper_claim**, `P06-AXIOM3-RELAXATION`: the paper retains Axioms 1 and 2 and explicitly relaxes Axiom 3 for the endpoint proxy.
8. **paper_claim**, `P06-WIDTH-NORMALIZATION-CHOICE`: $c = N$ has a clear average-capacity rationale and is explicitly proposed by the paper.
9. **counterexample**, `P06-ENDPOINT-PROXY-NONUNIQUENESS`: every $p > 0$ in the declared endpoint family shares scale neutrality and the same scoped extinction limit.
10. **verified_fact**, `P06-CROSS-LAYER-RANKING-SENSITIVITY`: the declared three-candidate fixture has different orders at $p = 0.5, 1, 2$.

6.20 What does not follow

- Symmetry and homogeneity alone uniquely force additive capacity.
- Pointwise positive scale-invariant c automatically guarantees an extinction barrier on every path.
- Geometric speed is identical to capacity drain on every deformation.
- The endpoint proxy preserves Axiom 3.
- The straight-path expression bounds every path or an end-to-end network metric.
- The retained endpoint properties uniquely select $p = 1$.
- Alternative exponents improve compression.
- Ranking changes occur within every layer or candidate set.

6.21 A constructive presentation repair

Evidence: **proposed_repair**.

Define

$$J_{\text{proxy}} = N\frac{D}{E_b}$$

directly as a normalized configuration-space projection proxy. Then prove its common-scale invariance, scoped extinction barrier, and locality. Keep the capacity-drain integral as idealized motivation. State that geometric speed, the straight path, the capacity floor, and width normalization complete the executable definition.

That revision leaves the score unchanged while aligning the claim with the properties the score actually has.

6.22 Source and evidence matrix

Statement	Evidence label	Anchor or proof
HOPE defines additive layer capacity	**paper_claim**	Section 6.1, PDF page 10; Lemma C.1, pages 42-43
Equal splitting selects $p = 1$ inside the L_p-style family	**verified_fact**	Eqs. 58-61, PDF page 50; calculations above
Axiom 3 and scale neutrality yield Eq. 43 with free c	**verified_fact**, reconstructed paper algebra	PDF page 43; `derivation_steps.csv`
Positive scale-invariant c need not imply the barrier	**counterexample**	calculations above, shape-factor construction
Eq. 44 replaces capacity rate with geometric speed	**paper_claim**	Section 6.2, page 11; Eq. 44, page 44
Eq. 44 is path dependent	**verified_fact**, bounded fixture	`path_integrals.csv`; calculations above
The selected straight path is bounded by cD/E_b under its premise	**verified_fact**, reconstructed paper algebra	Lemma C.3 and Theorem C.4, Eqs. 45-48, pages 44-46
The endpoint proxy retains Axioms 1 and 2 and relaxes Axiom 3	**paper_claim**	Section 6.2, page 12; Proposition C.5, page 46
The paper proposes $c = N$	**paper_claim**	Section 6.2 and Footnote 6, page 12
$J_p = N(D/E_b)^p$ is non-unique under retained properties	**counterexample**, independent family	`proxy_family.csv`; calculations above
The three-candidate order changes with p	**verified_fact**, bounded fixture	`ranking_fixture.csv`; calculations above
Another exponent improves a real model	**open_question**	Requires a separate compression study

6.23 Reproduction contract

The deterministic audit lives under `experiments/audits/part06_axioms/` and writes fresh results outside the frozen references. The canonical direct command is:

```
python3 experiments/audits/part06_axioms/run.py --result-dir <fresh-dir>
```

The audit README records the configuration and tolerances. The focused tests validate the six-output package, and the artifact checksum index authenticates the published result-file bytes.

6.24 Scope boundary

Part 6 conditionally grants the capacity and configuration-space geometry in order to audit the route from assumptions to endpoint score. It does not revisit Part 5, test physical structural execution, audit block actions, analyze optimizer complexity, evaluate DEFT, or reproduce a compression experiment.

6.25 The bounded conclusion

The additive capacity has a clean partition argument inside a declared L_p-style family. The continuous model yields a meaningful relative capacity-drain family, but leaves $c(\Phi)$ open. Section 6.2 changes the local objective to geometric speed, selects and bounds a straight path, explicitly relaxes Axiom 3, and chooses $c = N$ for cross-layer normalization.

That is enough to motivate

$$J_{\text{final}} = N\frac{D}{E_b}.$$

It is not enough to call it the unique consequence of all three axioms. Once path additivity is gone, the positive-exponent endpoint family passes the two retained tests under the same scoped extinction domain.

Separate what a theorem forces from what an algorithm designer chooses.

This article is published in the `v1.0.0` twelve-part package. Its conclusions remain bounded to the declared derivations, counterexamples, and controls.

Chapter 7

The zero-cost merge that removes one copy

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `twin_cases.csv`
- Evidence: `repair_comparison.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part07_merge_
semantics -p 'test_*.py' -v
```

Scope: Exact-twin and bounded synthetic fixtures separate paper-literal pricing, physical execution, and proposed repairs.

Figure status: This part's four PNGs are static editorial visualizations of the nearby derivations and authenticated tables, not computational outputs. Their exact published bytes are authenticated by `artifacts/SHA256SUMS`.

7.1 A correct projection can still price the wrong transition

Two neurons compute exactly the same nonzero function. A compression method replaces both with one parent. What should the parent compute?

If the layer adds neuron contributions, the answer is immediate: the parent must reproduce the sum, $2f$, if the local layer map is to remain unchanged.

HOPE v1 solves a different problem. Its merge objective asks one parent function to approximate each child coordinate separately:

$$[f, f] \longrightarrow [f_p, f_p].$$

That problem selects $f_p = f$ and reports zero distance. The physical update then keeps only one parent vessel. The executed local transition is

$$2f \longrightarrow f.$$

The projection is exact for the state it was asked to fit. The trouble is that the fitted state is not the state left by the surgery.

Part 7 makes this mismatch explicit, proves it symbolically, realizes the printed parent in two deterministic dense-BN-ReLU-dense forward fixtures, and compares two separately labeled repair objectives. The conclusion is narrow but exact:

> A zero duplicated-coordinate merge distance does not imply zero change in the executed local summed contribution.

Nothing here measures final logits, loss, accuracy, or the frequency of these configurations in trained networks.

7.2 Evidence labels used here

- **paper_claim** reports what HOPE v1 states, defines, or derives.
- **verified_fact** is supported by an independent derivation, direct source composition, or the deterministic Part 7 audit.
- **independent_inference** connects printed definitions without attributing the conclusion to the paper.
- **counterexample** is an explicit construction that refutes a general interpretation within its stated domain.
- **proposed_repair** changes the optimized object and is not paper-literal.
- **open_question** marks something this package does not establish.

The numerical tables are deterministic mathematical fixtures. They are not a paper experiment, a checkpoint study, or a random sample. The published result files provide the exact values; the audit README and focused tests define their tolerances and checks.

7.3 Begin with the strongest fair account of the paper

Evidence: **paper_claim**, `P07-DUPLICATED-TARGET`, and independently checked source composition for the later execution.

HOPE treats each neuron as a vector-valued function in a Hilbert space. When merging children i and j, it seeks one realizable parent direction and one positive parent scale. This is a meaningful constrained approximation problem. The paper also does substantial work

to turn the abstract parent back into physical incoming weights, outgoing weights, and BatchNorm state.

The relevant chain is spread across several places:

1. Eq. 6 prices the two child coordinates against two parent coordinates.
2. Eqs. 8 through 14 choose the parent direction and scale.
3. Eqs. 15 through 18 realize that one parent physically.
4. Eq. 86 defines a tuple's aggregate mapping by summing its components.
5. Algorithm 2 stores the parent in child i's vessel and removes child j.

Every individual step can be read coherently. The problem appears when the first and last objects are compared: two parent coordinates are priced, but one parent contribution remains.

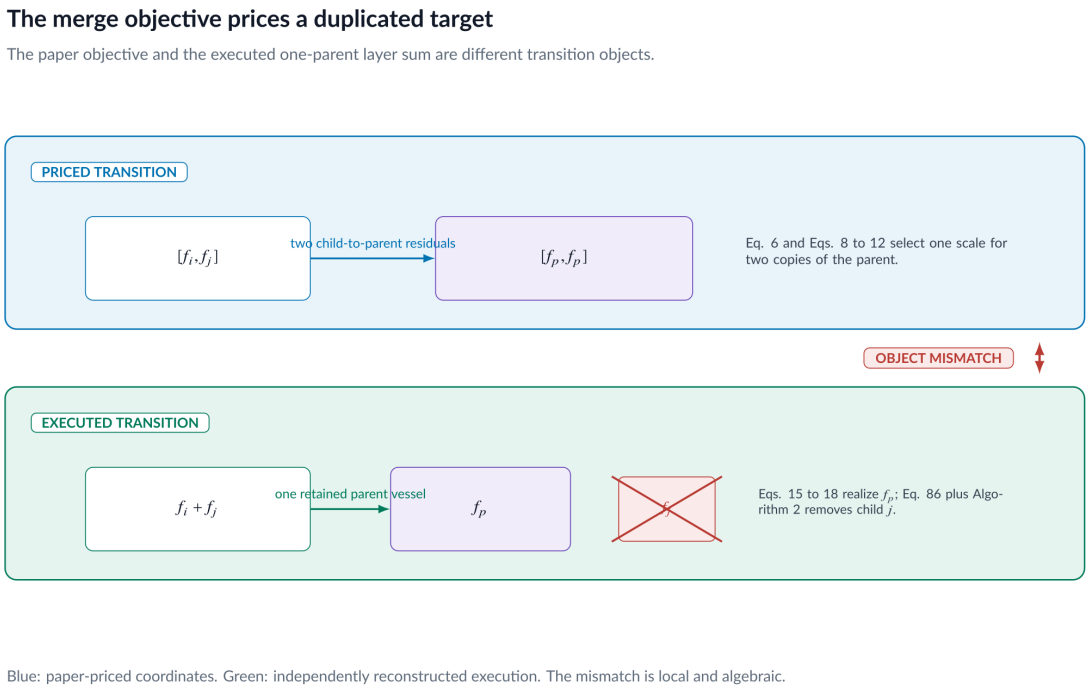


Figure 7.1: A map separating the priced duplicated-coordinate transition from the executed local sum transition.

Figure 1. Eq. 6 prices two parent coordinates. Eqs. 15 through 18 realize one parent, and Algorithm 2 leaves one parent vessel. Eq. 86 determines the local summed contribution on each side.

7.4 The paper-literal coordinate objective

Evidence: **paper_claim**, `P07-DUPLICATED-TARGET`, anchored to Eq. 6 and Eqs. 8 through 12 on physical PDF pages 13 through 15.

Fix a unit realizable parent direction ψ, and write

$$f_p = s\psi, \qquad \|\psi\|_{\mathcal{H}} = 1, \qquad s > 0.$$

Define

$$a = \|f_i\|_{\mathcal{H}}^2 + \|f_j\|_{\mathcal{H}}^2,$$

$$b = \langle \psi, f_i + f_j \rangle_{\mathcal{H}},$$

and let $R = E_{\text{rem}} \geq 0$ be the capacity outside the two children. The squared duplicated-coordinate distance is

$$\begin{aligned} D_{\text{dup}}^2(s) &= \|f_i - s\psi\|^2 + \|f_j - s\psi\|^2 \\ &= 2s^2 - 2bs + a. \end{aligned} \tag{1}$$

Ignoring the fixed live-width factor, the squared normalized objective is

$$Q_{\text{dup}}(s) = \frac{2s^2 - 2bs + a}{(R+s)^2}. \tag{2}$$

Under the paper's positive-alignment phase check, $b > 0$. Differentiating Eq. 2 gives the printed scale

$$s_{\text{paper}} = \frac{a + bR}{2R + b}. \tag{3}$$

This is the correct stationary scale for the duplicated-coordinate objective. Part 7 does not dispute that calculus.

7.5 The executed local object is a sum

Evidence: independent **verified_fact**, `P07-EXECUTED-SUM-TRANSITION`.

Eq. 86 states that the aggregate mapping of a tuple is the sum of its constituent operators:

$$\mathcal{F}(\Phi) = \sum_k f_k. \tag{4}$$

Before the merge, the two selected coordinates contribute

$$f_i + f_j.$$

Algorithm 2 lines 12 through 14 update capacity, place the parent in child i's vessel, and remove child j. After the merge, the selected region contributes

$$f_p = s\psi.$$

The local residual of the executed transition is therefore

$$D_{\mathrm{phys}}(s) = \|f_i + f_j - s\psi\|_{\mathcal{H}}. \tag{5}$$

If

$$c = \|f_i + f_j\|_{\mathcal{H}}^2,$$

then

$$D_{\mathrm{phys}}^2(s) = s^2 - 2bs + c. \tag{6}$$

Equation 1 and Eq. 6 are not two algebraic expansions of the same distance. They are distances between different objects.

7.6 The exact-twin counterexample

Evidence: **counterexample**, `MERGE-TWIN-COUNTEREXAMPLE`.

Let the children be nonzero exact twins:

$$f_i = f_j = f, \qquad q = \|f\|_{\mathcal{H}} > 0, \qquad \psi = f/q.$$

The three scalar quantities are

$$a = 2q^2, \qquad b = 2q, \qquad c = 4q^2. \tag{7}$$

Substitute a and b into the paper scale:

$$\begin{aligned} s_{\mathrm{paper}} &= \frac{2q^2 + 2qR}{2R + 2q} \\ &= q. \end{aligned} \tag{8}$$

This identity holds for every $R \geq 0$. The paper parent is

$$f_p = q\psi = f.$$

The duplicated-coordinate distance is then

$$D_{\text{dup}} = \sqrt{\|f - f\|^2 + \|f - f\|^2} = 0. \tag{9}$$

Because the numerator of Eq. 6 is zero and its denominator is positive, the printed merge cost is zero.

The executed local residual is different:

$$\begin{aligned} D_{\text{phys}} &= \|f + f - f\| \\ &= \|f\| \\ &= q > 0. \end{aligned} \tag{10}$$

Thus the exact same parent is perfect in the duplicated coordinate product space and imperfect for the one-parent summed layer map.

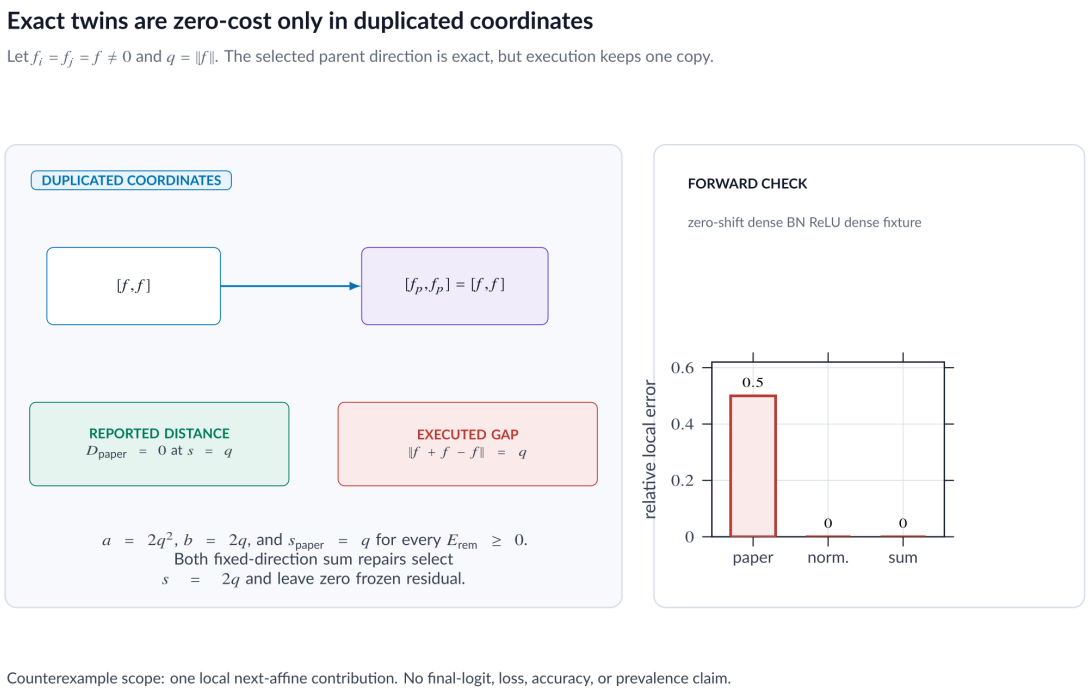


Figure 7.2: An exact-twin calculation showing zero duplicated-coordinate distance and a positive executed sum residual.

Figure 2. For exact twins, the paper scale is q for every remaining capacity. Two priced copies match, while one executed parent leaves residual q.

7.7 What the counterexample refutes

Evidence: **counterexample**, `MERGE-TWIN-COUNTEREXAMPLE`.

The witness refutes the universal implication

$$D_{\text{dup}} = 0 \quad \Longrightarrow \quad D_{\text{phys}} = 0.$$

It also refutes any universal interpretation of Eq. 6's merge numerator as the norm of the executed local aggregate change.

The counterexample does not require a failure of the Hilbert inner product, a bad parent direction, a floating-point instability, or a BatchNorm recovery error. It grants the geometry and gives the parent the most favorable possible direction: the exact child direction.

7.8 What the counterexample does not prove

Evidence: **open_question** boundary.

The witness does not establish:

- how often exact or near twins occur in a trained network;
- a bound on final logits after downstream nonlinearities;
- a loss or accuracy change;
- an end-to-end network norm bound;
- that every merge is harmful;
- that the parent-direction approximation fails; or
- that either proposed repair improves a trained model.

It proves a mismatch between a priced local object and an executed local object. That is enough to invalidate a universal executed-distortion reading, but not enough to quantify empirical consequences.

7.9 The physical realization confirms one copy

Evidence: independent **verified_fact**, `P07-PHYSICAL-FORWARD-TWIN`, anchored to Eqs. 15 through 18.

One possible response to the Hilbert-space proof would be that the physical parameter reconstruction secretly restores the missing factor of two. The audit tests that possibility directly.

It builds two deterministic exact-twin dense-BN-ReLU-dense fixtures. Each fixture contains two identical child paths, a nonzero outgoing vector, and seven fixed three-dimensional inputs. The first has

$$\beta = 0, \qquad \gamma = 1.2, \qquad q \approx 0.7776888838.$$

The second has

$$\beta = 0.35, \qquad \gamma = 1.4, \qquad q \approx 1.0026920316.$$

The reconstruction uses $\epsilon = 10^{-5}$ and the active boundary-safe variance relation. It follows the paper's input-output scale split and recovers the effective incoming affine map, outgoing vector, and BN state.

For the paper-literal scale $s = q$, the realized parent output equals one child contribution within 2×10^{-12} on every declared check. Against the reference sum of two child contributions, the relative Frobenius error is

$$\frac{\|f_p - (f_i + f_j)\|_F}{\|f_i + f_j\|_F} = \frac{1}{2}$$

in both fixtures.

The largest effective-weight, effective-bias, surrogate-scale, or expected-forward reconstruction discrepancy across all six physical rows is about 4.44×10^{-16}. The physical reconstruction is doing what its selected scale specifies. It does not create a second parent copy.

This forward check is local. The reference is the summed contribution into the next affine layer on seven fixed inputs, not final network output.

7.10 Near twins expose a bounded gap

Evidence: **counterexample**, `P07-CORRELATION-GAP`.

The exact-twin limit is decisive, but a single limiting case can hide how the two distances separate nearby. The audit therefore uses a bounded zero-bias fixture family.

The incoming directions have correlation

$$\rho \in \{0.7, 0.9, 0.95, 0.99, 0.999\}.$$

The children have equal capacities

$$q_i = q_j = 1/\sqrt{2},$$

and remaining capacity is fixed by

$$R = 32(q_i + q_j).$$

The parent direction follows the paper's rank-two direction rule and exact phase check. For aligned outgoing vectors, the ratio

$$\frac{D_{\mathrm{phys}}(s_{\mathrm{paper}})}{D_{\mathrm{dup}}(s_{\mathrm{paper}})}$$

takes these values:

Input correlation ρ	Duplicated distance	Physical residual	Ratio
0.700	0.371279	0.674453	1.8166
0.900	0.218264	0.693023	3.1752
0.950	0.155445	0.699458	4.4997
0.990	0.070178	0.705416	10.0518
0.999	0.022308	0.706929	31.6901

The denominator shrinks toward zero as the pair approaches an exact aligned twin, while the one-parent physical residual remains near one child scale.

Output alignment matters. Repeating the same input correlations with outgoing cosine 0.8 gives ratios

$$1.3903, 1.7547, 1.9087, 2.0697, 2.1128.$$

These are ten fixed rows, not a distribution over trained pairs. The audit supports the listed values and the existence of a gap on this declared family. It does not support a prevalence claim.

Figure 3. The true-to-reported distance ratio grows across the aligned-output fixture as input correlation approaches one. The partially aligned control shows that output geometry also matters.

7.11 The large-capacity limit reveals average semantics

Evidence: independent **verified_fact**, `P07-PAPER-SCALE-LIMIT`.

For fixed a and $b > 0$, Eq. 3 has the limit

$$\lim_{R\to\infty} s_{\mathrm{paper}} = \lim_{R\to\infty} \frac{a + bR}{2R + b} = \frac{b}{2}. \tag{11}$$



Figure 7.3: A bounded correlation sweep comparing the duplicated-coordinate distance with the executed local residual.

The factor of two comes from Eq. 1: the same parent scale appears in two coordinate residuals. In contrast, the orthogonal projection of the physical sum onto ψ has scale

$$\langle \psi, f_i + f_j \rangle = b.$$

The capacity sweep fixes the $\rho = 0.95$, aligned-output fixture and varies

$$R/b \in \{0, 0.25, 1, 4, 16, 64, 10^3, 10^6\}.$$

At $R/b = 10^6$, the observed s_{paper}/b differs from $1/2$ by about 6.19×10^{-9}. This is a finite check of the exact algebraic limit, not an empirical network result.

7.12 Near cancellation reveals the opposite mismatch

Evidence: **counterexample**, `P07-NEAR-CANCELLATION`.

The twin case shows undercharging of a physical sum change. A fixed near-cancellation fixture shows that duplicated coordinates can also remain large when the physical sum is small.

The two children share the same zero-bias input activation. Their outgoing vectors point in opposite directions, with magnitudes 1 and 0.99. Their summed function is only one percent of one child's signed contribution. Relative to the sum of their unsigned capacities, its norm fraction is

$$0.0050251256,$$

or about 0.5025%.

For the declared remaining-capacity rule, the paper-literal scale gives

$$D_{\mathrm{dup}} \approx 0.9951214, \qquad D_{\mathrm{phys}} \approx 0.0074569.$$

The duplicated-coordinate distance is about 140.7314 times the raw sum norm 0.0070711. This fixture does not say that cancellation pairs are common or desirable to merge. It shows that coordinate closeness and aggregate sum closeness are genuinely different objectives in both directions.

7.13 The downstream distortion symbol retains the same mismatch

Evidence: **independent_inference**, `P07-H4-DISTORTION-GAP`, anchored to Proposition H.4 and Eqs. 116 through 118 on physical PDF pages 66 through 67.

Proposition H.4 defines the merge distortion

$$\delta_k = \min_{f_p} \left(\sum_{m=1}^{M} \|f_m - f_p\|^2 \right)^{1/2}. \tag{12}$$

This is the duplicated-coordinate distance generalized to M children. The proposition bounds it above by projecting all children to zero, and Eq. 118 inserts the resulting δ_k values into a cumulative expression.

For two exact twins, $f_p = f$ gives

$$\delta_k = 0.$$

Algorithm 2 nevertheless leaves one parent contribution, so the local summed residual is

$$\|2f - f\| = q.$$

The independent conclusion is precise: the displayed δ_k does not charge the one-copy removal in this witness. Consequently, Eqs. 116 through 118 do not establish a bound on

that executed local residual merely by naming δ_k a merge distortion. This conclusion does not quantify final source-task degradation.

7.14 Repair 1: normalized physical-residual scale

Evidence: **proposed_repair**, `P07-NORMALIZED-SUM-REPAIR`.

Keep the selected unit parent direction ψ, but replace the duplicated-coordinate numerator with the physical sum residual:

$$Q_{\mathrm{norm}}(s) = \frac{s^2 - 2bs + c}{(R+s)^2}, \qquad c = \|f_i + f_j\|^2. \tag{13}$$

Differentiation gives

$$Q'_{\mathrm{norm}}(s) = \frac{2\left[s(R+b) - (bR+c)\right]}{(R+s)^3}. \tag{14}$$

For $R \geq 0$ and $b > 0$, the positive stationary scale is

$$s_{\mathrm{norm}} = \frac{bR+c}{R+b}. \tag{15}$$

For exact twins, $b = 2q$ and $c = 4q^2$, so

$$s_{\mathrm{norm}} = 2q$$

for every $R \geq 0$. The local physical residual is zero.

As $R \to \infty$, this repair tends to b, not $b/2$. On the declared capacity sweep, the observed s_{norm}/b at $R/b = 10^6$ differs from one by about 8.66×10^{-10}.

This is not Eq. 12. It changes both the numerator's semantic object and the scale formula.

7.15 Repair 2: unnormalized physical projection

Evidence: **proposed_repair**, `MERGE-SUM-REPAIR`.

A second option removes the capacity normalization from the scale selection and minimizes

$$Q_{\mathrm{unnorm}}(s) = s^2 - 2bs + c. \tag{16}$$

The derivative is $2(s-b)$, so

$$s_{\text{unnorm}} = b. \tag{17}$$

The remaining residual is

$$\sqrt{c - b^2}, \tag{18}$$

the norm of the component of $f_i + f_j$ orthogonal to ψ. For exact twins, $c = b^2 = 4q^2$, so $s = 2q$ and the local sum is preserved exactly.

The phrase "sum preserving" needs this qualifier. For a general pair whose sum does not lie in the fixed parent direction, $c - b^2 > 0$. The scale $s = b$ is the best unnormalized projection onto that direction, not a universal function-preserving merge.

7.16 The repairs solve different problems

Evidence: **proposed_repair** comparison, `P07-NORMALIZED-SUM-REPAIR` and `MERGE-SUM-REPAIR`.

The two alternatives agree for exact twins but are not the same method:

Method	Optimized quantity	Fixed-direction scale	Exact twins
Paper literal	duplicated coordinates normalized by terminal capacity	$(a + bR)/(2R + b)$	q
Normalized physical repair	physical sum residual normalized by terminal capacity	$(bR + c)/(R + b)$	$2q$
Unnormalized physical repair	physical sum residual	b	$2q$

On the three aligned-output fixtures with

$$\rho \in \{0.95, 0.99, 0.999\}, \qquad R = 32(q_i + q_j),$$

the unnormalized physical residual is smaller than the paper-literal physical residual by approximately

$$17.0143, \qquad 57.5442, \qquad 324.5049$$

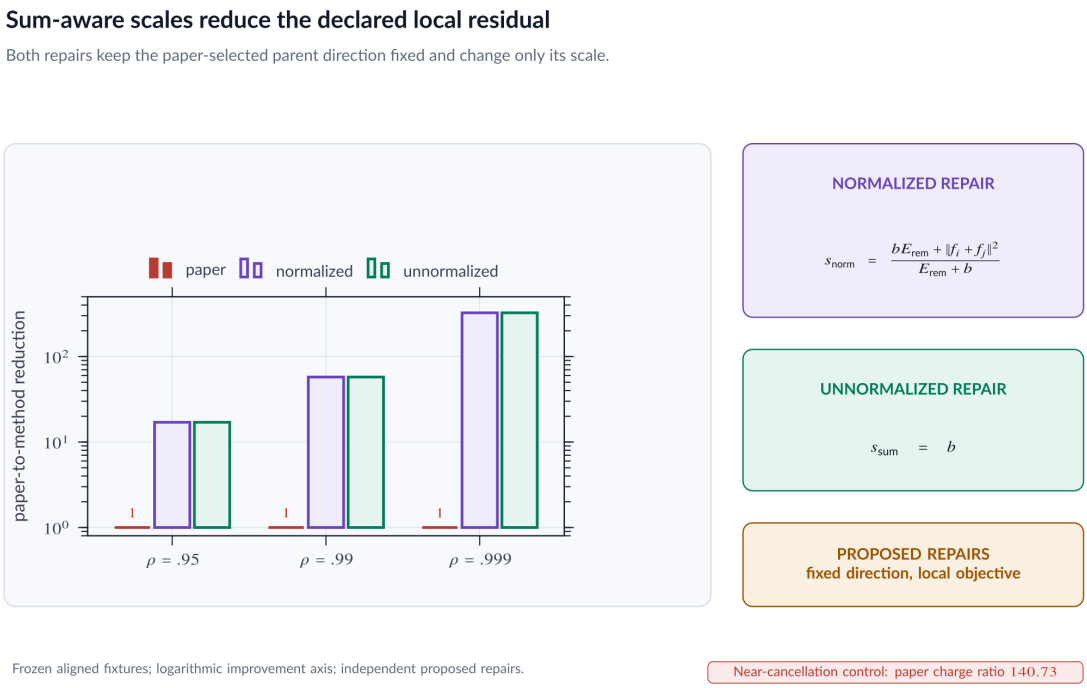


Figure 7.4: A comparison of the paper-literal scale with two distinct physical-residual repair scales.

times. The normalized repair gives nearly the same three finite values on this large-R fixture. These factors are not universal bounds. They apply only to the specified zero-bias, equal-capacity, aligned-output rows and compare local Hilbert residuals.

Figure 4. Both repairs choose $2q$ for exact twins. Away from the twin case, the normalized and unnormalized formulas remain distinct and neither guarantees exact preservation outside the selected direction.

7.17 Physical forward behavior under the repairs

Evidence: **proposed_repair** fixture results, `PO7-NORMALIZED-SUM-REPAIR` and `MERGE-SUM-REPAIR`.

The same Eqs. 15 through 18 realization machinery can be fed a different selected scale without changing the parent-direction fixture. For exact twins, both repairs request $s = 2q$.

Across the two deterministic physical fixtures and seven fixed inputs, the largest repair relative error against the two-child local reference is about

$$1.12 \times 10^{-16}.$$

That result verifies the implementation and the exact-twin homogeneity calculation. It is not evidence that either repair preserves arbitrary pairs, improves compression accuracy, or should be adopted without further tests.

7.18 A constructive correction to the claim boundary

Evidence: **proposed_repair** to presentation.

Even without changing the algorithm, the semantic boundary can be made accurate:

1. Call Eq. 6 a duplicated-coordinate projection proxy.
2. State that Eqs. 15 through 18 realize one selected parent operator.
3. Separately define and report the executed local aggregate residual $\|f_i + f_j - f_p\|$.
4. Do not use zero coordinate distance as a function-preservation certificate.
5. Keep any physical-residual scale in a clearly labeled non-paper-literal namespace.

That wording preserves the paper's valid coordinate optimization while removing the unsupported equivalence between priced and executed objects.

7.19 Source and evidence matrix

Statement	Evidence class	Claim ID	Support
The merge objective targets two copies of one parent	`paper_claim`	`P07-DUPLICATED-TARGET`	Eq. 6 and Eqs. 8 through 12, PDF pages 13 through 15
One parent is physically reconstructed	`paper_claim`	`P07-DUPLICATED-TARGET`	Eqs. 15 through 18, PDF pages 15 through 16
The local aggregate changes from child sum to one parent	`verified_fact`	`P07-EXECUTED-SUM-TRANSITION`	Eq. 86 plus Algorithm 2 lines 12 through 14
Exact twins have zero coordinate distance and residual q	`counterexample`	`MERGE-TWIN-COUNTEREXAMPLE`	Exact proof and `twin_cases.csv`
The paper scale tends to $b/2$	`verified_fact`	`P07-PAPER-SCALE-LIMIT`	Algebra and `capacity_sweep.csv`
Physical reconstruction gives one-copy behavior	`verified_fact`	`P07-PHYSICAL-FORWARD-TWIN`	`physical_forward.csv`
Near-twin ratios reach the listed bounded values	`counterexample`	`P07-CORRELATION-GAP`	`correlation_sweep.csv`

Statement	Evidence class	Claim ID	Support
The normalized physical scale is Eq. 15 above	`proposed_repair`	`P07-NORMALIZED-SUM-REPAIR`	Derivation and `repair_comparison.csv`
Unnormalized physical projection selects $s = b$	`proposed_repair`	`MERGE-SUM-REPAIR`	Derivation and `repair_comparison.csv`
A nearly canceled sum can retain a large coordinate distance	`counterexample`	`P07-NEAR-CANCELLATION`	`repair_comparison.csv`
Proposition H.4's δ_k misses the twin removal	`independent_inference`	`P07-H4-DISTORTION-GAP`	Eqs. 116 through 118 plus exact witness

7.20 Reproduction contract

The Part 7 audit uses Python and NumPy with float64 arithmetic. It uses no random sampling, dataset, checkpoint, Torch, or SciPy. It writes seven deterministic outputs and refuses to overwrite a nonempty result directory.

Run:

```
python3 experiments/audits/part07_merge_semantics/run.py \
  --result-dir artifacts/runs/part_07/manual
```

Verify the frozen package:

```
python -m unittest discover -s experiments/audits/part07_merge_semantics -p
'test_*.py' -v
```

The focused tests check the named metric ranges, row schemas, evidence classes, finite numbers, output identities, and deterministic rerun equality.

The seven result files separate source mapping, exact twins, bounded correlations, remaining-capacity behavior, repairs, and physical forward checks. The artifact checksum index authenticates their bytes.

7.21 Scope boundary

Part 7 concerns one local additive layer contribution before and after one merge. It includes:

- the paper's duplicated-coordinate algebra;
- exact and biased twin fixtures;
- zero-bias correlation and output-alignment sweeps;
- one remaining-capacity sweep;

- one near-cancellation fixture;
- the paper-literal physical parent reconstruction; and
- two fixed-direction repair comparisons.

It excludes trained-network incidence, final logits, loss, accuracy, end-to-end distortion, parent-direction optimality outside the declared fixtures, and any universal repair guarantee.

7.22 The bounded conclusion

HOPE's parent derivation solves the coordinate problem it writes down. Its physical reconstruction realizes the selected one-parent operator accurately on the declared fixtures.

Those facts do not bridge the missing semantic step.

The priced transition contains two copies of the parent. The executed local transition contains one. Exact twins make the gap impossible to dismiss:

$$D_{\mathrm{dup}} = 0, \qquad D_{\mathrm{phys}} = \|f\| > 0.$$

A repaired merge must decide explicitly whether it is fitting an average of coordinates or preserving an aggregate contribution. The two proposed scales in this package illustrate different physical-residual choices, but neither is promoted beyond its fixed-direction mathematical scope.

Chapter 8

Residual-block eviction: topology and a conditional Gaussian stress test

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `moment_validation.csv`
- Evidence: `ordering_fixtures.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part08_block_
eviction -p 'test_*.py' -v
```

Scope: The graph argument is bounded to eligible post-activation ResNet V1 identity shortcuts, not projection shortcuts. The numerical ratios additionally assume an explicit Gaussian law for the pre-ReLU coordinate; Figure 3 alone does not establish that law or identify its parameters with a visible BatchNorm.

Part 8 of HOPE Under the Microscope

Under an explicit Gaussian pre-activation model, a factor of $\sqrt{2}$ can influence which entire residual block gets deleted.

HOPE's block-level idea is excellent in spirit. A residual network offers an operation that ordinary channel pruning cannot express cleanly: remove a whole residual pathway and let the skip connection carry the representation forward. If neuron pruning, neuron merging, and block eviction are all going to compete in one optimizer, each action needs a comparable price.

The paper proposes such a price. The price depends on the capacity of the identity shortcut.

Figure 3 traces that shortcut input to the previous block's post-ReLU output, while Eq. 96 inserts an unrectified Gaussian second moment. That is a verified topology-versus-model mismatch, but topology alone does not determine the actual second moment of a trained ResNet activation.

The size of the mismatch can be stress-tested conditionally. If the previous pre-activation coordinate is explicitly modeled as $Y \sim \mathcal{N}(\beta, \gamma^2)$, then at zero shift the unrectified identity energy is larger than the post-ReLU energy by exactly $\sqrt{2}$. Under that same synthetic premise, negative shifts produce an unbounded ratio and an equal-payoff fixture can reverse two candidate rankings. These are conditional countermodels, not measured corrections for a trained ResNet-50.

There is also a separate issue. The score adds destruction terms from sequential internal layers, even though those layers are not additive components of the residual function. A ReLU-idempotence witness keeps the residual map exactly fixed while the local-layer sum grows with depth. That does not make the score useless. It means the score is an architecture-dependent heuristic rather than a projection distance determined only by the residual function.

8.1 The strongest fair version of the proposal

`paper_claim` | `P08-EVICTION-SCORE`

In the canonical bottleneck drawn in Figure 3, the residual branch computes $F(X)$, the skip carries X, and the block forms

$$Y = X + F(X), \qquad Z = \mathrm{ReLU}(Y).$$

Eviction forces $F(X)$ to zero, leaving the identity mapping. This can free the residual branch's parameters without disconnecting the network.

For the two internal layers, HOPE defines

$$J_{\mathrm{evict}} = \frac{\sum_{\ell=1}^{2} N_{\mathrm{active}}^{(\ell)} E_{\mathrm{active}}^{(\ell)}}{E_{\mathrm{identity}}}.$$

The numerator charges the active widths and capacities of the two internal layers. The denominator is meant to capture the survival capacity supplied by the skip connection. Appendix F decomposes that skip into coordinate identity operators and prints

$$E_{\mathrm{identity}}^{\mathrm{paper}} = \sum_{k} \sqrt{\gamma_k^2 + \beta_k^2}.$$

This is an appealing macro score: a branch becomes cheap to evict as its internal capacity shrinks, while a strong identity path prevents the connectivity denominator from collapsing.

The problem is not the desire for a survival term. The problem is that the actual boundary random variable and its second moment must be identified before a BatchNorm-parameter formula can be substituted for them.

8.2 Trace the tensor, not the block label

`verified_fact` | `P08-POST-RELU-BOUNDARY`

Figure 3 shows a post-activation ResNet V1 block. Its last operation is

$$Z = \mathrm{ReLU}(Y).$$

For two consecutive interior identity blocks in that V1 chain, the previous block's Z becomes the next block's X. The identity shortcut therefore carries a post-ReLU tensor.

That scope matters. This article does not say that every object called a "residual block" has the same boundary. A pre-activation V2 block has a different graph. A projection shortcut is a learned map, not the coordinate identity used in Eq. 95. A stride-changing shortcut changes the spatial operator. Those cases need their own derivations.

For the exact graph drawn in Figure 3, however, the interior identity input is post-ReLU. This topological fact neither makes the preceding sum Gaussian nor shows that Eq. 96's β, γ parameterize it. If a separate modeling or measurement argument establishes

$$Y \sim \mathcal{N}(\beta, \gamma^2),$$

then the skip coordinate in that conditional model is

$$X = \mathrm{ReLU}(Y),$$

not Y.

8.3 The two moments in the conditional Gaussian fixture

`independent_inference` | `BLOCK-IDENTITY-MOMENT`

Equation 96 uses the unrectified RMS associated with its stated Gaussian approximation,

$$R_{\mathrm{pre}} = \sqrt{\mathbb{E}[Y^2]} = \sqrt{\gamma^2 + \beta^2}.$$

Under the additional conditional model above, the post-ReLU boundary has

$$R_{\text{post}} = \sqrt{\mathbb{E}[\text{ReLU}(Y)^2]}.$$

The paper has already derived this second moment in Eq. 79:

$$\mathbb{E}[\text{ReLU}(Y)^2] = (\gamma^2 + \beta^2)\Phi\left(\frac{\beta}{|\gamma|}\right) + \beta|\gamma|\phi\left(\frac{\beta}{|\gamma|}\right).$$

Part 8 checks that conditional expression independently with deterministic quadrature and 500,000 seeded Monte Carlo samples. Across the declared validation cases, the largest quadrature error is 3.55×10^{-15} in second-moment units. The largest seeded Monte Carlo error is 0.01301. The analytic identity, not the random estimate, is the decisive result.

8.4 Under the Gaussian fixture, the zero-shift gap is exact

Within the declared synthetic model, set $\beta = 0$. Then

$$R_{\text{pre}} = |\gamma|, \qquad R_{\text{post}} = \frac{|\gamma|}{\sqrt{2}}.$$

So

$$\frac{R_{\text{pre}}}{R_{\text{post}}} = \sqrt{2}.$$

Now hold the eviction numerator and parameter payoff fixed. Because identity capacity is in the denominator,

$$J_{\text{post}} = \sqrt{2} J_{\text{paper}}.$$

Equivalently,

$$\frac{J_{\text{paper}}}{J_{\text{post}}} = \frac{1}{\sqrt{2}} \approx 0.7071.$$

The paper-literal cost is lower than this conditional-Gaussian comparator by

$$1 - \frac{1}{\sqrt{2}} \approx 29.289\%$$

relative to the conditional post-ReLU candidate cost.

Two qualifiers must remain attached: the exact $\sqrt{2}$ and 29.289 percent figures require both zero shift and the declared Gaussian pre-activation model. They are neither measured ResNet corrections nor worst-case bounds.

8.5 Conditional negative shifts make the energy ratio much larger

Write $\beta = -t\gamma$ with $t > 0$. The normalized post-ReLU moment is

$$(1 + t^2)\Phi(-t) - t\phi(t).$$

Gaussian-tail expansion gives

$$(1 + t^2)\Phi(-t) - t\phi(t) \sim \frac{2\phi(t)}{t^3}.$$

Meanwhile, the unrectified moment is $1 + t^2$. Therefore the RMS ratio grows without bound as t increases.

The frozen sweep is intentionally finite. At $\beta/|\gamma| = -4$, it records

$$\frac{R_{\text{pre}}}{R_{\text{post}}} = 2345.47.$$

That number is a synthetic fixture, not a population estimate. The asymptotic proof establishes unbounded behavior inside the Gaussian family. The row only makes that conditional scale visible.

The zero-variance boundary is even starker. If $\gamma = 0$ and $\beta < 0$, Eq. 96 assigns identity RMS $|\beta|$, while the deterministic post-ReLU coordinate is exactly zero.

8.6 A conditional two-candidate ordering reversal

`counterexample` | `P08-ORDERING-FLIP`

In the same synthetic Gaussian model, suppose two candidate records have equal parameter payoff. Candidate A has numerator 1, $\beta = -1$, and $\gamma = 1$. Candidate B has numerator 0.85, $\beta = 0$, and $\gamma = 1$.

Using Eq. 96:

$$\text{DR}_A^{\text{paper}} = 0.7071, \qquad \text{DR}_B^{\text{paper}} = 0.85.$$

Lower is better, so the paper moment selects A first.

Using the post-ReLU moment:

$$\text{DR}_A^{\text{post}} = 3.6432, \qquad \text{DR}_B^{\text{post}} = 1.2021.$$

Now B comes first.

This establishes a conditional sensitivity result: changing the modeled boundary moment can change the action order. It does not prove that either candidate occurs in a trained ResNet-50, that Eq. 96's parameters describe an actual post-add pre-activation, or that the alternative ranking produces higher accuracy.

8.7 A second issue: sequential layers are not additive branch functions

`counterexample` | `P08-SEQUENTIAL-DEPTH-GAP`

Appendix F notices an important difficulty. Sequential layers cannot simply be grouped into one additive macro state. Nulling an earlier layer can collapse the pathway, but an additive tuple would continue counting later layers.

The implemented score sidesteps that state construction by computing separate local-layer bounds and adding them. That produces a usable number, but it does not turn the sequential layers into additive components of the residual function.

Consider

$$h_d(x) = \underbrace{\mathrm{ReLU} \circ \cdots \circ \mathrm{ReLU}}_{d \text{ times}}(x).$$

ReLU is idempotent, so

$$h_d(x) = \mathrm{ReLU}(x)$$

for every $d \geq 1$. Every depth has the same residual map and the same function-space eviction change.

Assign the same local capacity to each charged stage. The Eq. 20-style sum is then proportional to d. The frozen fixture verifies exact function equality for depths one through four, while the proxy at depth four is four times its depth-one value.

This does not show that architecture should never matter. Parameter payoff, execution cost, and optimization risk can all depend on architecture. It shows something narrower and important: the local-layer sum is not a projection distance determined only by the residual branch function.

8.8 Where the repair is eligible

`verified_fact` | `P08-IDENTITY-ELIGIBILITY`

The coordinate decomposition in Eq. 95 assumes

$$I_k(x) = x_k e_k.$$

The Part 8 audit therefore distinguishes five cases:

1. An interior post-activation V1 identity bottleneck with equal shape and stride one is eligible for Eq. 95 and for auditing the post-ReLU boundary. The graph alone does not make the Gaussian closed form eligible.
2. A learned channel projection is not a coordinate identity.
3. A stride-changing learned projection also changes spatial sampling.
4. A V2 identity shortcut satisfies abstract identity algebra, but not the Figure 3 post-ReLU boundary used here.
5. A sequential block with no skip has no identity survival path.

Only the first case receives the scoped boundary repair below. Whether its closed form is Gaussian remains a separate empirical or modeling question.

8.9 The scoped correction

`proposed_repair` | `P08-RELU-MOMENT-REPAIR`

For a graph location proven to be an eligible post-ReLU identity shortcut, the primary repair is to use the actual boundary coordinate X_k:

$$E_{\text{identity}}^{\text{boundary}} = \sum_k \sqrt{\mathbb{E}[X_k^2]}.$$

That expectation may be measured on a declared calibration distribution or derived from a separately validated joint model of the preceding identity and residual branches. Only if one additionally establishes $X_k = \mathrm{ReLU}(Y_k)$ with $Y_k \sim \mathcal{N}(\beta_k, \gamma_k^2)$ does Eq. 79 give the conditional closed form

$$E_{\text{identity}}^{\text{conditional Gaussian}} = \sum_k \sqrt{(\gamma_k^2 + \beta_k^2)\Phi(\alpha_k) + \beta_k|\gamma_k|\phi(\alpha_k)}, \qquad \alpha_k = \frac{\beta_k}{|\gamma_k|}.$$

Within that conditional model, at $\gamma_k = 0$ use the deterministic boundary $\max(0, \beta_k)$.

The repair must not borrow β_k, γ_k from a branch BatchNorm without showing that they parameterize the post-add pre-activation entering the terminal ReLU. It does not repair the sequential-layer sum, derive projection shortcuts, validate a Gaussian surrogate in a checkpoint, or connect local cost to final accuracy.

8.10 What the evidence establishes

The frozen Part 8 package contains eight outputs and 20 fail-closed checks. It establishes:

- the exact Figure 3 boundary for consecutive interior post-activation V1 identity blocks;
- under the explicit synthetic Gaussian premise, the $\sqrt{2}$ zero-shift identity-energy ratio and the corresponding 29.289 percent cost difference;
- unbounded negative-shift behavior within that Gaussian family;
- one deterministic conditional equal-payoff ordering reversal;
- one exact ReLU-idempotence depth witness;
- and the ineligibility of projection and stride-changing shortcuts for the coordinate-identity derivation.

It does not establish that Eq. 96's β, γ describe the actual post-add pre-activation of a trained block. It also does not establish trained-network prevalence, ResNet-50 compression accuracy, downstream loss, latency, or the superiority of a corrected optimizer.

The broader lesson is simple: a macro score is only as sound as the tensor chosen at its boundary. Before pricing the deletion of an entire block, trace the activation that actually travels through the skip.

Chapter 9

Constant time per candidate is not a constant-time optimizer

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `candidate_counts.csv`
- Evidence: `refresh_comparison.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part09_constant_
time -p 'test_*.py' -v
```

Scope: Exact operation counts and deterministic cache fixtures are not wall-clock benchmarks.

Caching can make one price lookup constant-time. It cannot make millions of candidates disappear.

That distinction matters because HOPE contains two different complexity stories.

The careful story is in Section 10. It says that evaluating one cached candidate takes constant scalar work, while scanning all merge pairs in L layers of width N takes $O(LN^2)$ per greedy action.

The stronger story appears in Appendix B.4 and Algorithm 1. The appendix calls the scan over all remaining pairs an "$O(1)$ global scan," and Algorithm 1 labels its nested layer-and-candidate loop a "True $O(1)$ Greedy Scan."

Those statements cannot all describe the same unit of work.

This audit separates the units, counts a complete trajectory, and then asks a second question: after a physical channel is removed, which cached quantities are still valid?

9.1 Start with the strongest supported claim

HOPE's decoupled cache is a useful systems idea.

For one merge pair, the expensive geometry can be summarized by cached scalar constants such as a, b, and the warped correlation. Given those values and the live remaining layer capacity E_{rem}, Eq. 40 computes the parent magnitude

$$s^* = \frac{a + bE_{\text{rem}}}{2E_{\text{rem}} + b}.$$

The transition score can then be evaluated with a fixed number of scalar operations.

`paper_claim`: one already enumerated candidate can be repriced in $O(1)$ scalar arithmetic after its geometry has been initialized.

That is not the same as selecting the minimum candidate.

9.2 The paper's own main text gives the correct scan scope

At live width m, one layer has

$$m$$

solitary prune candidates and

$$\binom{m}{2} = \frac{m(m-1)}{2}$$

unordered merge candidates.

Algorithm 1 loops over every active layer and then every active action. Section 10 therefore states $O(LN)$ for pruning and $O(LN^2)$ for merging at one greedy step.

`verified_fact`: Section 10 and the pseudocode do not implement a constant-time global selection. They implement a quadratic merge scan whose individual candidate evaluations are constant-time.

The precise correction is important. It would be unfair to say that the paper never recognizes the quadratic scan. It does. The problem is that Appendix B.4 and the Algorithm 1 phase label contradict that recognition.

The clean vocabulary is:

- cached candidate lookup: $O(1)$ scalar arithmetic;
- one layer-wide merge scan: $\Theta(m^2)$ candidate evaluations;
- one network-wide scan: $\Theta(\sum_\ell m_\ell^2)$;
- one complete trajectory: the sum of those scans over every executed action.

9.3 Receding horizon repeats the scan

HOPE does not select one action and stop. It executes one action, updates the state, and scans again.

For a declared one-layer trajectory that removes one neuron at each step from width N down to width one, the merge-candidate evaluations are exactly

$$\sum_{m=2}^{N} \binom{m}{2} = \binom{N+1}{3} = \frac{N(N-1)(N+1)}{6}.$$

Therefore the cached merge-score scan alone is

$$\Theta(N^3).$$

If prune candidates are scanned alongside merge candidates, their trajectory count is

$$\sum_{m=2}^{N} m = \frac{N(N+1)}{2} - 1,$$

and the combined exact count is

$$\binom{N+2}{3} - 1.$$

`verified_fact`: constant arithmetic per candidate does not remove the cubic candidate-enumeration count of this complete single-layer trajectory.

The frozen width-4,096 fixture contains:

- 8,386,560 unordered merge pairs in the initial layer;
- 11,453,245,440 cached merge-score evaluations over the complete trajectory.

These are exact combinatorial counts, not timings and not estimates from a trained network.

9.4 Initialization is dimension-dependent

Appendix B.4 describes $O(N^2)$ initialization because there are quadratically many pairs. That statement suppresses the cost of constructing one pair's geometry.

Let d be incoming dimension and c the complete outgoing dimension. A pair calculation must at least visit coordinates from those vectors. The dimension-aware initialization scale is therefore

$$\Theta(N^2(d+c))\,,$$

before implementation-specific eigensolver, SVD, indexing, and allocation costs are counted.

`independent_inference`: $O(N^2)$ is an entry-count description, not a dimension-free end-to-end initialization bound.

The Part 9 operation-count fixture fixes $d = c = 64$. It compares:

1. recomputing every pair geometry at every live width;
2. initializing geometry once, scanning cached scalars, and refreshing only new-parent pairs within the same layer.

At width 4,096, the declared work-unit proxy favors the decoupled cache by about 108.64×. This is evidence that the cache can be valuable. It is not a wall-clock speedup claim because coordinate visits and scalar evaluations are only a deterministic workload model.

9.5 The cache has quadratic storage

A cache entry must exist for every live unordered pair if every pair is to be queried without rebuilding its geometry.

The entry count is

$$\binom{N}{2} = \Theta(N^2).$$

The frozen memory table counts only three float64 scalars per pair: a, b, and the warped correlation used for parent BatchNorm recovery. It deliberately excludes pair indices, object headers, hash tables, queues, alignment, and allocator overhead.

At width 4,096, those three raw scalar arrays alone occupy

$$8{,}386{,}560 \times 3 \times 8 = 201{,}277{,}440 \text{ bytes} = 191.953125 \text{ MiB}.$$

`verified_fact`: the paper's constant-time candidate lookup trades against a quadratic pair-cache entry count. The frozen byte figure is a lower-scope raw payload count, not a claim about a particular Python, C++, or accelerator implementation.

9.6 Static parameter payoff does not freeze functional state

Equation 23 divides distortion by a payoff computed on the initial network. Appendix B.3 argues that overlap between static and live payoffs acts like one uniform factor α, preserving candidate order.

But physical weights are shared across adjacent functional descriptions.

Consider a 1x1 Conv-BN-ReLU-Conv chain and remove one channel from its central layer. That one action does all of the following:

1. It removes one coordinate from every predecessor neuron's outgoing vector.
2. It removes one coordinate from every successor neuron's incoming vector.
3. It changes successor incoming norms and pairwise cosines.
4. It changes successor preactivation variances under the same declared input covariance.
5. It can invalidate block scores and any stored BatchNorm interpretation that assumed the old preactivation.

Static ΔP changes none of those facts. It freezes only a denominator used for action comparison.

9.7 A fixed dependency counterexample

The audit removes channel 1 from the following successor incoming matrix:

$$W_{\text{succ}} = \begin{bmatrix} -5 & 4 & -1 & -1 \\ 2 & -4 & 0 & -3 \\ -3 & -3 & 2 & -5 \end{bmatrix}.$$

Before removal, the three pair correlations are approximately

$$\rho_{01} = -0.6513, \qquad \rho_{02} = 0.1335, \qquad \rho_{12} = 0.5688.$$

The best pair under the monotone zero-bias ReLU kernel is $(1, 2)$.

After deleting the second coordinate, they become

$$\rho'_{01} = -0.3736, \qquad \rho'_{02} = 0.5620, \qquad \rho'_{12} = 0.4049.$$

Now the best pair is $(0, 2)$.

Algorithm 1's strictly intra-layer update leaves the successor cache at the old values and keeps pair $(1, 2)$ first. Full recomputation selects pair $(0, 2)$.

`counterexample`: "zero downstream recalculation" is false for this physical channel-surgery fixture. All three successor pair values change, and the stale cache selects a different next pair.

The same fixture changes all three predecessor capacities and all three successor preactivation variances. The result is not confined to one pair.

9.8 A dependency-aware refresh is a proposed repair

The repair is conceptually simple even if an efficient implementation requires careful graph bookkeeping.

After a physical action, refresh:

- current-layer candidates involving a synthesized or removed neuron;
- predecessor capacities and pairs whose outgoing vectors changed;
- successor geometry and kernels whose incoming coordinates changed;
- any block score spanning the affected tensors;
- any stored distributional premise invalidated by the changed preactivation.

`proposed_repair`: on the declared fixture, dependency-aware refresh reproduces the full-recomputation pair values and ranking exactly.

This audit does not claim that the repair has optimal asymptotic complexity on arbitrary graphs. Fan-out, shared weights, grouped convolutions, residual branches, and unresolved consumers change the dependency set.

9.9 Static payoff can reverse action order

The uniform-α claim also fails independently of the functional cache.

For a simple dense or 1x1 convolutional neuron with four BatchNorm values, write the parameter payoff as

$$\Delta P = d_{\text{in}} + d_{\text{out}} + 4.$$

The frozen equal-distortion fixture uses:

	$d_{\text{in}}^{\text{init}}$	$d_{\text{out}}^{\text{init}}$	$d_{\text{in}}^{\text{live}}$	$d_{\text{out}}^{\text{live}}$	ΔP^{init}	ΔP^{live}
A	8	8	1	1	20	6
B	6	6	5	5	16	14

With $J_A = J_B = 1$, static payoff ranks A first because $1/20 < 1/16$. Live payoff ranks B first because $1/14 < 1/6$.

The implied factors are

$$\alpha_A = 20/6 \approx 3.333, \qquad \alpha_B = 16/14 \approx 1.143.$$

They are not uniform.

`counterexample`: Eq. 39's ordering equivalence does not hold for general action histories. The audit also includes a true uniform-factor control and a distortion control; neither reverses.

Because HOPE executes only one action before replanning, a corrected optimizer can use the real immediate live payoff. It does not need to pretend that all overlap factors are equal.

9.10 What this audit establishes

The strongest fair conclusion is not that caching is useless. The opposite is true: the deterministic workload comparison shows why decoupling expensive geometry from live scalar capacity can save substantial repeated work.

The supported boundaries are:

- `paper_claim`: one cached candidate can be evaluated with constant scalar arithmetic.
- `verified_fact`: one global merge scan is quadratic in live width, exactly as Section 10 states.
- `verified_fact`: a complete single-layer merge trajectory performs cubic cached candidate evaluations.
- `verified_fact`: the pair cache has quadratic entries and dimension-dependent initialization.
- `counterexample`: physical channel surgery invalidates adjacent functional state and can reverse the next cached pair ranking.
- `counterexample`: static-to-live parameter payoff is not generally one uniform factor, so action ordering can reverse.
- `proposed_repair`: refresh the complete dependency neighborhood and use the immediate live payoff.

The audit does not report a trained-model runtime, GPU memory trace, accuracy curve, or end-to-end optimizer benchmark. Those remain empirical work.

The practical question to ask whenever a method says “constant time” is:

> Constant time per what?

Chapter 10

DEFT has multiple definitions of core

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `operator_drift.csv`
- Evidence: `bn_buffer_drift.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part10_deft_core -p
'test_*.py' -v
```

Scope: Deterministic NumPy fixtures expose specification boundaries; they do not measure transfer accuracy.

A frozen parameter is not necessarily a frozen neuron.

That distinction is the center of this audit.

HOPE uses its functional capacity to build Dispersed Elastic Fine-Tuning, or DEFT. The motivating idea is attractive: consolidate redundant source features, protect a high-capacity core, turn low-capacity structure into plastic slack, and block target-driven slack from feeding back into the core.

The paper then makes a strong promise. Once the structural mask has been applied, the core is said to experience zero dynamic deviation during target fine-tuning.

There are two separate questions hidden inside that promise:

1. Which neurons are core at the moment the promise is invoked?
2. What state must remain fixed for a core neuron, as defined by Eq. 2, to be the same operator?

The paper gives three different answers to the first question. Its proof gives a useful but narrower answer to the second.

The result is not that DEFT's structural mask does nothing. The mask supports a real routing guarantee. The result is that this guarantee protects a selected input route into downstream core neurons, not every component of every complete core operator.

10.1 Start with the strongest fair version

DEFT assigns each neuron a binary elasticity

$$E_i \in \{0, 1\},$$

where core neurons use $E_i = 0$ and slack neurons use $E_i = 1$. For a weight from upstream neuron k to downstream neuron j, Eq. 28 applies the structural mask

$$M_{j,k} = \begin{cases} 0, & E_{\text{in},k} > 0 \text{ and } E_{\text{out},j} = 0, \\ 1, & \text{otherwise.} \end{cases}$$

The target model starts from

$$W_0 = M \odot W_{\text{source}}.$$

Equation 29 then gates each weight gradient by the elasticity of its destination neuron:

$$g_t = E_{\text{out}} \odot \nabla_W L_{\text{target}}(W_t).$$

If an upstream slack activation changes but its route into a downstream core neuron is permanently zero, that changing signal contributes exactly zero to that downstream core input.

`verified_fact` (`P10-NARROW-ROUTING-GUARANTEE`): under the declared mask, destination gate, and routing assumptions, a severed slack-to-core weight stays zero in the deterministic audit. This is a genuine and useful decoupling property.

The scope words matter. The theorem would be stronger only if the mask and gradient rule also froze every state component belonging to the complete functional neuron in Eq. 2. They do not.

10.2 Before proving stability, define the partition

A partition must specify:

- the scalar being thresholded;
- the network state at which it is measured;
- whether the threshold is global or layer-specific;
- the treatment of equality;
- the treatment of merged, removed, inactive, and terminal-survivor slots;
- the time at which the resulting elasticity is attached.

DEFT does not give one answer to that list. It gives three.

10.2.1 Main-text trajectory-cost partition

Equations 24 through 27 assign a pruning cost when neuron i is removed during progressive encoding:

$$J_{\text{prune}}^{(i)} = \frac{N^{(i)}\|f_i\|_{\mathcal{H}}}{E_b^{(i)}}, \qquad E_b^{(i)} = E_a^{(i)} - \|f_i\|_{\mathcal{H}}.$$

The paper collects valid trajectory costs across the network, takes one global percentile, applies its numerical fallback, and defines

$$E_i = \begin{cases} 1, & J_{\text{prune}}^{(i)} < J_{\text{lock}}, \\ 0, & J_{\text{prune}}^{(i)} \geq J_{\text{lock}}. \end{cases}$$

`paper_claim` (`P10-TRAJECTORY-PARTITION`): the main-text rule thresholds a history-dependent destruction cost. Its scalar depends on the active width and remaining capacity at the step when that neuron is removed.

This is not a threshold on the neuron's current raw capacity. It is also not defined on an active terminal survivor that never receives a removal event, unless an additional convention is supplied.

10.2.2 Appendix H per-layer capacity partition

Definition H.1 instead starts from one threshold for each layer:

$$\mathcal{N}_{\text{core}}^{(\ell)} = \left\{ i : \|f_i\|_{\mathcal{H}^{(\ell)}} > \tau^{(\ell)} \right\},$$

$$\mathcal{N}_{\text{slack}}^{(\ell)} = \left\{ j : \|f_j\|_{\mathcal{H}^{(\ell)}} \leq \tau^{(\ell)} \right\}.$$

`paper_claim` (`P10-LAYER-THRESHOLD-PARTITION`): the Appendix H guarantees assume per-layer thresholds on current functional capacities. Equality belongs to slack.

The appendix does not prescribe how its layer-specific thresholds are obtained from the main-text global percentile. In the deterministic audit we set

$$\tau^{(A)} = 2, \qquad \tau^{(B)} = 4.$$

These values are an explicit audit instantiation allowed by Definition H.1. They are not thresholds reported by the paper and not an inferred conversion from the paper's percentile hyperparameter.

10.2.3 Algorithm 3 global current-capacity partition

Algorithm 3 uses a third sequence:

1. merge redundancies;
2. compute capacities of the current active neurons;
3. take one global capacity percentile;
4. freeze a neuron only if it is active and its current capacity is above that global threshold.

`paper_claim` (`P10-CURRENT-CAPACITY-PARTITION`): Algorithm 3 thresholds current functional capacities globally after merging. That differs in both scalar and scope from the other two rules.

The pseudocode does not specify a percentile interpolation convention. At the declared 50th percentile, the audit uses a float64 median and averages the two central values for an even count. That implementation choice is part of our fixture, not a paper result.

10.3 One fixture, three different rule outputs

The smallest useful witness has two layers and six active neurons:

layer A	$A0$	$A1$	$A2$
current capacity	1	2	8

layer B	$B0$	$B1$	$B2$
current capacity	3	4	5.

The declared encoding trace removes two neurons from each layer and retains one terminal survivor.

For layer A:

$$J(A0) = \frac{3 \cdot 1}{10} = 0.3, \qquad J(A1) = \frac{2 \cdot 2}{8} = 0.5.$$

For layer B:

$$J(B0) = \frac{3 \cdot 3}{9} = 1, \qquad J(B1) = \frac{2 \cdot 4}{5} = 1.6.$$

At percentile 50,

$$J_{\text{lock}} = \text{median}(0.3, 0.5, 1, 1.6) = 0.75.$$

The trajectory rule therefore calls A0 and A1 slack, and B0 and B1 core. A2 and B2 have no removal event in the declared trace, so the audit marks them unassigned rather than inventing an Eq. 24 value.

The instantiated Definition H.1 rule calls only A2 and B2 core.

Algorithm 3 takes the global current-capacity threshold

$$\tau_{\text{global}} = \text{median}(1, 2, 3, 4, 5, 8) = 3.5,$$

and calls A2, B1, and B2 core.

The pairwise disagreement counts are:

comparison	disagreements	neurons
trajectory vs. layer threshold	2	$B0, B1$
trajectory vs. global current capacity	1	$B0$
layer threshold vs. global current capacity	1	$B1$.

The first two comparisons use the four neurons assigned by the trajectory rule. The trajectory rule also leaves two terminal survivors outside its observed-event domain.

`verified_fact` (`DEFT-PARTITION-MISMATCH`): the three printed specifications threshold different quantities at different scopes. They disagree pairwise on the four-neuron common assigned domain, while the trajectory rule leaves two terminal survivors unassigned.

This is not a claim that one of the three rule outputs is empirically best. It is a reproducibility result: Appendix H's guarantees cannot automatically transfer to an inequivalent main-text or Algorithm 3 partition.

10.4 What the structural proof actually protects

Theorem H.2 expands the input to a downstream core neuron into contributions from upstream core and upstream slack subsets. Its dynamic argument uses two facts:

1. destination-core weights and destination-core trainable parameters are frozen;
2. upstream-slack to downstream-core weights were set to zero and remain zero.

That proves a narrow statement about the signal entering downstream core.

The deterministic routing fixture makes the boundary visible. Let rows be downstream destinations (core, slack) and columns be upstream sources (core, slack):

$$W_{\text{source}} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}, \qquad M = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

After initialization,

$$W_0 = M \odot W_{\text{source}} = \begin{bmatrix} 2 & 0 \\ 5 & 7 \end{bmatrix}.$$

Use raw gradient

$$G = \begin{bmatrix} 99 & 99 \\ -2 & -2 \end{bmatrix}$$

and learning rate $\eta = 0.5$. Destination elasticity zeroes the first row and permits the second row to update:

$$W_1 = \begin{bmatrix} 2 & 0 \\ 6 & 8 \end{bmatrix}.$$

Now let the upstream activation vector change from

$$h_0 = [1, 1] \quad \text{to} \quad h_1 = [1, 4].$$

Before masking, the source core input is 5. Immediately after masking it is 2, so the structural initialization shock has magnitude 3. During adaptation, the downstream core input remains exactly 2 even though the upstream slack activation changes.

The fixture therefore separates two quantities that should never be combined:

- static intervention shock: 3;
- post-mask dynamic slack-to-core interference: 0.

Theorem H.1 addresses the first with a capacity bound. Theorem H.2 addresses the second under its assumptions. Neither number says that all state owned by the upstream core operator is immutable.

10.5 A core source can keep a plastic outgoing edge

Equation 2 defines a functional neuron using its complete outgoing vector:

$$f_i(x) = g_i(x)\, w_{\text{out},i}.$$

In the routing fixture, the upstream core neuron's outgoing vector is the first column of the weight matrix. It changes from

$$w_{\mathrm{out},c}^{(0)} = [2, 5] \quad \text{to} \quad w_{\mathrm{out},c}^{(1)} = [2, 6].$$

The core-to-core component stays frozen because its destination is core. The core-to-slack component changes because Eq. 29 assigns elasticity by the slack destination.

For a core activation with second moment $\mathbb{E}[g_c^2] = 1$, the operator capacities are

$$\|f_c^{(0)}\| = \sqrt{29} = 5.385164807134504,$$

$$\|f_c^{(1)}\| = \sqrt{40} = 6.324555320336759.$$

The exact operator-difference norm is 1, and the relative capacity change is about 0.174440439.

`counterexample` (`P10-OUTGOING-OPERATOR-DRIFT`): destination-owned gradient gating permits a core source neuron's outgoing component into slack to change. Because Eq. 2 includes that component, the complete core source operator is not immutable.

This does not contradict the narrow routing result. Both are true at once: slack cannot enter downstream core through the severed route, while core can still feed and adapt through downstream slack.

10.6 Frozen trainables do not freeze BatchNorm state

BatchNorm running mean and variance are buffers. They change during training mode through forward-pass state updates, not through the gradient multiplication in Eq. 29.

The audit freezes every trainable scalar in a one-channel state machine:

$$(w, b, \gamma, \beta) = (1, 0, 1, 0).$$

It starts with source buffers

$$(\mu_0, v_0) = (0, 1)$$

and uses old-state momentum $r = 0.5$:

$$\mu_{t+1} = r\mu_t + (1 - r)m_t, \qquad v_{t+1} = rv_t + (1 - r)q_t.$$

Two target batches provide declared moments

$$(m_1, q_1) = (4, 4), \qquad (m_2, q_2) = (6, 9).$$

The live buffers become

$$(\mu_2, v_2) = (4, 5.75).$$

On fixed source probes $z = [-1, 0, 1]$ with $\epsilon = 10^{-5}$, the source-buffer output is approximately

$$[-0.999995, 0, 0.999995],$$

while the target-updated live-buffer output is approximately

$$[-2.085142, -1.668114, -1.251085].$$

Their root-mean-square difference is 1.7346960857596274. The trainable change norm remains exactly zero.

`counterexample` (`P10-BN-BUFFER-DRIFT`): gradient masking alone does not make the BatchNorm inference map immutable.

Appendix G.5 confirms the operational importance of this state: its source retention protocol explicitly restores source BatchNorm moving statistics before source evaluation.

`proposed_repair` (`P10-BN-STATE-REPAIR`): on the declared source probes, freezing source buffers during target updates gives zero BatchNorm-output RMSE. Explicit frozen or dual-tracked buffer ownership is a proposed policy; that local control does not establish target-domain or trained-network performance.

10.7 The complete Eq. 2 immutability claim fails twice

The two counterexamples are independent.

1. A core-to-slack outgoing weight can change because the destination is elastic.
2. A core BatchNorm buffer can change through target forward passes even when every trainable parameter is fixed.

`verified_fact` (`DEFT-CORE-DRIFT`): Theorem H.2's proof protects the selected slack-to-core route, but does not freeze the complete Eq. 2 operator through either of these state channels.

The precise correction is therefore:

> The structural mask can make downstream core inputs dynamically independent of upstream slack along the declared parameterized routes. It does not, by itself, make every complete core functional neuron immutable.

That narrower statement preserves the useful part of the theorem without assigning it a larger state domain than the proof controls.

10.8 A purged child is not yet a reusable vessel

DEFT also needs a state transition from compression to target adaptation.

The released-capacity narrative says that merging redundant core neurons consolidates their feature into one parent and releases the other children as plastic target capacity. Figure 4 depicts empty slots. Axiom 3 assigns released children full plasticity. Proposition H.4 counts the freed parameter volume.

Algorithm 2, however, says that child j is purged by removing it from the layer.

`verified_fact` (`P10-PURGED-RELEASED-CONFLICT`): purging a child from the active layer and exposing a reusable fixed-width plastic vessel are different states. The paper does not specify the transition that recreates or retains the slot, reconnects it, initializes it, masks it, and registers it with the optimizer.

This is an under-specification, not a proof that released capacity cannot be implemented. The missing state transition can be repaired explicitly.

10.9 Keeping a completely zero slot is not enough

One tempting interpretation is to preserve tensor width but zero every parameter belonging to the released child. That interpretation is not printed by the paper, so the audit labels it as an independent test rather than a paper-literal implementation.

Consider the scalar path

$$s = \gamma u x + \beta, \qquad a = \mathrm{ReLU}(s), \qquad \hat{y} = v a, \qquad L = \frac{1}{2}(\hat{y} - y)^2.$$

Use $x = y = 1$ and the common convention $\mathrm{ReLU}'(0) = 0$. If

$$u = \gamma = \beta = v = 0,$$

then $a = 0$, the loss is 0.5, and every gradient is exactly zero. Direct SGD leaves the slot at zero forever.

`counterexample` (`P10-DEAD-ZERO-SLOT`): a completely zeroed ReLU vessel is not automatically plastic. Elasticity one cannot create a gradient where both sides of the path are dead.

The result is scoped to the declared path and derivative convention. Bias, normalization, skip routes, or explicit reinitialization can change the result. That is precisely why the released-slot transition must be specified.

10.10 A fixed-width repair can wake without an initial contribution

A minimal proposed repair keeps the logical slot but initializes its two sides asymmetrically:

$$u = 1, \qquad \gamma = 1, \qquad \beta = 0, \qquad v = 0.$$

At initialization, the slot contributes zero because its outgoing coefficient is zero. Its internal activation is live, so

$$\frac{\partial L}{\partial v} = -1.$$

After one SGD step with learning rate 0.25, $v = 0.25$. A second backward pass gives

$$\frac{\partial L}{\partial v} = -0.75,$$

and

$$\frac{\partial L}{\partial u} = \frac{\partial L}{\partial \gamma} = \frac{\partial L}{\partial \beta} = -0.1875.$$

`proposed_repair` (`P10-FIXED-WIDTH-REPAIR`): on the declared scalar fixture, a live incoming feature plus a zero outgoing coefficient preserves zero initial contribution while allowing the slot to wake in two gradient steps.

A complete implementation still has to:

- keep every released-slot route into downstream core hard-masked;
- allow only slack or target-head destinations to become active;
- clear stale optimizer state for the reused slot;
- define BatchNorm affine and buffer initialization;
- record active/inactive status separately from elasticity;
- keep tensor shape and parameter ownership consistent across the transition.

Those are proposed implementation rules. They are not results reported by HOPE v1.

10.11 Source retention uses restored source state

Appendix G.5 does not evaluate source retention by sending source examples through the untouched live target model.

It:

1. grafts the original source classification head back onto the adapted backbone;
2. masks connections from slack features into the source head;

3. restores source BatchNorm moving means and variances.

Algorithm 3 similarly constructs a source-evaluation state through `MaskSlackDrift` before computing source validation accuracy.

`paper_claim` (`P10-SOURCE-STATE-RESTORATION`): the reported source protocol is task-conditional state restoration, not a one-state evaluation of the live target-adapted model.

The deterministic audit holds the source head and slack mask fixed and varies only the BatchNorm policy:

source-evaluation buffer policy	source-probe RMSE
live target-updated buffers	1.7346960857596274
restore source buffers at evaluation	0
freeze source buffers during adaptation	0.

The zeroes are exact fixture equalities, not accuracy results. The comparison shows why the policy changes what function is evaluated.

Restoration is a legitimate experimental protocol if it is reported as such. But it carries task-specific state: a source head, a source mask, and source BN buffers. It therefore cannot simultaneously serve as evidence that source retention requires no task-specific evaluation state.

Freezing core buffers or maintaining explicitly named source and target buffer tracks are possible repairs. Their target-domain consequences require real experiments and are not inferred from this fixture.

10.12 What survives, what fails, and what needs repair

The audit supports a more precise DEFT state machine.

What survives:

- `paper_claim`: DEFT proposes capacity-informed stability and plasticity.
- `verified_fact`: Eq. 28 can permanently sever a declared slack-to-core route.
- `verified_fact`: with destination-core trainables frozen, changing upstream slack does not dynamically enter the downstream core through that route.
- `paper_claim`: source evaluation restores source-specific state.

What does not survive as written:

- `verified_fact`: the three printed partition procedures are inequivalent.
- `counterexample`: frozen trainables do not freeze BatchNorm buffers.
- `counterexample`: destination gating permits core-to-slack outgoing drift.
- `verified_fact`: a child purged from the active layer is not yet an executable reusable slot.
- `counterexample`: a completely zeroed ReLU slot can be gradient-dead.

What remains proposed:

- choose one partition rule and bind the guarantee to it;
- keep fixed-width logical slots through transfer;
- initialize released slots with a live incoming side and zero permitted outgoing side;
- keep slack-to-core routes structurally zero;
- clear masked optimizer state;
- freeze or dual-track BatchNorm buffers;
- report live one-state and restored-state source evaluation separately.

The audit uses no trained network. It does not say how frequently the two drift channels dominate, whether the repair improves H-score, or which partition gives the best transfer tradeoff. Those are Part 11 and Part 12 questions.

10.13 The central conclusion

DEFT's most defensible theorem is a routing theorem.

Under its stated structural assumptions, a severed upstream-slack to downstream-core edge stays zero, so dynamic slack variation cannot cross that edge. That is useful.

But HOPE's Eq. 2 neuron owns more than that protected route. It includes a complete outgoing vector, and its inference map includes BatchNorm state. Destination elasticity can change core-to-slack outgoing components. Target forward passes can change running buffers. The reported source evaluation then restores task-specific state when measuring source retention.

The correct boundary is therefore:

> Narrow routing is protected. Complete Eq. 2 operator immutability is not.

Once that distinction is explicit, DEFT becomes implementable in a way that can be tested: one partition, one state-transition table, explicit released slots, explicit buffer ownership, and separate live-state and restored-state evaluation.

Chapter 11

What the experiments establish

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `table2_recalculation.csv`
- Evidence: `compression_digitization.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part11_
experimental_evidence -p 'test_*.py' -v
```

Scope: The package transcribes and audits the published record; it does not reproduce either paper experiment.

A correct H-score can still answer an underspecified question.

HOPE v1 presents its experiments as proofs of concept. That is the right place to begin: one compression plot asks whether the functional score can identify useful structure in a pretrained network, and one cross-domain benchmark asks whether DEFT can trade target plasticity against source accuracy.

The published results are interesting. In Table 2, DEFT has the highest displayed mean H-score. The paper describes the HOPE compression curve as superior to the three magnitude baselines over the plotted trajectory.

Neither observation is an exact reproduction, a statistical significance result, or evidence for one universal framework. The table omits the scenario-level values and the meaning of its plus/minus terms. The compression plot omits author-reported numbers, repetitions, uncertainty, and parameter or compute budgets. The source side of the DEFT score is measured with restored task-specific state, and the paper prints two different checkpoint-selection rules.

Part 11 makes those boundaries executable. It does not train ResNet-50, VGG-8, DEFT, or any baseline. It authenticates the exact v1 paper, reproduces the arithmetic that can be recovered from the printed summaries, constructs small counterexamples for the missing aggregation and checkpoint rules, and records what a decisive experiment would still need.

11.1 Start with the reported record

Section 11.2 evaluates 20 CIFAR-100 to SVHN transfer scenarios: four trials, each containing five source tasks. Every source task contains the 20 fine classes belonging to four sampled CIFAR-100 superclasses. Every target task uses all ten SVHN digits.

Table 2 reports:

Method	Target accuracy	Source accuracy labeled retention	H-score
DEFT	89.79 +/- 0.84	52.14 +/- 5.29	65.82 +/- 3.96
Head-Only	36.11 +/- 2.79	63.13 +/- 4.62	45.79 +/- 2.05
Full FT	94.09 +/- 0.28	7.52 +/- 1.63	13.88 +/- 2.84
EWC	93.94 +/- 0.22	6.74 +/- 1.74	12.54 +/- 2.99
PEFT	81.91 +/- 0.49	5.44 +/- 0.98	10.18 +/- 1.63

paper_claim (P11-TABLE2-RECORD): under the reported protocol, DEFT has the highest displayed mean H-score, Full FT has the highest displayed target accuracy, and Head-Only has the highest displayed source accuracy.

That ordering is a faithful reading of the table. The word displayed is doing real work. The repository has no task-level predictions, checkpoints, seeds, or metric records from which to reconstruct the experiment.

11.2 The displayed means do not reconstruct the displayed H-scores

Appendix G.5 defines validation H-score as

$$H(T, S) = \frac{2TS}{T + S},$$

where T and S are target and source accuracies.

Applying that equation directly to the two displayed accuracy means gives:

Method	H from displayed means	Displayed H	Difference
DEFT	65.9713	65.82	0.1513
Head-Only	45.9416	45.79	0.1516
Full FT	13.9269	13.88	0.0469
EWC	12.5776	12.54	0.0376
PEFT	10.2024	10.18	0.0224

The deterministic audit also treats each displayed accuracy as a conventional two-decimal rounding bin. Because the harmonic mean increases in both positive arguments, the lowest and highest possible H values come from the two lower and two upper bin endpoints. For all five methods, that entire derived interval is disjoint from the displayed H-score's own rounding interval.

verified_fact (P11-HSCORE-AGGREGATION-GAP): Eq. 101 applied to the displayed aggregate accuracy means cannot produce any of the five displayed H-scores, even after allowing for ordinary two-decimal rounding.

The discrepancy requires an unreported computation or aggregation convention to reconcile. One plausible workflow is to compute H separately for every scenario and then average those H values. The paper does not say whether it averages by task, by trial, by nested trial and task, or in some other order, and it does not publish the records needed to distinguish that possibility from a transcription, reporting, or table error.

open_question (P11-HSCORE-AGGREGATION): what experimental unit and aggregation order produced each H-score mean and plus/minus term?

11.3 Aggregate means do not identify mean per-scenario H

The distinction is structural, not cosmetic. Consider two records with the same target and source means:

$$\bar{T} = 50, \qquad \bar{S} = 50.$$

In a balanced fixture, both scenarios are $(T, S) = (50, 50)$. Their mean per-scenario H-score is 50.

In a crossed fixture, the scenarios are $(90, 10)$ and $(10, 90)$. The target and source column means are still 50, but each H-score is 18, so their mean is 18.

The same two aggregate accuracy means therefore support a 32-point range between these two valid examples. No operation on the printed column means can recover which taskwise tradeoffs occurred.

This matters because H is intended to penalize imbalance. Aggregating away the pairing before publishing the score removes the very information that gives H its interpretation.

11.4 An undefined plus/minus does not establish significance

Table 2 says that all metrics are averaged across tasks and trials, but it does not define the reported plus/minus statistic. It could denote, among other possibilities:

- a standard deviation across 20 scenario values;
- a standard deviation across four trial means;
- a standard error across either experimental unit;
- a confidence-interval half-width;
- a quantity produced by a nested aggregation.

Those interpretations are not interchangeable. The same printed value implies different sampling units, standard errors, and intervals under each one. Paired comparisons would additionally require the scenario pairing or the covariance of method differences.

The exact v1 paper reports no definition of the plus/minus term and no hypothesis test. Its prose nevertheless says DEFT significantly outperforms the baselines.

verified_fact (P11-UNCERTAINTY-UNDEFINED): the printed record does not define the uncertainty statistic or report an inferential test.

The bounded conclusion is not that the methods are equal. It is that a formal statistical-significance statement cannot be reconstructed from Table 2.

11.5 Source accuracy is not normalized retention

The Table 2 source column is labeled Source Retention (CIFAR). Appendix G.5 describes it operationally as CIFAR-100 test accuracy after target adaptation. The paper does not report the corresponding pre-transfer source accuracy for each scenario.

If normalized retention is defined as

$$R = \frac{A_{\text{after}}}{A_{\text{before}}},$$

then the denominator is necessary. Holding the displayed DEFT source value 52.14 fixed, a hypothetical pre-transfer accuracy of 60 gives 86.9 percent retention, while a hypothetical pre-transfer accuracy of 90 gives about 57.9 percent. These are deliberately independent fixtures, not estimates of the paper's missing baselines.

verified_fact (P11-ABSOLUTE-SOURCE-ACCURACY): Table 2 identifies absolute post-transfer source accuracy. It does not identify normalized retention from each source model's own pre-transfer performance.

This does not make the absolute source score useless. It makes the word retention ambiguous unless the baseline is published beside it.

11.6 The H-score combines two evaluation states

Appendix G.5 does not evaluate source accuracy on only the live target-adapted model. For source evaluation it:

1. grafts the original source classification head back onto the backbone;
2. masks source-head connections originating from target-adapted slack;
3. restores the original source BatchNorm moving means and variances.

paper_claim (P10-SOURCE-STATE-RESTORATION): this task-specific restoration is part of the reported source protocol.

independent_inference (P11-CROSS-STATE-HSCORE): the target term and source term therefore describe performance under different task-conditional states. The reported H-score is a tradeoff between those two evaluation procedures, not a single live model evaluated unchanged on both domains.

The distinction should be visible in the result name. A useful future table would report at least:

- target accuracy in the live target state;
- source accuracy in that same live state;
- source accuracy after head and mask restoration;
- source accuracy after full head, mask, and BN-state restoration.

That decomposition would show which part of apparent retention belongs to the learned backbone and which part belongs to task identity and stored state.

11.7 The paper prints two checkpoint rules

Algorithm 3 evaluates target validation accuracy and restored source validation accuracy every epoch, forms their H-score, and returns the epoch with highest validation H-score.

Appendix G.5 instead says the final run identifies the epoch with peak target validation accuracy. Table 2's caption repeats the target-validation rule.

These selectors are not equivalent. In the deterministic two-epoch fixture:

Epoch	Target validation	Source validation	H-score
1	80	80	80
2	90	10	18

Target-only selection chooses epoch 2. H-score selection chooses epoch 1.

verified_fact (P11-CHECKPOINT-RULE-CONFLICT): Algorithm 3 and Appendix G.5/Table 2 specify different checkpoint objectives, and a fixed example makes them select different epochs.

The fixture does not tell us which epoch would win in the paper's training runs. It proves that the implementation choice can change the reported model. The exact checkpoint rule used to produce Table 2 remains unresolved.

11.8 Data-free capacity is not a data-free benchmark protocol

Table 1 marks DEFT as source-data independent, but does not define the scope of that label. The narrow capacity calculation reads checkpoint and BatchNorm state instead of source examples when estimating neuron capacity.

The reported model-selection protocol has a different scope. Appendix G.5 uses both source and target validation accuracy to choose hyperparameters. Algorithm 3 requires source validation examples and evaluates them at every epoch. Separately, source evaluation owns task-specific head, mask, and BN state; that stored state is not itself source data.

independent_inference (P11-SOURCE-DATA-SCOPE): data-free capacity evaluation does not make the reported tuning and checkpoint-selection workflow independent of source examples.

This is not an accusation that source validation is illegitimate. It is a requirement to name what is independent of source examples and what is not.

11.9 Neuron density is not the parameter objective

Section 11.1 compares compression methods at model density, defined as the ratio of active to initial neurons across the network. Equation 21 formulates the compression objective using released parameters and a final parameter budget.

Those axes are generally different. In a weight-only sequential fixture with widths

$$100 \longrightarrow 2 \longrightarrow 4 \longrightarrow 10,$$

the three matrices contain 248 weights. Removing one neuron from the width-2 hidden layer releases 100 incoming plus 4 outgoing weights, or 104. Removing one neuron from the width-4 hidden layer releases 2 incoming plus 10 outgoing weights, or 12. Both actions reduce the hidden-neuron density from 6/6 to 5/6.

independent_inference (P11-DENSITY-BUDGET-MISMATCH): equal active-neuron density does not fix equal parameter release and therefore does not directly test Eq. 21 at a common parameter budget. It fixes neither FLOPs nor latency.

The fixture does not claim that the plotted method ordering would reverse at equal parameters. The author data needed to test that question are absent.

11.10 What the compression raster can and cannot say

PDF page 20 contains one unnumbered 2929 by 1724 raster. It plots test accuracy against neuron density for HOPE, incoming L1, joint incoming-outgoing L1, and BN scale on a public Keras ImageNet-pretrained ResNet-50 checkpoint.

paper_claim (P11-COMPRESSION-PLOT): the paper describes the HOPE curve as superior to all three baseline curves over the plotted trajectory.

The Part 11 data file is a digitization, not an author numeric table. Every row is labeled digitized_approximation and author_reported_numeric=false. Its provenance binds:

- the exact v1 PDF SHA-256;
- the embedded RGB raster and soft-mask pixel hashes;
- the axis calibration pixels;
- the raw pixel range used for every extracted value;
- an accuracy interval rather than an invented midpoint when a trace is locally vertical.

The joint-L1 curve has a conspicuous vertical discontinuity near density 0.50. The digitization preserves the vertical range. Other visible vertical spans are treated the same way.

At the sampled compressed densities from 0.30 through 0.90, the HOPE interval is separated above every baseline interval. At density 1.00, the digitized HOPE and BN-scale intervals overlap. The digitization therefore does not independently certify strict endpoint superiority across the full trajectory.

verified_fact (P11-COMPRESSION-EVIDENCE-BOUNDARY): the plot provides no exact author numeric table, repeated trials, uncertainty, parameter/FLOP/latency budget, or complete reproduction protocol. The vertical label Test Accuracy does not specify top-1 versus top-5.

The inherited adapted VGG-8 compression run cannot fill this gap. Its recorded baseline accuracy does not match the surviving checkpoint, so its metrics remain excluded under LEGACY-COMPRESSION-PROVENANCE.

11.11 Baseline conclusions need the searched tradeoff

Appendix G.5 tunes one representative Task 0 and locks the winning hyperparameters for all 20 scenarios. The EWC grid ends at $\lambda = 50$. The reported EWC point is close to Full FT: high target accuracy and very low source accuracy.

independent_inference (EWC-TRUNCATED-GRID): that capped grid does not establish EWC's best attainable stability-plasticity tradeoff. This is not a claim that EWC would win, only that the relevant part of its tradeoff curve was not shown.

The same fairness principle applies across methods. A convincing comparison should disclose winning settings, search budgets, selection units, and full Pareto curves where a method has an explicit stability knob.

11.12 What this evidence establishes

The strongest defensible conclusion is narrower and more useful than either uncritical acceptance or blanket dismissal.

paper_claim:

- HOPE v1 describes one compression trajectory as superior to three magnitude baselines; the independent digitization does not certify strict ordering at the overlapping density-1.00 endpoint.
- HOPE v1 reports one CIFAR-100 to SVHN protocol where DEFT has the highest displayed mean H-score.

verified_fact:

- the exact Table 2 values and their arithmetic are faithfully reproduced;
- aggregate displayed accuracies do not determine the displayed H column;
- the plus/minus statistic and inferential unit are undefined;
- source retention is reported as absolute restored-state accuracy;
- the checkpoint rules conflict;
- the compression record is a raster without author numeric data or uncertainty.

independent_inference:

- the reported H-score combines task-conditional evaluation states;
- neuron density is not a controlled parameter or compute budget;
- the reported protocol supports proof-of-concept evidence, not universal compression superiority, general continual learning, or demonstrated parameter efficiency.

open_question (P11-PROTOCOL-IDENTITY): exact task identities, seeds, split seeds, winning hyperparameters, aggregation unit, uncertainty definition, partition rule, checkpoint rule, and source-state policy per baseline remain necessary to reproduce the reported benchmark.

11.13 The next experiment

The missing-experiment matrix in the frozen artifact is a proposed design, not a set of observed outcomes. At minimum, a stronger compression study should report exact numbers and uncertainty at equal neuron, parameter, FLOP, and latency budgets; disclose action types and BN handling; and include prune, merge, eviction, and combined ablations.

A stronger DEFT study should report:

- pre-transfer source accuracy and normalized retention;
- source accuracy in live and restored evaluation states;
- per-scenario target, source, and H-score records;
- a defined uncertainty statistic and paired inferential unit;
- one declared checkpoint objective;
- trainable and frozen parameter counts, merge counts, and released slots;
- multiple source-target directions and sequential tasks;

- broader regularization, projection, masking, replay, adapter, and low-rank adaptation baselines;
- sensitivity to percentile, merge budget, and source-state policy.

independent_inference (P11-EVIDENCE-SCOPE): proof-of-concept evidence can justify these next experiments. It cannot substitute for them.

11.14 Reproduction boundary

The frozen Part 11 package is an independent deterministic evidence audit. It uses no dataset, checkpoint, training run, random sample, Torch model, or wall-clock benchmark. It reproduces published table transcription and metric arithmetic, not the paper's trained experimental results. Compression values are explicitly digitized approximations from the authenticated raster. Every missing experiment is recorded as unknown rather than assigned an outcome.

Chapter 12

Building a corrected HOPE

Code and reproducibility

This page is paired with the immutable `v1.0.0` release.

- Exact audit package
- Focused tests
- Artifact checksums
- Evidence: `design_contract.csv`
- Evidence: `integration_ablations.csv`

From the repository root:

```
python -m unittest discover -s experiments/audits/part12_corrected_
hope -p 'test_*.py' -v
```

Scope: HOPE-R is an independently proposed reference design tested on deterministic fixtures, not author code or a benchmark result.

The most useful criticism ends with a runnable alternative.

Parts 1 through 11 found a valuable local idea inside HOPE v1: a neuron can be treated as a complete function-valued operator, not merely as an incoming filter. They also found that the published pipeline asks several incompatible objects to carry that idea. Pair-local surrogates are assembled as though they came from one distribution. A symbolic merge target is not always the physical one-parent map that Algorithm 2 executes. Local cache refresh omits graph dependencies. DEFT freezes destinations while the complete operator is owned by its source. The experiments do not resolve those specification gaps.

Part 12 builds forward from that record.

HOPE-R is a framework-independent proposed reference design with four joined contracts:

1. Every neuron in one layer is evaluated under one declared weighted calibration measure.
2. A physical merge fits one realizable parent to the executed child sum and reports the residual it actually leaves.

3. Every physical channel mutation has an explicit graph dependency closure, live resource account, and fail-closed block boundary.
4. Transfer uses fixed-width slots, complete source-owned masks, explicit optimizer state, and task-declared BatchNorm and evaluation state.

These are proposed repairs, not an author implementation and not results reported by HOPE v1. The supporting audit uses deterministic float64 fixtures. It trains no model, reads no benchmark dataset, and makes no claim about accuracy, convergence, latency, or how often a local failure occurs in a trained network.

12.1 Correction begins by separating the branches

A corrected method cannot quietly replace a paper equation and then continue calling the result a reproduction. HOPE-R therefore preserves three distinct labels:

Branch	Meaning	Permitted conclusion
Paper-literal	The equation or algorithm as printed in HOPE v1	What the paper specifies on the declared input
Deterministic fixture	A finite NumPy construction used to test one boundary	What happens on that fixture within enforced tolerances
Proposed HOPE-R repair	A new rule chosen after the audit	What the implemented rule guarantees in its declared domain

The separation is visible in code. Paper-literal modules remain under `hope_repro.paper_literal`. The new modules live under `hope_repro.proposed_repairs.hope_r`. The Part 12 audit imports the latter as proposed repairs and retains paper-literal calculations only as named comparisons or negative controls.

This distinction prevents four common overclaims:

- calibrated exactness is not population exactness;
- a positive-semidefinite fixture Gram is not proof that the paper's pair-assembled Gram is always positive semidefinite;
- an exact-twin merge is not universal one-neuron preservation;
- a zero-drift optimizer fixture is not evidence of benchmark retention.

The purpose of HOPE-R is narrower: make the desired invariants explicit enough that an implementation can either satisfy them or fail before changing a model.

12.2 The exact paper anchors stay attached

The corrected pipeline answers specific parts of the authenticated HOPE v1 paper. It does not replace their provenance. The Part 12 source map uses these exact anchors from

`paper/equation-map.json`:

HOPE-R boundary	Exact paper anchor used for comparison
Functional atom and kernels	Eqs. 1-5, PDF pages 5-9
Printed physical merge	Eq. 6 and Eqs. 7-18, PDF pages 13-16
Residual eviction	Eqs. 19-20, PDF page 17
Planning and payoff	Eqs. 21-23, PDF pages 18-19
DEFT partition and routing	Eqs. 24-29, PDF pages 21-23
Common expectation geometry	Eq. 30 and Proposition A.1, PDF page 35
Cached payoff and pair geometry	Appendix B.2-B.4 and Eqs. 36-40, PDF pages 38-40
Self and cross-kernel comparison branches	Eqs. 73-85, PDF pages 52-56
Executed aggregate map	Eq. 86, PDF page 57
Identity-shortcut assumptions	Eqs. 93-97, PDF page 58
Core ownership and released vessels	Definition H.1 and Eqs. 102-118, PDF pages 62-67
Physical surgery, compression, and transfer	Algorithms 1-3, PDF pages 68-70

The anchors matter because HOPE-R changes different things for different reasons. It retains Eq. 30's common-expectation logic, but changes how the measure is obtained. It retains Eq. 86's sum-map target, but changes how a realizable physical parent is selected. It retains the useful routing idea of Eqs. 28-29, but changes parameter ownership and slot lifecycle.

`source_map.csv` is the canonical machine-readable route from each Part 12 claim to these paper locations and to the independent implementation branch.

12.3 One layer gets one declared calibration measure

The first repair resolves the geometry mismatch from Parts 3 through 5.

Let a layer receive declared calibration rows

$$x_1, \dots, x_m$$

with nonnegative weights

$$q_1, \dots, q_m, \qquad \sum_{n=1}^{m} q_n = 1.$$

For neuron i, fold its inference-time affine state and evaluate

$$a_i(x_n) = \mathrm{ReLU}(w_i^\top x_n + b_i).$$

The calibrated cross-moment is then

$$K_{ij}^{\mathrm{cal}} = \sum_{n=1}^{m} q_n a_i(x_n) a_j(x_n).$$

Every entry uses the same rows and the same normalized weights. No pair may silently choose a new covariance, a new marginal law, or a new sample subset. The label attached to the calibration measure is carried into downstream merge results.

proposed_repair (P12-CALIBRATED-COMMON-MEASURE): HOPE-R evaluates all neurons in one layer under one declared weighted calibration measure, so the exact activation moments share one distribution.

The word calibrated is an essential limitation. This mode consumes input rows. It is not strictly data-free, even when the rows are unlabeled. A user must declare their source, weighting, preprocessing, and sampling policy. A different calibration measure defines a different geometry.

This is preferable to preserving a data-free label by hiding incompatible pairwise laws. It also creates an experimental obligation: a future trained study must test calibration-source and calibration-size sensitivity rather than treating the selected rows as an invisible constant.

12.4 The common Gram is positive semidefinite by construction

Collect the calibrated activations in a matrix

$$A_{n,i} = a_i(x_n)$$

and let $D = \mathrm{diag}(q_1, \ldots, q_m)$. The whole-layer Gram is

$$G = A^\top D A = (D^{1/2}A)^\top (D^{1/2}A).$$

For any vector c,

$$c^\top G c = \|D^{1/2} A c\|_2^2 \geq 0.$$

This is the shared-distribution guarantee anticipated by Eq. 30 and Proposition A.1 on PDF page 35. It does not need separately warped pairwise correlations to happen to assemble into one feasible matrix.

The implementation still validates the output. It checks finite values, symmetry, and the smallest eigenvalue against a declared float64 roundoff tolerance. Material asymmetry or

negative curvature raises an error. Small accepted numerical negativity remains visible in the returned certificate rather than being silently projected away.

verified_fact (P12-COMMON-GRAM-PSD): on the declared Part 12 weighted calibration fixtures, the exact activation table produces a positive-semidefinite Gram by construction and the fail-closed spectral check finds no material negative eigenvalue.

The result is local in two ways. It concerns the declared activation rows, and it concerns the Part 12 fixtures. It is not a claim that arbitrary empirical calibration represents the target population, nor that the paper-literal pair construction has acquired this property retroactively.

The diagonal of this common Gram supplies each calibrated self-moment. HOPE-R then computes the functional capacity of neuron i as

$$\sqrt{G_{ii}}\,\|v_i\|_2,$$

where v_i is the complete outgoing vector. Positive rescaling of the incoming affine coordinates with reciprocal outgoing rescaling leaves the represented contribution and the tested capacity unchanged.

12.5 BatchNorm and kernel limits are explicit

Equation 1 on PDF page 5 motivates folding BatchNorm into the functional affine map. HOPE-R performs that fold with the supplied finite epsilon intact. For raw affine state w_0, b_0, BatchNorm scale γ, shift β, running mean μ, running variance σ^2, and epsilon ϵ, the folded values are

$$s = \frac{\gamma}{\sqrt{\sigma^2 + \epsilon}}, \qquad w = sw_0, \qquad b = \beta + s(b_0 - \mu).$$

The denominator must be strictly positive. Negative variance, negative epsilon, non-finite input, and non-finite output fail closed.

The zero-scale boundary is handled before unsafe multiplication. If $\gamma = 0$, the folded affine map is the deterministic constant β, even when an irrelevant raw subtraction would overflow. The corresponding zero-variance ReLU self-moment is exactly

$$\mathrm{ReLU}(\beta)^2.$$

It is not replaced by a tiny positive variance and it is not sent through a formula whose derivation assumes positive scale.

HOPE-R also keeps two cross-kernel APIs separate:

- `calibrated_cross_kernel` is exact with respect to the declared weighted empirical measure;

- `zero_bias_cross_kernel_approximation` is explicitly the zero-bias angular comparison associated with Eqs. 84-85 on PDF page 56.

The exact biased pair moment in Eqs. 80-83 on PDF pages 54-55 remains a paper-literal comparison branch. It is not used to smuggle pair-specific surrogates into the common calibrated Gram.

proposed_repair (P12-KERNEL-BOUNDARY-CONTRACT): HOPE-R folds finite-epsilon BatchNorm explicitly, implements the deterministic zero-scale ReLU limit, and keeps calibrated exact cross-moments separate from the paper-literal zero-bias approximation.

`common_gram_cases.csv` records the deterministic Gram, scale-symmetry, fold, and boundary cases. Those rows establish API behavior, not a population model and not a benchmark result.

12.6 The physical merge target is the executed child sum

Equation 86 on PDF page 57 defines the aggregate tuple map as a sum of functional contributions. If two child neurons contribute vector-valued maps f_i and f_j, the object that one retained physical vessel must replace is

$$F(x) = f_i(x) + f_j(x).$$

That target is different from preserving two duplicated parent coordinates in an abstract comparison space. Algorithm 2 on PDF page 69 retains one child vessel and purges the other. The realized post-surgery network therefore has one scalar ReLU activation and one outgoing vector where it previously had two.

HOPE-R scores that actual one-parent problem. For every calibration row it forms the child sum $F(x_n)$. A candidate parent has augmented affine direction $d = [w; b]$, activation

$$h_d(x_n) = \mathrm{ReLU}(d^\top [x_n; 1]),$$

and outgoing vector v. For fixed d, it solves the weighted least-squares problem

$$v_d^* = \arg\min_v \sum_n q_n \|F(x_n) - h_d(x_n) v\|_2^2.$$

The denominator-zero case has an explicit zero-output solution. Otherwise the outgoing vector is the closed-form weighted least-squares fit. The returned record includes the selected direction, fitted outgoing vector, child-sum norm, realized parent norm, absolute residual, relative residual, candidate count, span rank, and calibration-measure label.

The incoming search is deterministic and realizable. It stays within the at most two-dimensional augmented affine span of the children. It always includes the signed child

directions as anchors. A rank-two case adds a fixed full-circle angular candidate set. HOPE-R selects the smallest measured residual among that declared finite set.

proposed_repair (P12-SUM-MAP-MERGE): HOPE-R ranks a physical one-parent merge by its measured residual against the executed child sum and searches a deterministic realizable direction set.

This is why the method is named `sum_map_optimized`, not universally sum-preserving. The optimization is exact in the outgoing vector for each candidate and deterministic over the candidate set. It is not a proof of a global optimum over every possible ReLU parent, and it does not say that an arbitrary non-twin pair has a zero-residual one-neuron representation.

12.7 Exact twins receive an anchor, not a universal theorem

Exact twins are the realizable boundary a corrected merge should not miss. Two positively rescaled parameterizations can describe the same functional contribution because ReLU is positively homogeneous. Their augmented directions are collinear, and a retained child direction is already present in the candidate set.

On the declared zero-bias and biased exact-twin fixtures, the fitted outgoing vector doubles the realized anchored contribution. The one retained parent therefore matches the two-child sum within the enforced numerical tolerance. The same fixture remains invariant when the child parameterizations are changed by positive incoming scaling and reciprocal outgoing scaling.

verified_fact (P12-TWIN-PRESERVATION): on the declared exact-twin fixtures, the corrected physical merge retains an anchored child direction, doubles its realized contribution through the fitted outgoing vector, and matches the original child sum to numerical tolerance.

The non-twin control is equally important. A noncollinear pair returns a measured residual rather than being relabeled as exact. That residual is evaluated under the same calibration measure as the candidate fit. A zero value would certify equality only on those declared rows; a nonzero value is a direct local record of what the physical merge failed to preserve.

`merge_cases.csv` preserves both boundaries. It does not license a claim of universal non-twin preservation, final-logit control, or downstream loss control.

12.8 Physical surgery invalidates a graph closure

Algorithms 1 and 2 on PDF pages 68-69 mutate physical channels. Once a channel is removed or two children are replaced, the change is not confined to one cached pair in one layer.

HOPE-R requires a complete explicit graph declaration before surgery. Every physical channel has a stable `ChannelRef`. Every live input coordinate names its source. Every

producer names every direct consumer, including fan-out branches. A terminal channel may have no consumer; an unresolved nonterminal channel may not masquerade as one.

For a declared prune or merge, the dependency plan includes:

- every changed-layer capacity, outgoing, incoming, BatchNorm-assumption, and affected pair key, including removed children and the retained merge-parent vessel;
- every resolved predecessor outgoing dependency;
- every affected predecessor capacity and pair dependency;
- every resolved successor incoming and capacity dependency;
- every successor pair and BatchNorm assumption;
- every residual-block score spanning an affected layer.

The plan is constructed before physical mutation and serialized in canonical order. A merge must explicitly name which child vessel receives the synthesized parent; both child records are invalidated, the removed child is discarded, and the retained vessel is recomputed. Stable channel IDs are identity tokens, not a promise that surviving channels will be renumbered densely.

proposed_repair (P12-DEPENDENCY-CLOSURE): after physical channel surgery, HOPE-R invalidates the changed layer, all resolved predecessor and successor dependencies, every affected successor BatchNorm assumption, and every spanning block score.

This is a deliberately fail-closed contract. Grouped and depthwise convolution, shared parameters, unknown operators, incomplete consumer maps, and unresolved fan-out are rejected. The reference module does not claim to trace a framework model automatically. A caller must supply and validate the graph it wants to mutate.

The deterministic branched fixture now constructs its oracle independently of the invalidation plan. `cache_universe(graph)` enumerates 37 pre-surgery cache records from the declared graph. A separate topology surgery produces the 32-record post-prune graph, and graph-derived structural digests identify 25 removed or changed prune keys. The merge oracle independently identifies 29 keys because it also refreshes the retained physical parent. Both plan key sets match their oracles exactly, with zero missing and zero extra keys.

The refresh comparison starts from the complete pre-surgery snapshot, removes the plan's invalidated keys, recomputes surviving affected keys from the post-surgery graph, and compares the resulting dictionary with the independently built post-surgery snapshot. A changed-layer-only negative control leaves 20 dependent records stale.

verified_fact (P12-REFRESH-EQUIVALENCE): on the declared abstract branched fixture, dependency-aware structural refresh agrees with the independently enumerated post-surgery cache snapshot, while the deliberately local negative control does not. This is a structural cache-contract test, not a numerical capacity or framework-runtime equivalence claim.

`invalidation_cases.csv` records the resolved branch, closure categories, refresh comparison, and fail-closed cases. This fixture is not evidence that an undeclared production graph has been traced completely.

12.9 Resource accounting names the unit before counting it

Section 10 and Appendix B.4 distinguish constant scalar arithmetic for one cached candidate from the scan needed to choose among live candidates. HOPE-R keeps those units separate.

At every declared state, the resource record can report:

- live widths;
- live prune, merge, and eligible-block candidates;
- exhaustive candidate evaluations in the current scan;
- the full declared sequence of trajectory scan states;
- pair-cache entries and their scalar and index payload bytes;
- canonical key-record bytes, separately excluding container overhead;
- dependency-plan refresh evaluations and unique refreshed keys;
- calibration examples, input values, output elements, and branch-cost evaluations.

Immediate physical payoff is also recomputed from the live graph. Removing a channel counts its generating weights, optional bias, declared BatchNorm values, and the corresponding input coordinate in every explicit downstream consumer. Fan-out is counted branch by branch. The paper's initial static parameter payoff remains available as a separate baseline; it is not silently substituted for the live release after prior surgery.

proposed_repair (P12-LIVE-RESOURCE-ACCOUNTING): HOPE-R records initialization, live candidates, exhaustive scans, trajectory evaluations, cache storage, dependency refreshes, calibration evaluations, and immediate live physical payoff without converting those counts into a wall-clock claim.

The count record is not a benchmark. It does not include Python container overhead unless explicitly added, allocator behavior, kernel fusion, device traffic, parallel scheduling, or elapsed time. It also does not turn parameter count into FLOPs or latency. Those require separately instrumented measurements.

`resource_accounting.csv` is the canonical count record. It keeps the initial static comparator and live physical values in different fields.

12.10 Residual-block handling has a narrow eligibility gate

The paper's Eqs. 19-20 on PDF page 17 sum local internal-layer quantities and normalize by an identity-path capacity. Parts 6 and 8 showed why that construction does not automatically become a function-level bound for a sequential nonlinear residual branch.

HOPE-R does not generalize the claim. It accepts only a declared:

- ResNet V1 block;
- post-activation boundary;
- identity shortcut;
- unit stride;
- equal input and output shape.

For that eligible boundary, the calibrated repair directly compares the retained output

$$\mathrm{ReLU}(x + F(x))$$

with the evicted output

$$\mathrm{ReLU}(x)$$

on declared calibration arrays. The score is the weighted mean squared boundary displacement. It measures the executed block change rather than summing local sequential proxies.

Projection shortcuts, stride changes, ResNet V2 ordering, missing skips, shape changes, and unresolved cases are rejected with explicit reason codes. Eqs. 93-97 on PDF page 58 remain the exact identity-shortcut comparison anchor; they are not treated as authority for unsupported block families.

proposed_repair (P12-BLOCK-ELIGIBILITY): HOPE-R accepts only resolved same-shape stride-one post-activation ResNet V1 identity shortcuts for this calibrated direct branch score and fails closed outside that domain.

`block_eligibility.csv` records the accepted boundary and each named rejection. It does not show how often any block type occurs in a benchmark architecture, nor whether evicting an eligible block preserves accuracy.

12.11 DEFT keeps stable physical slots

The Part 10 audit found three distinct partition procedures across Eqs. 24-27, Definition H.1, and Algorithm 3. HOPE-R chooses one instead of trying to prove all three equivalent.

For each layer, the caller declares a fixed ordered tuple of stable slot IDs and an integer core count k_ℓ. HOPE-R ranks current functional capacity within that layer by:

1. capacity descending;
2. stable slot ID ascending for an exact tie.

Exactly k_ℓ slots become core. The remaining active slots become slack. The core count, tie rule, capacities, ranked IDs, and final assignment are returned together. Mapping insertion order and physical array order do not silently resolve ties.

The slot lifecycle has four explicit states:

State	Meaning
`core`	Source-owned complete operator protected from target updates
`slack`	Active plastic slot

State	Meaning
`released`	Fixed physical position retained but inactive and not trainable
`reinitialized`	Reused position with a live incoming path and initially silent outgoing path

A slack slot may be released without changing tensor width or stable identity. Core slots have no release transition after the partition and remain protected. Only a released slot may be reinitialized. A reinitialized slot may then enter ordinary slack, or be released again. Direct transitions that would create a reinitialized state without parameters are rejected.

Reinitialization is asymmetric on purpose. The deterministic initializer gives the slot a nonzero, normalized incoming vector, affine scale one, affine shift zero, and an all-zero outgoing vector. On the declared all-ones probe, the incoming ReLU activation is live while the initial vector contribution is exactly zero.

That creates a two-stage gradient path in the scalar squared-loss fixture. On the first backward pass, the outgoing parameter receives a nonzero gradient while the internal parameters remain blocked by the zero outgoing value. After the first outgoing update, the second backward pass reaches the incoming and affine parameters. The fixture demonstrates local gradient liveness; it does not prove convergence or useful feature learning.

proposed_repair (P12-FIXED-WIDTH-DEFT): HOPE-R uses stable fixed-width slots, one explicit layer-wise top-k current-capacity partition, live released-slot initialization, source-owned masks, cleared optimizer state, and declared BatchNorm ownership.

12.12 Core invariance needs two different masks

The structural routing mask and parameter ownership mask serve different purposes.

For a route indexed as `[source_slot, destination_slot]`, the structural mask permanently zeros every plastic-to-core edge. Slack, released, and reinitialized sources are all treated as plastic-side states for this rule. That preserves the narrow routing property associated with Eqs. 28-29 on PDF pages 22-23.

The source-owned update mask protects the complete Eq. 2 operator. For each core source slot, it zeros updates to:

- every incoming weight;
- the incoming bias;
- BatchNorm scale and shift;
- every outgoing route, including routes into slack destinations.

That final row-wide rule is the correction to destination-only ownership. A core-to-slack outgoing edge belongs to the upstream core operator even though its destination is plastic. Released slots are also nontrainable. Slack and properly reinitialized slots are trainable, subject to the separate structural route mask.

Gradient masking alone is still insufficient when optimizer state already exists. HOPE-R clears masked momentum and accepts a separate reset mask for reused entries. Its deterministic momentum plus decoupled-weight-decay step applies gradient, momentum, and weight decay only where the update mask is one. Masked values are copied directly from the original parameter tensor, and their returned momentum is exactly zero. Per-tensor diagnostics report masked element count, maximum masked drift, maximum masked optimizer state, and the exact unchanged predicate.

BatchNorm ownership is explicit as well. Scale and shift for channel i are owned by that channel's stable source slot. Running buffers use one of two declared policies:

- `frozen`: source and target evaluation select the source track, and target adaptation does not update it;
- `dual_track`: source and target own distinct running-mean and running-variance tracks, and an update changes only the selected task track.

Dual tracking preserves the source-evaluation map by retaining its source track. It does not claim that one live set of buffers is simultaneously ideal for both tasks. It stores task-specific state and must be reported as such.

Finally, evaluation cannot select state implicitly. A `StateDeclaration` names the task, head, BatchNorm track, stable slot IDs, and slot mask. An `EvaluationDeclaration` names the dataset, metric, aggregation, checkpoint rule, and required state. `TaskStateSelector` rejects a task or buffer-track mismatch. This makes the state restoration and checkpoint semantics exposed in Parts 10 and 11 part of the result definition rather than hidden setup.

verified_fact (P12-CORE-STATE-INVARIANCE): in the declared fixed-width fixtures, source-owned update masks plus the declared BatchNorm buffer policy freeze the complete core operator, including every outgoing route, under momentum and decoupled weight decay, while the reinitialized slot becomes gradient-live in two stages.

`deft_transitions.csv` records the partition, lifecycle, gradient, structural route, complete ownership, optimizer, BatchNorm, and evaluation-state transitions. The result is a deterministic state-machine witness, not source retention accuracy.

12.13 The modules compose in one order

HOPE-R is most useful as an ordered contract rather than a bag of independent patches.

1. Declare the calibration measure, graph, supported block boundaries, and task-state policy.
2. Fold finite inference-time BatchNorm state and evaluate all layer neurons on the common calibration rows.
3. Build and validate one common Gram, then compute capacities from its diagonal and each complete outgoing vector.
4. Enumerate live actions. Score a merge against the executed child sum and an eligible block against direct calibrated boundary displacement.
5. Record the current exhaustive scan, live physical payoff, cache storage, and calibration work before selecting an action.

6. Resolve the full dependency closure before physical surgery. If graph ownership or fan-out is unresolved, stop.
7. Execute the selected supported action, invalidate every returned key, and refresh those dependencies before the next scan.
8. At transfer, keep physical widths fixed, partition once by the declared layer-wise rule, initialize reusable slots, clear their optimizer state, and combine structural routing with complete source-owned update masks.
9. Evaluate only through a declaration that names the task head, slot mask, BatchNorm track, metric aggregation, and checkpoint rule.

`design_contract.csv` makes these preconditions, outputs, and fail-closed boundaries machine-readable. `integration_ablations.csv` records deterministic controls that isolate repaired module boundaries and their interaction. Those controls are implementation ablations, not trained-network or accuracy ablations.

The integration order also exposes a dependency that an isolated equation can hide. A merge changes physical outgoing and incoming coordinates. That change invalidates capacities, pair geometry, successor assumptions, and block scores. A later DEFT partition is meaningful only after those quantities have been refreshed in the same declared geometry. Likewise, a source accuracy is meaningful only after its evaluation state has been selected explicitly.

12.14 What the corrected design establishes

The Part 12 claims fall into three evidence classes.

proposed_repair:

- P12-CALIBRATED-COMMON-MEASURE chooses one declared weighted empirical law for every neuron in a layer and gives up a strict data-free claim.
- P12-KERNEL-BOUNDARY-CONTRACT makes finite-epsilon folding, zero-scale limits, and exact-versus-approximate kernel branches explicit.
- P12-SUM-MAP-MERGE optimizes a realizable physical parent against the executed child sum and reports its measured residual.
- P12-DEPENDENCY-CLOSURE invalidates the complete independently enumerated cache set for the declared resolved graph contract or fails before surgery.
- P12-LIVE-RESOURCE-ACCOUNTING records live candidates, scans, trajectory, cache, refresh, calibration, and physical payoff without a timing claim.
- P12-BLOCK-ELIGIBILITY restricts direct calibrated eviction scoring to the implemented identity-shortcut domain.
- P12-FIXED-WIDTH-DEFT specifies stable slots, one partition, reusable-state transitions, complete masks, optimizer state, and BatchNorm ownership.

verified_fact:

- P12-COMMON-GRAM-PSD holds on the declared common-measure fixtures and their fail-closed spectral validation.
- P12-TWIN-PRESERVATION holds on the declared biased and zero-bias exact-twin physical merge fixtures within numerical tolerance.

- P12-REFRESH-EQUIVALENCE holds on the declared branched cache fixture against full recomputation.
- P12-CORE-STATE-INVARIANCE holds on the declared fixed-width optimizer, buffer, and two-stage gradient fixtures.

independent_inference (P12-GUARANTEE-SCOPE): the integrated fixtures establish local algebra, geometry, graph, state, and accounting contracts for HOPE-R. They do not establish trained-network prevalence, optimizer convergence, final-logit or loss control, accuracy, latency, author-code reproduction, or universal compression superiority.

That last boundary is not a disclaimer attached after the fact. It is part of the design. Every result row carries a claim ID and evidence class so a local fixture cannot silently become a benchmark conclusion.

12.15 What remains unsupported

HOPE-R is not a completed empirical answer to the HOPE research program.

First, the common geometry is calibrated. Its sensitivity to calibration distribution, sample size, weights, preprocessing, and domain shift is open. The Part 12 fixtures show algebraic consistency, not population fidelity.

Second, the merge search is finite. Exact twins are anchored, but a general non-twin pair may leave substantial residual, and a denser direction search may select a different candidate. Even zero calibration residual does not guarantee equality outside the calibration rows.

Third, a local residual is not a network guarantee. Later nonlinearities, normalization, fan-out, and task loss can amplify or attenuate a local change. No Part 12 test bounds final logits or loss across an arbitrary network.

Fourth, the graph implementation has a deliberately narrow operator domain. Grouped and depthwise convolution, projection shortcuts, shared weights, attention, LayerNorm, GroupNorm, non-PH-1 activations, framework tracing, and unresolved fan-out are unsupported or fail closed.

Fifth, operation and byte counts are not measurements of elapsed time, energy, or peak device memory. A systems claim requires an implementation on declared hardware with synchronized measurement and a stated workload.

Sixth, fixed-width DEFT retains task-specific state under dual tracking and task-state selection. It has not shown that the selected partition is optimal, that a released slot learns useful target structure, or that source accuracy is preserved in a trained network.

Finally, this repository does not contain an authenticated author implementation against which HOPE-R could claim code equivalence. The paper-literal and proposed branches answer different questions by design.

12.16 The decisive next experiment

A credible empirical evaluation should test the repaired boundaries alone and together without changing the result definition between methods.

For compression, publish:

- the calibration-row source and weighting rule;
- exact action traces and measured per-action physical residuals;
- results at equal live parameters, FLOPs, and measured latency;
- literal versus common-Gram, literal versus sum-map merge, local versus dependency-closure refresh, and internal-score versus direct-block-cost controls;
- final-logit, loss, and accuracy changes after each supported action;
- unsupported-action counts and fail-closed reasons.

For transfer, publish:

- the exact layer-wise core counts and stable slot assignments;
- source-owned and structural masks;
- cleared optimizer-state records;
- frozen versus dual-track BatchNorm policy;
- live and restored source-state evaluation separately;
- per-scenario target, source, and H-score records with one aggregation and checkpoint rule;
- parameter, optimizer-state, and task-state storage budgets.

The essential ablation is cumulative. A common Gram cannot rescue an unrealizable physical merge. A better merge cannot rescue stale downstream caches. Correct cache refresh cannot freeze a core operator whose outgoing row and BatchNorm buffers remain plastic. Each invariant must survive after the next module is enabled.

The outcomes of those experiments are unknown. Part 12 supplies the reference contracts needed to run them without first reopening the specification.

12.17 Reproduction boundary

The canonical Part 12 numerical result contract contains exactly ten files under `artifacts/part_12/results/`:

- `summary.json`
- `design_contract.csv`
- `common_gram_cases.csv`
- `merge_cases.csv`
- `invalidation_cases.csv`
- `resource_accounting.csv`
- `block_eligibility.csv`
- `deft_transitions.csv`
- `integration_ablations.csv`
- `source_map.csv`

The files contain deterministic local fixtures and evidence metadata. They do not contain benchmark measurements. The focused tests validate the numerical contract, and the artifact checksum index authenticates the published files.

Run a disposable independent audit with:

```
python3 experiments/audits/part12_corrected_hope/run.py \
  --result-dir artifacts/runs/part_12/manual
```

Run the focused framework-independent module tests with:

```
python3 -m unittest \
  tests.test_part12_common_merge \
  tests.test_part12_graph \
  tests.test_part12_deft -v
```

Run the focused audit tests with:

```
python -m unittest discover -s experiments/audits/part12_corrected_hope -p
'test_*.py' -v
```

The runner refuses to overwrite a nonempty result directory. These local fixtures do not turn HOPE-R into author code or reproduce the paper's CIFAR-100 to SVHN or ImageNet experiments.

The final conclusion is deliberately bounded:

> HOPE's functional-neuron idea survives as a testable reference design when geometry, physical realization, graph mutation, resource units, optimizer ownership, and evaluation state are all declared. Whether that design wins on a trained benchmark remains an experiment, not a consequence of the local proofs.